Phinma University of Iloilo
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SA LIKOD NG BAHA

Submitted by:

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Abstract:

Sa Likod ng Baha is a compelling tragedy and political thriller that probes the terrible human cost of systemic corruption. The story is about Tinoy, a hardworking construction worker whose life is shattered when his wife, Linda, is swept away by a violent flash flood. In the depths of his sorrow, Tinoy discovers a harsh reality. The disaster could have been avoided if not for a crooked mayor who embezzled the money meant for building the community's vital drainage system. Fueled by a potent cocktail of love, grief and a fierce need for justice, Tinoy becomes the public face of a dangerous campaign against the town's powerful elite. He makes a dangerous alliance with a mysterious government official, and succeeds in unmasking the mayor, only to realize too late that he has been used as a pawn in a far bigger, crueler political game. Sa Likod ng Baha is an intimate, humanized portrait of grief and courage, and ultimately, a cautionary tale of the illusion of justice and the vicious cycle of political exploitation in the margins of society.

The narrative unfolds not only as a fight for truth but also as an exploration of the resilience of the human spirit amid overwhelming adversity. Tinoy's journey is a testament to the power of ordinary people who dare to confront systemic wrongs, even when faced with threats that extend beyond their personal loss. The story also delves into the complexities of power, manipulation, and the blurred lines between ally and adversary, highlighting how those who promise change may harbor their own ambitions.

Beyond the political intrigue, Sa Likod ng Baha vividly captures the socio-economic struggles of marginalized communities, illustrating how corruption exacerbates poverty and vulnerability. The vivid portrayal of the flood's devastation serves as a metaphor for the broader societal collapse caused by greed and neglect. Ultimately, the film challenges viewers to question the true cost of justice and the sacrifices made in its pursuit.

Chapter I:

Introduction:

Sa Likod ng Baha is a thrilling tale of grief, rage, and the pursuit of justice. It centers on Tinoy, a diligent construction worker whose life is turned upside down when his wife Linda perishes in a horrific storm. As he grieves, he learns that if dishonest authorities hadn't embezzled the funds intended to keep the community safe, the damage from the flood may have been less.

Determined to seek justice, Tinoy faces a world of lies, corruption, and powerful people who will do anything to keep their secrets. He learns that the real danger is not just the flood, but the people hiding behind it. This story is about courage, revenge, and hard truths. It shows that fighting for what is right is not easy, and not everyone who stands up is a hero but some will risk everything to make a change.

Beyond the personal tragedy, Sa Likod ng Baha exposes the deep-rooted challenges faced by marginalized communities, where systemic corruption worsens the cycle of poverty and despair. Tinoy's journey is not just a quest for justice, but a powerful reflection on the cost of speaking out and the sacrifices demanded by those who dare to challenge the status quo. It is a gripping reminder that true change often comes at a steep price, and that the fight against injustice is ongoing and complex.

The story also highlights the nuanced nature of power and politics, where alliances are fragile and motives are often hidden behind layers of manipulation. Tinoy's alliance with a government official reveals the uneasy partnerships that emerge in struggles for reform—partnerships that can both empower and endanger. Sa Likod ng Baha challenges audiences to look beyond surface narratives and to question who truly holds power, who benefits from the system, and what it really takes to break its grip on vulnerable communities.

Problem Statement:

Beyond the issue of corruption itself, "Sa Likod ng Baha" seeks to address the problem of systemic indifference toward vulnerable populations. In many communities affected by

poverty and recurring disasters, suffering often becomes routine, receiving attention only when tragedy reaches catastrophic proportions. Government agencies, institutions, and even society at large may gradually become desensitized to the conditions faced by marginalized citizens. This indifference creates an environment in which preventable problems remain unresolved for years, allowing infrastructure to deteriorate and communities to remain exposed to danger. The film examines how the normalization of hardship contributes to a culture in which the needs of ordinary people are continually postponed, overlooked, or sacrificed in favor of political interests and short-term gains.

The project also investigates the problem of unequal access to justice. While legal systems are intended to protect all citizens equally, the reality experienced by many disadvantaged communities is often very different. Individuals with wealth, influence, or political connections frequently possess greater access to legal resources and institutional protection than ordinary citizens. In contrast, those who lack power may encounter significant barriers when attempting to seek accountability for wrongdoing. Through Tinoy's journey, the film explores how justice can become inaccessible to those who need it most, forcing victims to navigate systems that are often intimidating, complex, and vulnerable to manipulation. This imbalance raises important questions regarding fairness, representation, and the effectiveness of institutions tasked with protecting the public.

Another major problem highlighted by the film is the disconnect between political rhetoric and lived reality. Public officials often make promises regarding development, disaster prevention, public safety, and social welfare. However, when these promises are not accompanied by genuine action, communities continue to suffer despite repeated assurances of improvement. "Sa Likod ng Baha" examines the consequences of this disconnect by portraying a community that remains vulnerable despite the existence of projects intended to protect it. The unfinished drainage system becomes a powerful symbol of broken promises and failed governance, illustrating how words alone cannot safeguard lives when action is absent. Through this narrative, the project questions the credibility of political commitments and the mechanisms through which leaders are held accountable for their responsibilities.

The film additionally addresses the problem of exploitation during periods of vulnerability. Individuals experiencing grief, trauma, or desperation are often more susceptible to manipulation by those who possess authority, influence, or strategic interests. Tinoy's emotional state following the loss of his wife makes him particularly vulnerable to individuals who present themselves as allies. The story explores how personal suffering can be transformed into a political tool, exposing the ethical dangers of exploiting victims for personal advancement. This issue extends beyond the narrative itself, reflecting broader

concerns regarding the ways in which public tragedies are sometimes used to gain influence, shape public opinion, or advance political agendas.

Another issue explored by the project is the concentration of attention on individual offenders rather than systemic failures. Public discourse frequently focuses on identifying and punishing specific corrupt officials, yet less attention is often given to the institutional conditions that allowed corruption to occur in the first place. While holding individuals accountable is important, the removal of a single leader does not necessarily eliminate the structures that encourage abuse of power. Through its portrayal of the transition from Mayor Ricardo to the Government Official, the film demonstrates how systems can remain fundamentally unchanged even when leadership changes. This problem highlights the need to examine corruption not only as an individual moral failing but also as a structural issue requiring comprehensive reform.

The project further investigates the problem of political apathy and social resignation. Repeated exposure to corruption scandals, failed reforms, and recurring injustices can lead individuals to believe that meaningful change is impossible. This sense of hopelessness may discourage civic participation and reduce public willingness to demand accountability. Through the experiences of Tinoy and his community, the film seeks to explore how despair develops and how it affects collective action. The narrative raises important questions about what motivates people to continue fighting for justice despite repeated disappointments and whether societies can overcome the fatigue produced by long-standing systemic problems.

Furthermore, "Sa Likod ng Baha" examines the issue of intergenerational consequences resulting from corruption and neglect. The effects of stolen public funds extend far beyond a single incident or disaster. Poor infrastructure, inadequate public services, and weak institutional accountability can influence the quality of life experienced by future generations. Children growing up in vulnerable communities may inherit the consequences of decisions made long before they were born. By portraying a community trapped in recurring cycles of flooding and neglect, the film highlights how corruption can create lasting disadvantages that persist across decades. This problem emphasizes the importance of long-term accountability and sustainable governance practices.

The project also addresses the issue of moral ambiguity within systems of power. Individuals who appear to support justice may possess hidden motives, while those who oppose corruption may themselves become susceptible to compromise under certain circumstances. The Government Official embodies this complexity, demonstrating how power can conceal itself behind the language of reform and public service. This aspect of the narrative challenges audiences to question simplistic assumptions regarding morality, leadership, and

institutional trust. By presenting characters with conflicting motivations and ethical contradictions, the film encourages viewers to engage with the complexities of real-world social and political issues.

Another important problem examined by the film is the failure of societies to adequately value human life when economic and political interests are prioritized. Throughout the narrative, public resources intended to protect communities are treated as opportunities for personal enrichment rather than tools for safeguarding citizens. As a result, lives become secondary to profit and political ambition. The death of Linda serves as a representation of countless invisible victims whose lives are affected by decisions made without regard for their welfare. The project seeks to challenge this devaluation of human life by emphasizing the emotional, social, and moral costs of corruption. It reminds audiences that the consequences of institutional failure cannot be measured solely in financial terms because they involve the loss of relationships, dreams, opportunities, and futures.

Finally, "Sa Likod ng Baha" seeks to address the broader problem of societal complacency toward structural injustice. Many forms of suffering persist because they have become deeply embedded within social, political, and economic systems. Communities may adapt to these conditions without questioning why they continue to exist, while institutions may focus on managing symptoms rather than addressing root causes. The film challenges this complacency by exposing the interconnected nature of corruption, poverty, vulnerability, and exploitation. Through its tragic narrative, it asks audiences to consider whether true justice can ever be achieved without confronting the systems that perpetuate inequality and suffering. In doing so, the project aims to foster critical awareness and encourage meaningful reflection on the responsibilities shared by citizens, leaders, and institutions in creating a more equitable society.
This problem includes:
Highlighting the impact of natural disasters on communities
Exploring personal loss and the quest for justice
Revealing political corruption and its consequences
showing the moral complexity of heroes and villains

Objective:

Another significant objective of "Sa Likod ng Baha" is to examine the concept of environmental justice and the unequal burden that disasters place upon marginalized communities. While natural calamities can affect entire regions, their impact is rarely distributed equally among all members of society. Those living in poverty often reside in areas with inadequate infrastructure, limited access to emergency resources, and fewer

opportunities for recovery. The film seeks to demonstrate how socioeconomic inequality magnifies the consequences of environmental disasters, transforming what might otherwise be a manageable crisis into a life-altering catastrophe. Through the experiences of Tinoy and his community, the narrative reveals how vulnerable populations are frequently forced to endure the harshest effects of both natural and man-made failures. By highlighting these realities, the film encourages audiences to consider the connections between poverty, public policy, environmental management, and social responsibility, emphasizing that the protection of communities requires more than emergency response—it requires long-term investment, planning, and accountability.

The film also aims to explore the relationship between power and invisibility. Throughout the story, many of the individuals responsible for the suffering of ordinary citizens remain physically distant from the consequences of their actions. Decisions are made in offices, meetings, and political circles, while the resulting hardships are experienced in flooded homes, crowded evacuation centers, and devastated neighborhoods. This separation allows those in positions of authority to overlook or underestimate the human impact of their choices. "Sa Likod ng Baha" seeks to close this distance by forcing audiences to confront the lived experiences of those most affected by corruption and neglect. Through its intimate portrayal of grief, loss, and resilience, the film gives visibility to voices that are often ignored within larger political discussions. The objective is to challenge viewers to recognize the humanity behind public issues and to understand that governance is not merely a matter of policy but a matter of human lives.

A further objective of the film is to examine how communities respond to collective trauma. Disasters often leave behind not only physical destruction but also emotional scars that shape relationships, identities, and social dynamics for years to come. The narrative seeks to portray the ways in which individuals attempt to rebuild their lives while carrying memories of loss and uncertainty. Through scenes of mourning, reflection, and resistance, the film illustrates how communities struggle to preserve hope in the face of overwhelming adversity. At the same time, it acknowledges the emotional fatigue that can emerge when promises of reform repeatedly fail to materialize. By presenting these realities, the story encourages audiences to appreciate the strength required to endure hardship while recognizing the psychological toll that prolonged injustice can inflict upon individuals and communities alike.

The film additionally seeks to investigate the moral ambiguity that often exists within systems of power. Rather than presenting a simplistic conflict between good and evil, "Sa Likod ng Baha" portrays a world in which motivations are complex and appearances can be deceiving. The Government Official represents a particularly important aspect of this exploration. He initially appears to be an ally, a figure committed to exposing corruption and

seeking justice for victims. However, his eventual betrayal reveals how easily noble causes can be appropriated for personal gain. This narrative choice serves to challenge audiences' assumptions and encourages a deeper examination of the ethical complexities that characterize political and social struggles. The objective is to foster critical thinking by demonstrating that individuals should be evaluated not solely by their words or public image but by the systems they create, the actions they take, and the consequences of those actions.

Another key objective is to highlight the long-term consequences of corruption across generations. The effects of stolen public funds do not end with a single disaster or a single victim. They can influence the quality of education, healthcare, infrastructure, and economic opportunities available to future generations. Through the symbolism of the unfinished drainage project and the recurring floods that continue to plague the community, the film illustrates how corruption creates cycles of disadvantage that persist over time. The suffering experienced by Tinoy and Linda is therefore not an isolated incident but part of a broader pattern that threatens the future of countless families. By emphasizing these long-term consequences, the film encourages audiences to think beyond immediate events and to consider how present decisions shape the lives of future generations.

"Sa Likod ng Baha" also aims to explore the fragility of trust within societies affected by corruption. Trust is essential for healthy relationships between citizens, communities, and institutions. However, when leaders repeatedly betray public confidence, trust becomes difficult to maintain and even more difficult to restore. Throughout the narrative, characters struggle to determine whom they can believe and where they can find support. This uncertainty reflects a broader social reality in which promises of change are often met with skepticism due to repeated disappointments. The film seeks to capture this atmosphere of doubt while encouraging viewers to reflect on the importance of transparency, accountability, and integrity in rebuilding public confidence. The objective is to demonstrate that trust is not merely a political issue but a deeply human one that influences people's sense of security, belonging, and hope for the future.

Furthermore, the film seeks to investigate the role of memory in shaping collective consciousness. Communities often carry memories of disasters, injustices, and struggles long after the events themselves have passed. These memories become part of a shared identity, influencing how people understand their history and imagine their future. Through Tinoy's memories of Linda and the community's recollection of the flood, the narrative illustrates how remembrance can serve as both a source of pain and a source of strength. The objective is to underscore the importance of preserving these stories, ensuring that victims are not forgotten and that the lessons of the past continue to inform future generations. By

emphasizing memory, the film positions storytelling itself as an act of resistance against indifference and historical erasure.

The film also aims to challenge the normalization of suffering that often occurs within disadvantaged communities. When poverty, corruption, and disasters become recurring realities, there is a risk that society begins to view such hardships as inevitable. "Sa Likod ng Baha" rejects this perspective by emphasizing that the conditions faced by Tinoy and his neighbors are not natural or unavoidable. They are the result of choices, priorities, and systems that can be changed. The objective is to encourage audiences to question assumptions about what is considered normal and to recognize that every individual deserves access to safety, dignity, and opportunity. By exposing the preventable nature of much of the suffering depicted in the story, the film seeks to inspire a greater sense of urgency regarding social reform and public accountability.

As a work of cinema, the film also aspires to demonstrate the power of storytelling as a tool for social awareness and emotional engagement. Art possesses a unique ability to communicate complex issues through personal experiences that resonate with audiences on a deeply emotional level. Through its characters, imagery, and narrative structure, "Sa Likod ng Baha" seeks to transform abstract discussions about corruption, governance, and inequality into relatable human experiences. The objective is not merely to inform audiences but to encourage empathy, reflection, and dialogue. By combining emotional storytelling with social critique, the film hopes to contribute to broader conversations about justice, responsibility, and the values that shape society.

Ultimately, the film seeks to leave audiences with an enduring sense of reflection regarding the relationship between individuals and institutions. It challenges viewers to consider what genuine justice looks like, what responsibilities accompany positions of authority, and what sacrifices are often demanded of those who pursue truth. More importantly, it asks whether society is willing to address the root causes of suffering rather than merely reacting to its symptoms. Through Tinoy's story, the film underscores the idea that meaningful change cannot be achieved through symbolic victories alone. It requires sustained commitment, collective action, ethical leadership, and a genuine respect for human dignity. In presenting these themes, "Sa Likod ng Baha" aspires to become not only a compelling narrative but also a powerful reflection on the social realities that continue to shape the lives of countless individuals and communities..

Simple Visual Mix: To make the project look unique and intriguing without requiring a large budget, use 2D drawings for individuals and 3D drawings for the environment.

Relatable Sadness: Tell a tale that evokes some of the annoyance and "pagod" (exhaustion) that ordinary people experience when they deal with misfortune and dishonest employers.

Basic Atmosphere: Use simple 3D rain, shadows, and local sounds to make the setting feel like a real flooded neighborhood that the audience recognizes.

Clear Message: Make sure the audience gets the point: that sometimes, the "new guy" in charge is just as bad as the old one, and life for the poor stays the same.

Scope and Limitation of the Study

As "Sa Likod ng Baha" begins its journey to create an emotionally moving animated short film, we must define its scope and acknowledge its limitations. This section outlines the boundaries of our work and the constraints that will shape the final story.

Project Scope

2D Animation Production: The main goal is to create a short film that puts flat 2D characters into a 3D world, covering everything from drawing the people to building the 3D sets.

Visual Contrast: We will focus on how 2D drawings interact with 3D rain and shadows to show how "fragile" regular people are compared to a cold, heavy system.

Atmospheric Storytelling: The project uses dark visuals and gritty sounds to make the audience feel the mood of the flood and the sting of the final betrayal.

Project Limitations

Only Three Locations: To keep the 3D work simple, the story only happens in the slums, the worksite, and the town hall.

No Complex Movement: We will skip difficult 3D character acting and big action scenes to focus on simple 2D expressions and the environment.

Focused Story: The plot stays strictly on our character; Tinoy’s perspective, avoiding a big political world to keep the "human cost" as the main point. An additional scope of "Sa Likod ng Baha" involves the exploration of social realism through the medium of animation. While animated films are often associated with fantasy, entertainment, or stylized storytelling, this project seeks to demonstrate how animation can effectively communicate serious social issues and emotionally grounded narratives. The film aims to utilize visual symbolism, environmental storytelling, and character-driven drama to depict themes of corruption, loss, and injustice. However, the project is limited to artistic interpretation rather than complete realism. Certain visual elements, environmental designs, and character portrayals may be stylized to support the emotional and thematic goals of the narrative. As a result, the film

should be understood as a creative representation of social realities rather than a direct reconstruction of actual events or locations.

The scope of the project also includes the deliberate use of symbolism as a storytelling device. Elements such as floodwaters, rainfall, darkness, abandoned structures, and damaged personal belongings are intended to function as visual metaphors for larger social issues. These symbols help communicate complex themes without relying solely on dialogue or exposition. However, the interpretation of symbolic imagery is inherently subjective and may vary among audience members. Consequently, the project cannot guarantee that all viewers will derive identical meanings from the film's visual language. This limitation is acknowledged as a natural characteristic of artistic expression and audience interpretation.

Furthermore, the project focuses specifically on the experiences of individuals belonging to working-class communities. Through Tinoy's perspective, the film examines how ordinary citizens are affected by corruption and institutional neglect. While this focus allows for a more intimate and emotionally resonant narrative, it also limits the range of perspectives represented within the story. The film does not extensively explore the viewpoints of policymakers, disaster management professionals, journalists, law enforcement agencies, or other stakeholders who may be involved in similar situations. Instead, the narrative remains intentionally centered on the experiences of those directly affected by the consequences of systemic failures. This creative decision supports the project's objective of emphasizing the human cost of corruption while acknowledging the limitations of a single-perspective narrative.

Another significant limitation concerns the project's production timeline and available resources. As an animated short film, the project must operate within practical constraints related to scheduling, rendering capacity, software performance, and asset creation. Due to these limitations, certain visual elements may be simplified to ensure project completion within the allotted production period. Complex environmental simulations, advanced fluid dynamics, highly detailed character rigs, extensive crowd animations, and cinematic-scale destruction sequences may not be fully realized. Instead, the project prioritizes visual storytelling, emotional clarity, and thematic consistency over technical spectacle. This approach ensures that production resources are allocated efficiently while maintaining the integrity of the narrative.

The project is also limited in its representation of disaster response and recovery processes. While the story portrays the immediate consequences of flooding and the emotional aftermath experienced by survivors, it does not attempt to provide a comprehensive examination of disaster management systems, emergency response protocols, humanitarian assistance

programs, or long-term rehabilitation efforts. These topics are beyond the primary focus of the film and would require a significantly broader scope of research and narrative development. Consequently, the project concentrates on the personal and societal consequences of the disaster rather than the technical or administrative aspects of disaster response.

From an audience perspective, the project is intended to appeal primarily to viewers interested in socially relevant narratives, dramatic storytelling, and contemporary societal issues. While the themes explored in the film possess universal relevance, audience reception may vary depending on cultural background, personal experiences, political perspectives, and familiarity with the issues depicted. The project acknowledges that viewers may interpret characters, events, and thematic messages differently based on their own beliefs and experiences. Therefore, the film does not seek to impose a singular viewpoint but rather encourages discussion, reflection, and critical engagement with the ideas presented.

Additionally, the scope of the project includes the examination of emotional realism through character relationships and personal loss. Significant emphasis is placed on portraying grief, vulnerability, determination, and betrayal in a manner that feels authentic and relatable. However, due to the limited runtime of a short film, character arcs and interpersonal relationships may not receive the same depth of exploration typically found in feature-length productions. Certain emotional developments may be condensed or communicated through visual cues, symbolism, and brief interactions rather than extended dialogue sequences. This limitation is an inherent aspect of the short-film format and requires careful balancing between narrative efficiency and emotional impact.

The project is further limited in terms of geographical representation. While the setting is inspired by communities vulnerable to flooding and infrastructure-related issues, the film does not aim to depict every environmental, cultural, or socioeconomic condition present in such areas. The location functions primarily as a narrative space that supports the story's themes and emotional objectives. As a result, certain environmental details may be simplified, generalized, or adapted for storytelling purposes. The film's setting should therefore be interpreted as a symbolic representation of vulnerable communities rather than a precise depiction of a specific location.

Another important limitation involves the measurement of the film's impact. Although the project aspires to raise awareness regarding corruption, accountability, and social inequality, its effectiveness in changing attitudes, beliefs, or behaviors cannot be guaranteed. Audience reactions are influenced by numerous external factors beyond the control of the filmmakers, including personal values, educational background, prior knowledge, and social context.

Consequently, any conclusions regarding the film's influence will be limited to observed audience responses and feedback rather than definitive claims about long-term social or political impact.

Moreover, the project intentionally avoids providing simplistic solutions to the complex issues it portrays. Corruption, poverty, environmental vulnerability, and systemic inequality are multifaceted problems that cannot be adequately resolved within the framework of a short narrative film. Instead of presenting definitive answers, the story seeks to encourage reflection and dialogue regarding these challenges. This limitation reflects a conscious creative choice to prioritize critical thinking over prescriptive messaging, allowing audiences to engage with the themes on a deeper and more personal level.

Finally, "Sa Likod ng Baha" is limited by the nature of storytelling itself. While the film endeavors to portray human suffering, resilience, and injustice with honesty and sensitivity, no single narrative can fully capture the diversity of experiences associated with disaster, corruption, and social inequality. Countless individuals and communities possess unique stories, perspectives, and struggles that extend beyond the boundaries of this project. The film therefore serves as one narrative among many, offering a focused exploration of these issues through the experiences of its characters. Despite these limitations, the project aspires to contribute meaningfully to conversations about accountability, empathy, social responsibility, and the importance of recognizing the human lives that exist behind every statistic, policy failure, and public tragedy.

Significance of the study

In "Sa Likod ng Baha," we are using a unique 2D animation style to tell a grounded, gritty story. Here is why our study matters:
Visual Innovation (2D/3D Hybrid): We are exploring how putting flat 2D characters into 3D worlds can create a new kind of "Pinoy Noir" aesthetic. This helps future animators see how to mix different styles to show the contrast between people and their environment.

Highlighting Social Realities: Our study uses animation to show the "human cost" of disasters and corruption. By focusing on our character; Tinoy’s story, we want to show that animation is a powerful tool for talking about real-life struggles that many Filipinos face.

Inspiring Local Creators: We want to prove that you don't need a massive budget for complex 3D acting to tell a heavy, emotional story. By sharing how we used lighting, weather, and

simple 2D expressions, we hope to inspire other students to create high-impact films with limited resources.

4. The completion of this study, alongside the rigorous conceptualization and production of the animated short film "Sa Likod ng Baha," holds profound theoretical, practical, cultural, and socio-economic value. By bridging the critical gap between innovative digital production techniques and urgent socio-political discourse, this research transcends the traditional boundaries of standard student productions. It positions animation not merely as visual art, but as an active agent for societal reflection. Specifically, this study provides vital insights and serves as a foundational reference for the following diverse sectors:

5. For Local Animators, Independent Filmmakers, and the Animation Economy
This study acts as a critical, liberating turning point for grassroots creators by actively dismantling the widespread, discouraging misconception that high-impact, emotionally heavy storytelling is a luxury reserved exclusively for major studios backed by massive financial capital. In the local independent landscape, creators frequently find themselves severely bottlenecked by inadequate hardware and a lack of access to high-end industry tools. This research pulls back the curtain on a highly viable alternative: a comprehensive, meticulously documented production blueprint that maximizes atmospheric lighting, dynamic weather systems, and hand-drawn 2D expressions within accessible 3D spaces.

6. Furthermore, this study champions the shift of the Philippine animation industry from being primarily a service-oriented outsourcing hub for foreign studios into a powerhouse of Original Intellectual Property (IP) creation. By proving that economic design choices can yield deeply moving, cinematic experiences without the need for expensive motion-capture technology, this research empowers local animators to tell homegrown stories that are globally competitive but distinctly Filipino in flavor and execution.

7. For the Evolution of Philippine Animation and the "Pinoy Noir" Aesthetic
On a technical and stylistic level, this research pioneers a distinct visual identity within the local industry by successfully marrying flat, stylized 2D characters with gritty, deeply textured three-dimensional environments. In doing so, it formalizes a fresh benchmark for the "Pinoy Noir" aesthetic—an emergent sub-genre that takes the classic, shadow-drenched undertones of traditional film noir and roots them firmly in the unique socio-political, architectural, and meteorological realities of the Philippines.

8. The stark visual contrast between the flat, inherently vulnerable 2D characters and the heavy, uncaring 3D world they inhabit serves as a deliberate, baked-in visual metaphor. It mirrors the crushing isolation of the human condition and the helplessness of the ordinary

citizen when pitted against vast, deeply embedded systemic issues. This study encourages local creators to break away from safe, commercially driven westernized imitations, urging them to explore experimental, avant-garde visual expressions that can proudly expand the definition of Philippine digital arts.

9. For Academic Institutions, Educators, and Cross-Disciplinary Theorists
For the academic community, this study addresses a notable gap in local literature regarding the integration of hybrid dimensions (2D inside 3D spaces) specifically utilized for dramatic and tragic narratives. Most local animation literature focuses heavily on traditional folklore or highly commercialized 3D works. By documenting the technical compositing methods and artistic theories used in "Sa Likod ng Baha," this research provides a fresh academic reference for film and animation curricula across the country.

Moreover, its value extends beyond multimedia arts into sociology, political science, and media studies. It gives educators a concrete, localized case study to teach students how art direction can directly serve subtext, themes, and political commentary, thereby elevating the standard of cross-disciplinary academic discourse surrounding local media.

10. For Environmental Advocates and Disaster Risk Reduction Management (DRRM)
The Philippines consistently ranks as one of the most vulnerable nations in the world to climate change and extreme weather events. Therefore, this study functions as an important piece of eco-critical media. By intimately showcasing the terrifying reality of a localized flood, it brings a deeply human face to the often-abstract statistics of climate vulnerability and disaster preparedness. It serves environmental advocates and DRRM sectors by illustrating the catastrophic intersection where natural calamities meet man-made institutional failure, reinforcing the urgent message that while we cannot stop the rain, the resulting devastation is often entirely preventable.

11. For Psychological Practitioners and the Processing of Collective Trauma
At its emotional core, "Sa Likod ng Baha" is an exploration of grief, powerlessness, and the psychological toll of systemic betrayal. For psychologists, sociologists, and trauma counselors, this study illustrates the therapeutic and cathartic power of narrative media. Tinoy’s journey represents the collective trauma of thousands of Filipinos who have lost homes, livelihoods, and loved ones to preventable floods. By seeing this specific, localized grief validated on screen, audiences are provided a space for collective mourning and catharsis. The study highlights how animation can be used as a psychological tool to process complex emotions that are often too painful to confront in live-action formats.

12. For Social Advocates, Non-Governmental Organizations (NGOs), and the General Public

This project aggressively redefines the cultural boundaries of animation in the Philippines, challenging the deeply ingrained societal notion that the medium is strictly meant for children's entertainment or lighthearted escapism. By grounding the narrative in the raw, lived experiences of marginalized Filipinos, the study demonstrates that animation is a potent vehicle for hard-hitting social commentary. Through the lens of Tinoy's tragedy, abstract headlines about embezzled infrastructure funds, unfinished drainage projects, and nameless death tolls are transformed into a visceral human tragedy. It equips social advocates and the public with an emotionally resonant tool to spark vital conversations about institutional accountability and the dangerous cycle of political gaslighting.

13. For Local Government Units (LGUs) and Policy Makers

Though an artistic endeavor at its core, this study holds profound indirect significance for policymakers and local leaders by serving as an empathetic mirror to the real-world consequences of administrative negligence. By illustrating the fatal domino effect of corruption—where a stolen drainage fund directly translates to a citizen drowning in their own home—the film strips away the bureaucratic abstraction of local governance. It underscores the urgent, life-or-death need for absolute transparency, strictly monitored infrastructure development, and genuine empathy in public service, reminding those in power that budget allocations dictate the literal survival of their constituents.

14. For Future Media Historians and Cultural Archivists

Finally, this research serves as a crucial historical and cultural time capsule, capturing the contemporary anxieties, fears, and frustrations of the Filipino working class during an era heavily marked by climate vulnerability and political disillusionment. By rigorously archiving the creative and conceptual process behind "Sa Likod ng Baha," future media historians will gain a much clearer, more intimate understanding of how young Filipino creators utilized emerging digital storytelling mediums to dissent, reflect, and cope with the defining systemic challenges of their generation. It cements the film not just as an aesthetic milestone, but as an active, enduring artifact in the long history of Philippine protest art.

Chapter II

Related Literature:

The study "Sa Likod ng Baha" draws inspiration from various literature sources that shed light on the significance of 2D and 3D animation and its influence on modern digital art.

(1)The name of the study is; A MEDIA FRAMING ANALYSIS OF THE FLOOD CONTROL ISSUE IN THE
PHILIPPINES: CURRENT NARRATIVES AND IMPLICATIONS FOR POLICY, from
International Journal of Engineering Technology Research & Management

The purpose of this study was to examine how the Philippines' flood management problem is framed in the media, as well as its contemporary narratives and policy consequences. It incorporates the four discursive dimensions of Framing theory in content analysis—problem identification, cause attribution, moral judgment, and conclusion—and is based on both Agenda-Setting theory and Framing theory. This qualitative study looks at how flood control is portrayed in Philippine media.

The results of this study included information from local and national media sources, including print, radio, television, and the internet. The findings emphasize the predominance of political issues as well as the presentation of qualitative evidence in problem identification, causality, morality, and conclusion. Based on these results, the report offers policy recommendations for boosting media coverage in the Philippines and strengthening governance through cooperation with the business sector, civil society, media professionals, and academic institutions.

Introduction:

The Philippines is recognized as one of the most disaster-prone countries in the world, mainly due to its unique geographical location (Gallman, 2021). The most common disasters are flooding and landslides, which impose a significant burden on the national economy (Sotto, 2022; Tolentino et al., 2025). Moreover, frequent disasters challenge the Philippines' sustainable development by increasing socioeconomic and humanitarian impacts. For example, recent Typhoon Tino (Kalmaegi) and Super Typhoon Uwan (Fung-Wong) left devastation across central and northern Luzon. Typhoon Tino caused 232 deaths, 100 missing persons, and affected over 4.1 million people, with 135,000 homes damaged or destroyed

(Philippine News Agency, 2025). Following closely, Super Typhoon Uwan struck the country again, displacing over 1.4 million people and resulting in at least 18 deaths (Oxfam Pilipinas, 2025). This highlights the pressing need for robust governance to foster resilience against disasters. However, the Philippines' vulnerability to disasters is not solely due to natural factors; the failure of flood control projects has also worsened disaster risk nationwide (Adekola & Lamond; Sotto, 2022; Tolentino et al., 2025). The country's vulnerability is linked to corruption and government inefficiency, highlighting allegations of misused public funds, ghost projects, and substandard infrastructure that costs more than it should (Gera, 2025; Greenpeace Philippines, 2025; OneNews.PH, 2025). This undermines the effectiveness of the country's disaster response framework (Gera, 2025). Furthermore, the media significantly shapes public understanding and influences government policy priorities. Gallman (2021), Moncada et al. (2024), and OSOU (2013) noted that media use framing techniques to craft narratives that direct both public and government attention to viral issues (Moncada et al., 2024; Raza et al., 2025). Media coverage emphasizes casualties and government accountability during disasters, highlighting responsibility for failed flood control projects (Adekola & Lamond, 2018; Raza et al., 2025). Additionally, the media discusses tensions among government agencies, the private sector, and civil society regarding corruption

(https://ijetrm.com/issues/files/Dec-2025-29-1766978085-DEC82.pdf)

ISSN/ISBN: 2456-9348

(2)FLOOD RISK ASSESSMENT IN POST ANTIAO RIVER CONTROL PROJECT IN CATBALOGAN CITY, PHILIPPINES

Cities in underdeveloped nations frequently experience flooding, mostly as a result of inadequate planning, designing, and carrying out projects. Floods are caused by factors including as population growth, an antiquated drainage system that isn't working, and a badly silted With smaller rivers, inadequate waste disposal, a lack of erosion prevention strategies, decreased vegetation and the exceptionally high precipitation. In reaction to The local administration requested assistance from the Department of Highways and Public Works, which subsequently created a Flood Management and Master Plan for the Drainage System, which incorporates a River Control Project building river walls and desilting a nearly two-

kilometer section of the river. River discharge was calculated using a hydrologic modeling system (HEC-HMS) version 4. computed for a minimum of four extreme occurrences. Findings indicate that the project won't address the issue of floods brought on by excessive precipitation. The walls of the river are more essential upstream than the river's downstream section, yet the. The river's disiltation is insufficient to transport stormwater. Consequently, there is an Building river walls upstream is critical, along with other treatments. outlined in the master plan to address Catbalogan's flooding issues.

Introduction

The changing climate has increased the intensity of precipitation which has exceeded capacities of natural and humanmade drainage systems (www.eoearth.org, 2014, www.apf,.info, 2012). The rapid growth in population and urbanization has forced communities to change land use. These changes in ground cover from vegetation to impervious layers like paved surfaces and buildings has reduced infiltration capacity and increased volume of run-off (www.epa.gov). Natural disasters in the Philippines resulted into PHP 5B to PHP 15B direct damages which are about 9.5% of the country's Gross Domestic Product (GDP) and 972 casualties per year from 1970-2006 (Porcil, 2009). About 20 typhoons cross the Philippines yearly, 5 of which are destructive (ibid). Storms carry with them heavy precipitation swelling canals, streams and rivers. In September 2009, Metro Manila in the Philippines experienced a recordbreaking flood due to a 448.5mm of rain in 12 hours placing the over 11 million citizens to a virtual standstill (Lagmay, undated). Catbalogan has a fair share of floods due to unusually heavy rain causing the river to
ISSN 2449-457

https://www.researchgate.net/profile/Ronald-Orale/publication/318494852_Flood_Risk_Assessment_in_Post_Antiao_River_Control_Project_in_Catbalogan_City_Philippines/links/596de364aca272d552fe2410/Flood-Risk-Assessment-in-Post-Antiao-River-Control-Project-in-Catbalogan-City-Philippines.pdf

(3)Flood Projects and Filipino Protests: The Call to Improve Anti-Corruption Law and Infrastructure in the Philippines

Protests broke out in September 2025 in Manila, the political center of
The Philippines, in reaction to recent claims of embezzling money meant for infrastructure initiatives. It was discovered that an estimated $9.5 billion ($552 billion Philippine pesos) had been stolen. 9,855 flood control initiatives. President Marcos Jr. oversaw the probe, which examined flood projects and found uneven flood management audits, then developed the The Independent Commission for Infrastructure (ICI) will look into corruption cases in

more detail. The anger over repeated floods has led to recent demonstrations against government malfeasance. and earthquakes, as well as the inability to bounce back from them. Communities in the tropical The nation has frequently experienced flooding during the habagat (wet monsoon) season. Scenes from
Filipinos wading or swimming over the sea and property destruction have become commonplace in news networks in the Philippines. The notable difference in living conditions between the general population and the elite shows structural injustices that are directly related to the standard of infrastructure that would shield communities from severe flooding. extreme economic disparity and misallocation of money is a reflection of the treachery and corruption of the Philippine political elite by enabling the rich to
prosper while millions endure poverty and systemic inequality. Following anti-corruption rallies and ongoing natural calamities, significant legal reforms in the Philippines should concentrate on political openness, appropriate funding distribution for infrastructure projects, and heightened regulation of utilization of the public revenues.

DOI: 10.57912/31828555
file:///C:/Users/User/Downloads/Flood%20Projects%20and%20Filipino%20Protests.pdf

(4)GOVERNING FLOOD RISK IN THE PHILIPPINES: MECHANISMS OF DISASTER RISK PRODUCTION IN FLOOD CONTROL INFRASTRUCTURE

Flooding is still a major problem in the Philippines despite decades of investment in flood control infrastructure, indicating significant difficulties for Disaster Risk Reduction (DRR). This research contends that normal governance practices actively increase flood risk, despite the fact that most explanations point to climatic and technical variables. We go beyond criticisms of institutional failure by using political ecology as a diagnostic lens to pinpoint and demonstrate three distinct causal mechanisms of disaster risk creation: (1) corrupt implementation that creates subpar infrastructure; (2) the deliberate exploitation of institutional fragmentation to facilitate rent-seeking and diffuse accountability; and (3) exclusionary planning that shifts risk onto marginalized communities. The paper uses a qualitative case study technique to provide an organized investigation of secondary evidence from three representative examples, such as Commission on Audit reports, investigative journalism, and project documentation. This methodical approach provides a novel explanation of risk formation based on mechanisms. The results show that these dynamics are ingrained in the political economics of infrastructure and are not incidental. Therefore, we contend that a fundamental change from infrastructural solutions to governance interventions that interfere with these risk-producing pathways is necessary for effective catastrophe risk reduction. In order to provide a tangible link between political ecology theory and policy practice, we suggest specific, implementable reforms such as the establishment of an

independent infrastructure ombudsman, mandatory social audits, and legally required co-design procedures.

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6213037
Lucero, L., Lagunda, S. J., Catamco, D., & Sy, F. A. GOVERNING FLOOD RISK IN THE PHILIPPINES: MECHANISMS OF DISASTER RISK PRODUCTION IN FLOOD CONTROL INFRASTRUCTURE. SSRN 6213037.
From vulnerability to resilience: Addressing the causes, impacts, and solutions for recurrent flash floods in the Philippines

The Philippines is one of the countries most vulnerable to natural disasters. Its islands are frequently hit by floods, typhoons, landslides, earthquakes, volcanic eruptions, and droughts due to their location along major tectonic plates and in a typhoon belt .The country is blessed with oceans, rivers, lakes, and streams, but weather events can release massive amounts of water, causing flooding. This study looks into the causes, consequences, and remedies of flash floods in the Philippines. It focuses on large floods, particularly those brought on by monsoons and typhoons, as well as vulnerable areas and ASEAN preventative strategies. This study supports Ary's idea that descriptive research aims to capture the existing condition of circumstances by using a descriptive-evaluative research approach. To inform the study's conclusions, it involves data mining from trustworthy, relevant sources. According to analysis, all Philippine areas are at significant risk from recurring flash floods, with Region III being especially vulnerable. Super Typhoon Yolanda's devastating flooding was the Philippines' worst flooding incident between 2010 and 2020.

https://www.sciencedirect.com/science/article/pii/S2225603225000372
Rivera, A. T., & Vega, J. B. G. D. (2025). From vulnerability to resilience: Addressing the causes, impacts, and solutions for recurrent flash floods in the Philippines. Tropical Cyclone Research and Review.

(6)A MEDIA FRAMING ANALYSIS OF THE FLOOD CONTROL ISSUE IN THE PHILIPPINES: CURRENT NARRATIVES AND IMPLICATIONS FOR POLICY

The purpose of this study was to examine how the Philippines' flood control problem is framed in the media, its prevailing narratives, and policy ramifications. It incorporates the four principles of agenda-setting theory and framing theory. Framing theory's discursive aspects in content analysis include problem identification, causal attribution, and moral assessment and conclusion. This qualitative study looks at how the Philippine media presents the topic of flood prevention. Information from local and international media sources, such as print, radio, television, and websites National were portrayed as the study's conclusions. The

findings demonstrate how qualitative data is presented in problem identification, causality, morality, and conclusion, in addition to the predominance of political issues. The report offers policy recommendations for improving governance through cooperation based on these findings. with academic institutions, the commercial sector, civil society, and media professionals, as well as for enhancing the efficacy of Philippine media coverage.

Introduction

The Philippines is recognized as one of the most disaster-prone countries in the world, mainly due to its unique geographical location (Gallman, 2021). The most common disasters are flooding and landslides, which impose a significant burden on the national economy (Sotto, 2022; Tolentino et al., 2025). Moreover, frequent disasters challenge the Philippines' sustainable development by increasing socioeconomic and humanitarian impacts. For example, recent Typhoon Tino (Kalmaegi) and Super Typhoon Uwan (Fung-Wong) left devastation across central and northern Luzon. Typhoon Tino caused 232 deaths, 100 missing persons, and affected over 4.1 million people, with 135,000 homes damaged or destroyed (Philippine News Agency, 2025). Following closely, Super Typhoon Uwan struck the country again, displacing over 1.4 million people and resulting in at least 18 deaths (Oxfam Pilipinas, 2025). This highlights the pressing need for robust governance to foster resilience against disasters. However, the Philippines' vulnerability to disasters is not solely due to natural factors; the failure of flood control projects has also worsened disaster risk nationwide (Adekola & Lamond; Sotto, 2022; Tolentino et al., 2025). The country's vulnerability is linked to corruption and government inefficiency, highlighting allegations of misused public funds, ghost projects, and substandard infrastructure that costs more than it should (Gera, 2025; Greenpeace Philippines, 2025; OneNews.PH, 2025). This undermines the effectiveness of the country's disaster response framework (Gera, 2025). Furthermore, the media significantly shapes public understanding and influences government policy priorities. Gallman (2021), Moncada et al. (2024), and OSOU (2013) noted that media use framing techniques to craft narratives that direct both public and government attention to viral issues (Moncada et al., 2024; Raza et al., 2025). Media coverage emphasizes casualties and government accountability during disasters, highlighting responsibility for failed flood control projects (Adekola & Lamond, 2018; Raza et al., 2025). Additionally, the media discusses tensions among government agencies, the private sector, and civil society regarding corruption

https://ijetrm.com/issues/files/Dec-2025-29-1766978085-DEC82.pdf
Campana, C. P., & Manguiob, J. P. A MEDIA FRAMING ANALYSIS OF THE FLOOD CONTROL ISSUE IN THE PHILIPPINES: CURRENT NARRATIVES AND IMPLICATIONS FOR POLICY.

(7)Political Economy: The Truth Behind Politics and the Challenges in Advancing Growth and Development of the Philippine Economy

The interconnection of political, economic, and legal institutions in determining a country's course of development is highlighted by political economy. The "power of the purse," or legislative control over public spending, is crucial in determining the country's economic goals in the Philippines. However, recent disclosures on the execution of flood control projects revealed widespread corruption among legislators, public servants, and private contractors. These irregularities erode public confidence, worsen climate-related vulnerabilities, and jeopardize the integrity of infrastructure
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https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5807262
Piano, F. J., Santos, F. D., Barcelona, V., & Barboza, N. (2025). Political Economy: The Truth Behind Politics and the Challenges in Advancing Growth and Development of the Philippine Economy. N/A.

The Dynamics of "Grand Corruption"
A system of widespread corruption has long existed in the Philippines. However, the incidents of
The biggest anti-graft demonstrations this nation has witnessed in almost a ten years. Millions in taxpayer loses to "ghost" are the main source of the outrage. infrastructural initiatives like flood prevention in the provinces most at danger during the season of typhoons. Even more annoying, individuals are left wondering since, as an archipelago, encounter 20 tropical cyclones on average each year, making our country extremely susceptible to flooding. Then there were reports and other instances of horrifying information from Senate investigations. There According to reports, former government engineers testified that the projects were purposefully maintained at inadequate or nonexistent levels. But the accountants had channeled distributing the money to lawmakers and bureaucrats in the form of large bribes. The public's outrage stems from concrete, since it highlights the terrible irony of people's homes being damaged by floods.
whereas it is alleged that the money that could have been utilized to defend them was stolen.

https://www.researchgate.net/profile/Shad-Zoe-Kaiya-Dimapilis/publication/395934597_The_Dynamics_of_Grand_Corruption/links/68d8f83ed221a404b2a30e3c/The-Dynamics-of-Grand-Corruption.pdf

DIMAPILIS, S. Z. K. The Dynamics of" Grand Corruption.

(9)FLOODWATER GHOSTS AS SOCIAL CRITICISM: A horror film and Tropical Storm Ondoy.

More than ten years ago, Typhoon Ketsana, sometimes referred to as Tropical Storm Ondoy in the Philippines, struck the country's capital. Floodwaters up to 20 meters high engulfed Metro Manila and its surrounding rural areas, causing $1.15 billion in damages and 921 fatalities. The majority of those who made it through the severe weather event had a variety of Ondoy tales to share. In the 2011 episode Rain, Rain, Go Away (RRGA) of the popular Philippine horror anthology film series Shake, Rattle & Roll, Ondoy victims were represented as floodwater ghosts. The floodwater ghosts serve as terrifying reminders of a number of interrelated problems that the nation faces, including child labor, inadequate infrastructure, fraud and corruption, a lack of appropriate weather forecasting tools and methods, and many more. The ghosts also serve as a metaphor for how the most vulnerable groups are affected by water disasters, and they encourage potential discussions on the necessary steps to mitigate the negative consequences of extreme weather events on the Philippine islands.

ISSN 1834-6049

(10)Flood Control Management in Built Environment

Due to its dynamic impact on the lives of people, animals, properties, and other assets, the term "flooding" is now widely used and a major cause of concern in many regions of the world. A flood is caused by damp environmental circumstances that raise the soil's moisture content much above normal levels. As a result, water is not absorbed and collects above ground. The author's article "Does Climate Change Impact our Environment?" discusses this.ResearchGate/LinkedIn platforms, June 2021. "With tropical areas like the Philippines with archipelagic locations, it is indeed that Climate Change impacts and suffers the entire environment," it states. And as a result of this phenomena, the low pressure that has evolved in the Pacific Ocean, particularly whenever typhoons occur, makes flooding more regular and severe throughout the country's archipelagic topography. Due to water entering the drainage system from various sources of drainage connections and being released into rivers, streams, lakes, and the sea, these factors, along with heavy rainfall, cause considerable flooding in cities and municipalities. As a result, the water levels in those also rise and frequently surpass the bank levels.

Gomeseria, R. (2021). Flood Control Management in Built Environment. Available at SSRN 3926560.

(11) Flood Control Measures in One Municipality in Camarines Sur, Philippines: Bases for Community-Based Flood Control Interventions

Flooding and other hydrometeorological threats are unavoidable and have an impact on agriculture. could have a long-term effect on the economy. This investigation was carried out. to ascertain and evaluate the Nabua municipality's flood control systems in the Philippines' Camarines Sur Province as a foundation for creating community-based flood control measures. Applying the method of descriptive evaluative surveys with a validated questionnaire being the main tool for collecting data, collected information was calculated, tallied, examined, and evaluated. It has demonstrated that the level of implementation Flood management methods are highly recommended for both structural and non-structural floods. administration. Rejecting the null hypothesis. The variation in the answers of the participants was noteworthy. The degree to which the causes directly influence permits the flood control measures to be put into action. Thus, it may be said that the The Municipality of Nabua mainly uses its flood control infrastructure to prevent flooding. structures that are positioned strategically within the municipality, human resources, and cooperation and connections. Additionally, the respondents' responses are significant. distinctly different. Enhancement and modernization of its physical infrastructure and apparatus to gather information in real time; stepping up its campaign and communication campaign to create a stronger connection between the municipality and its residents, and information Disseminating information via social media and communications technologies can boost Thus, it is advised to be aware of, mitigate, regulate, and respond to flooding. Community-based flood control initiatives can act as a roadmap and a watchdog. opening for the Nabua local government unit to issue and implement municipal regulations pertaining to flood management.

Sotto Jr, R. B. Flood Control Measures in One Municipality in Camarines Sur, Philippines: Bases for Community-Based Flood Control Interventions.

(12)A Comparative Study of the Flood Control Programs of Malabon City and Candaba, Pampanga in the Philippines

This research sought to compare the flood control program of Malabon City and Candaba, Pampanga, in Philippines. The study used a case study design. It involved community stakeholders and local government officials such as the officials of Municipal Disaster Risk Reduction and Management Office (MDRRMO), barangay captain, counselors and members of the community selected using convenience sampling in each of the locale. Coding of themes was done to analyze the collected data. Results showed that flooded area became a natural environment for the people of Candaba. Improper waste disposal is rampant that possess the biggest threat in flooding problem in Malabon. Early warning framework, concrete evacuation plan, order and security management, relief system, DRRM and reporting system, rescue and recovery plan are the contingency plans to delimit the effect of flood in these places. Systematic review on existing policies, sustainability measures and community involvement will provide more productive and significant results in the future. https://po.pnuresearchportal.org/ejournal/index.php/apherj/article/view/1546

ISSN: 2408-350x

(13)Flood Control Management in Built Environment

Due to its dynamic impact on the lives of people, animals, properties, and other assets, the term "flooding" is now widely used and a major cause of concern in many regions of the world. A flood is caused by damp environmental circumstances that raise the soil's moisture content much above normal levels. As a result, water is not absorbed and collects above ground. The author's article "Does Climate Change Impact our Environment?" discusses this.ResearchGate/LinkedIn platforms, June 2021. "With tropical areas like the Philippines with archipelagic locations, it is indeed that Climate Change impacts and suffers the entire environment," it states. And as a result of this phenomena, the low pressure that has evolved in the Pacific Ocean, particularly whenever typhoons occur, makes flooding more regular and severe throughout the country's archipelagic topography. Due to water entering the drainage system from various sources of drainage connections and being released into rivers, streams, lakes, and the sea, these factors, along with heavy rainfall, cause considerable flooding in cities and municipalities. As a result, the water levels in those also rise and frequently surpass the bank levels.

SSRN:3926560

(14)Investigation of Flash Floods on Early Basis: A Factual Comprehensive Review

Ultimate extreme flash floods can be acknowledged as a main reason of high casualties and infrastructure loss in many countries like Pakistan, Malaysia, Philippines, Southern France, India, Bangladesh, China, Nepal, Canada, United States of America and others. Run offs can devastate huge buildings and personal belongings within fraction of seconds. Flash floods usually occurs due to many reasons like higher precipitation velocity, melting of ice debris in ocean, high wave current at sea shore, broken reservoir (dam), Cloud to ground flashes, thunderstorm and hurricane inside the ocean. More than one hundred and twenty thousand casualties resulted due to the flash floods during the 1992 and 2005. According to the literature review deadliest flash floods have been observed in past history. Many approaches have been completed to investigate the flash floods accurately and precisely with less false alarm rate. Disaster management authorities are unable to forecast the natural disasters accurately and precisely like tsunami, flash floods, hurricanes and seismic events due to the poor efficiency of the sensors and transmission of missed information. It has also been observed that during the wireless data transmission of sensors to the controller unit some bits of the data are missed, due to these phenomena data is not transmitted properly or indicate the wrong observations. Several diversified approaches have been made to identify the run offs more accurately and precisely. Generally, the approaches can be classified into two categories a) Engineering Based b) Non-Engineering Based. Engineering techniques based on the construction of the dams and reservoirs to store the excess water which causes severe run offs. Designing of various Artificial Intelligence Based competent algorithms to predict the flash floods vigorously can be considered as non-engineering based approaches. Authors have tried their best to summarize and portray all the successful techniques that can be used for the early prediction of flash floods. Scientists can be benefited by this research paper as this research paper is the detailed capsulization of all the approaches that has been carried out for the robust investigation of flash floods. Extensive literature review has been done to observe the comparative analysis for the investigation of flash floods identification accurately. Literature review has been categorized into following types; 1. Sensory Fusion based 2. Artificial Intelligence Based methods 3. Radar and Satellite based approaches 4. Modeling and Nowcasting. According to the exhaustive literature review it can be concluded that swarm intelligence weights optimization for multi-layer perceptron neural network configuration performed better among all the forecasting approaches and recommended as the future enhancementhttps://ieeexplore.ieee.org/abstract/document/8962024
ISSN: 2169-3536

(15)Political Economy: The Truth Behind Politics and the Challenges in Advancing Growth and Development of the Philippine Economy

Political economy highlights the interdependence of political, economic, and legal institutions in shaping a nation's development trajectory. In the Philippines, the legislative control over public budgeting referred to as the "power of the purse" plays a critical role in steering national economic priorities. However, recent revelations on the implementation of flood control projects exposed deep-seated corruption involving lawmakers, government officials, and private contractors. These anomalies compromise infrastructure integrity, exacerbate climate-related vulnerabilities, and undermine public trust.

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5807262

ISSN:5807262

(16). Ethics in the Rubble: Disaster Governance, Corruption, and the Moral Cost of Calamities in the Philippines.

In the Philippines, natural disasters often expose not only environmental vulnerabilities but also deep ethical failures in governance. This paper examines the moral dimensions of disaster response through the lens of the 2023-2025 flood control scandal, which involved thousands of substandard or fictitious infrastructure projects. Drawing on testimonies, Senate inquiries, and Filipino ethical concepts such as loob and pakikipagkapwa, the study argues that corruption in disaster governance constitutes a "secondary disaster"--a man-made crisis that compounds suffering, erodes public trust, and violates principles of justice and human dignity. The analysis situates corruption as both a structural and moral failure, revealing how disaster profiteering undermines resilience and deepens inequality. By integrating indigenous moral frameworks with global disaster ethics, the paper offers a culturally grounded critique of governance and proposes reforms centered on ethical leadership, transparency, and communitybased resilience. Ultimately, the study reframes disaster response not merely as a logistical challenge but as a moral imperative, calling for a transformation in how public service is understood and practiced in times of crisis.

ISSN 1173-2571

(17). Authoritative Pragmatism? Examining The Influence of the Lee Kuan Yew Leadership Paradigm on Philippine Governance Amidst Flood Control Corruption.

This article explores whether the leadership principles of Singapore's Lee Kuan Yew-marked by meritocracy, zero tolerance for corruption, and long-term pragmatic governance-have influenced Philippine governance, particularly in the context of corrupt flood-control projects. Using a qualitative multiple-case approach, including reformist cities and flood-control controversy narratives, the paper argues that adaptation of LKY-style governance, contextualized with the nuances of Philippine bureaucracy, will transition this systemically inefficient government into accountable and transparent administration. However, entrenched

patronage and systemic corruption-exemplified in flood-control anomalies-limit broader application. Finally, the article concludes that while LKY's model inspires reform, meaningful impact in Philippine governance requires systemic changes in public accountability, merit-based recruitment, and transparency reforms.

Pajimola, A. H. (2025). Authoritative Pragmatism? Examining The Influence of the Lee Kuan Yew Leadership Paradigm on Philippine Local Government Amidst Flood Control Corruption. Examining The Influence of the Lee Kuan Yew Leadership Paradigm on Philippine Local Government Amidst Flood Control Corruption (August 31, 2025).

(18). Governing the "Golden Age of Infrastructure": Assessing Transparency Innovations in Philippine Infrastructure Development

Amidst rising infrastructure investment across the Asia-Pacific, glaring accountability deficits have raised questions about governments' capacity to contain corruption in infrastructure development in the region. Recent developments in the Philippines, however, indicate the presence of challenges related to the ability of digitally enhanced transparency measures to bridge such accountability deficits. This article presents the shifting emphasis in transparency and accountability reforms related to Philippine infrastructure development beginning from the 1990s and assesses transparency innovations under the Duterte administration. While milestone measures such as the establishment of an electronic freedom of information (eFOI) platform have provided convenient access to public information, major hurdles remain in obtaining critical documents concerning infrastructure projects. As borne out in an exercise to request the feasibility studies of 48 flagship infrastructure projects, access to information is still obstructed by factors ranging from technical constraints, uneven service delivery, coordination failures, as well as active legal restrictions against the public's right to know.

(19) FLOODWATER GHOSTS AS SOCIAL CRITICISM: A horror film and Tropical Storm Ondoy.

Typhoon Ketsana, known in the Philippines as Tropical Storm Ondoy, hit the nation's capital more than a decade ago. Metro Manila and its neighbouring rural areas were submerged in floodwaters up to 20 metres high resulting in 921 fatalities and $1.15 billion in damages. Most of those who survived the extreme weather event have various Ondoy stories to tell. In 2011 the popular Philippine horror anthology film series Shake, Rattle & Roll portrayed Ondoy victims as floodwater ghosts in a narrative included in instalment 13, in an episode entitled Rain, Rain, Go Away (RRGA). The floodwater ghosts are chilling reminders of various intersecting issues plaguing the country such as child labour, poor infrastructure, graft and corruption, the lack of proper weather forecasting techniques and equipment, and many others. The ghosts also represent how the most vulnerable sectors suffer from such water disasters, and invite possible discourses on actions that need to be taken against the detrimental effects of extreme weather events on the Philippine islands.

ISSN 1834-6049

(20) Strategic Antagonism and the Flood Control Scandal: Blame-Seeking, Performative Accountability, and Institutional Routing in the Philippines

This article examines strategic blame-seeking in democratic governance through the case of the 2025 Philippine flood-control corruption scandal. Drawing on blame-avoidance and blame-seeking literatures, populist performance theory, and scandal politics, the study asks: (RQ1) Why would an incumbent seek blame? (RQ2) What tactics enable this strategy? (RQ3) How did blame-seeking operate in the Philippine case? Using qualitative process tracing of presidential speeches, Independent Commission for Infrastructure (ICI) briefings, administrative records, investigative reporting, and Ombudsman/DOJ filings (July–November 2025), I identify a patterned strategy of strategic antagonism—moral denunciation, naming/shaming, selective enforcement, and narrative amplification—that converted scandal salience into short-term political insulation while limiting structural reform. The paper contributes to accountability theory by showing how performative moral signaling and institutional routing can make blame a political resource in polarized, clientelist contexts. Policy implications include statutory timelines for probes, open contracting, and strengthened investigatory independence to reduce the political returns to performative accountability

ISSN: 2581-3889

(21) How do cartoon destination pictures inspire cartoon-induced tourism? Based on the associative network model and dual-coding theory

While research on film-induced tourism has extensively explored the emotional connections audiences establish with live-action films (Kim, 2012; Pham & Hwang, 2022; Wu & Lai, 2023), most studies have focused on live-action films, neglecting another important film genre – cartoons (Kirillova et al., 2019; Ono et al., 2020). Cartoon-induced tourism increases tourism demand and has emerged as a powerful economic driver in destinations worldwide (Florido-Benítez, 2023). For example, Disney cartoon characters attract over 52 million visitors worldwide to Disney World, which generates more than $14 billion in revenue

annually (Gitnux Report, 2024). In Japan, Hello Kitty attracts more than 2 million visitors to Sanrio Puroland each year (Tripadvisor, 2024), and Ghibli lovers immerse themselves in real-life locations in Japan (Japan Endless Discovery, 2024). This success highlights the need to study deeper into the specific influence of animated content.

Despite the recent growing research in cartoon-induced tourism, existing studies have primarily focused on motivational factors (e.g., Lin et al., 2021; Liu et al., 2021; Ono et al., 2020), the impact of sociodemographic factors on motivation (e.g., Kirillova et al., 2019), and cartoon tourists' experiences (e.g., Liu, Liu, & Rahman, 2022). There remain many research opportunities regarding the impact of animations and cartoon content on promoting cartoon tourist attractions (Florido-Benítez, 2023; Liu, Lai, & Li, 2022; Tung et al., 2019; Wong et al., 2024), especially the research gap in the development of strategies to sustain and reignite audience emotions and travel motivations towards cartoon-based destinations over time. Similar to live-action films, audiences may have constraints such as time and other limitations that do not allow them to visit the film destination following the film watching, leading to the gradual fading of these emotions and impulses over time. Therefore, cartoon-post-released destination marketing is crucial for the promotion of cartoon-induced tourism.

Cognitive psychologists posit that human knowledge is stored in memory and organized in the form of associative networks (Collins & Loftus, 1975). The associative network model has been extensively applied in marketing research to understand the connections between brands (Wang et al., 2020). It proposes that information in people's memory is interconnected through a network of nodes, with various elements connected and influencing each other. Referring to the associative network model, cartoon destination marketers showcase the filming locations depicted in the cartoon alongside real-life destinations in promotional photographs. They expect that memory recalling in comparing cartoon pictures with real photographs can stimulate audiences' emotional intensity towards cartoon destinations because recalling positive experiences from the past increases one's positive emotions (Speer et al., 2014). Then, viewers may have travel impulses. This comparison (one cartoon picture and one real picture of the destination) makes viewers feel a little familiar but alienated because the plot connection between the cartoon movie and the destination is weak. Russell (2002) stated that the key elements of the film can enhance memorability because they play a significant role in the plot. The cartoon character is a type of key element which has a high plot connection, such as Pokémon (Tung et al., 2019). Apart from cartoon characters, the classic quotes also play a crucial role in driving the story, eliciting emotional resonance among the viewers, and leaving a lasting impression (The Hollywood Report, 2016). Therefore, in designing cartoon-inducted tourism advertising, cartoon characters serve as visual stimuli, while classic quotes represent verbal or textual stimuli.

According to dual-coding theory, a nonverbal system deals with information about objects and events, and a verbal system deals with information related to language (Paivio, 2007). The stimulated outcome will be stronger when these two systems are interconnected (Yoo &

Kim, 2014). Advertising slogans (famous quotes) have been investigated in advertising research (e.g., Burgers et al., 2015). Compared to advertising slogans, cartoon quotes have the advantage of seamlessly integrating into the plot, characters, and place (destination). Therefore, classic quotes establish an emotional connection and resonance between the destination and the audience, thereby creating a deeper impact on the audience. However, no studies were found on the impact of classic movie quotes in film-induced tourism research. Therefore, research on including cartoon characters and cartoon quotes in destination advertising is important for stakeholders such as destination marketers, local governments, and tourism boards to understand how the reach of cartoons can be prolonged. Stakeholders can develop marketing strategies to maintain audiences' minds on cartoons and encourage their visits, driving the sustainable development of the cartoon-induced tourism industry. Based on the associative network model and dual-coding theory, this study aims to examine how cartoon destination pictures arouse viewers' emotional intensity and stimulate their cartoon-induced tourism impulse. Specifically, this study provides answers to the questions below: (1) whether the presence of cartoon characters in cartoon destination pictures enhances viewers' emotional intensity and cartoon-induced tourism impulse, (2) whether the emotional intensity mediates the relationship between cartoon destination pictures and cartoon-induced tourism impulse, and (3) whether the presence of classic quotes in cartoon destination pictures moderates the effects of the pictures on viewers' emotional intensity and cartoon-induced tourism impulse. The study employed a between-subjects factorial design to examine the impact of cartoon destination pictures with different cartoon elements on viewers' emotional intensity and cartoon-induced tourism impulse. Two cartoon elements were investigated: cartoon characters (presence vs. absence) and classic quotes (presence vs. absence). The research consisted of two separate studies. Study 1 assessed the influence of cartoon characters on viewers' emotional intensity and their cartoon-induced tourism impulse. Study 2 explored the interactive effects of cartoon characters and classic quotes on viewers' emotional intensity and cartoon-induced tourism impulse.

The study contributes to cartoon-induced tourism literature by incorporating the associative network model and dual-coding theory to explain the underlying mechanism of how cartoon elements can effectively trigger connections and associations in viewers' minds, leading to a stronger emotional bond with the cartoon and a tourism impulse. Unlike most film-induced tourism research that focuses on how movies affect audiences' attitudes and behaviour toward a destination, this study investigates how film-post-released marketing takes place after a film has been released for some time. It provides researchers with a new direction to perform film-induced tourism research. It also contributes to tourism advertising research by proposing the investigation of how advertising content affects impulsive travel to a destination. Additionally, the study also offers practical suggestions for cartoon-induced tourism destination marketers on how to develop and implement effective marketing strategies that capitalize on these powerful connections and associations.

(22) The Impact of Artificial Intelligence on Animation Filmmaking: Tools, Trends, and Future Implications

Artificial intelligence (AI) is transforming the animation film industry. This research explores AI's impact at all levels: from AI-driven tools and techniques to changes in creative processes and professional relationships. It will analyze existing and emerging AI-driven film-making techniques, such as procedural generation, AI-enhanced motion capture, and deep learning implementation. Both benefits and limitations will be examined, focusing on technical capabilities and creative capacities. Each new technique will be compared to its traditional counterpart regarding cost-effectiveness, time efficiency, and creative enhancement. Additionally, the research will consider AI integration's societal effects: whether it increases human productivity or dehumanizes jobs by narrowing stylistic variety. Ethical issues related to the extreme use of machinery, such as biases, will also be addressed. This work aims to navigate the fine line between human ingenuity and advanced machinery, presenting imaginative narrative possibilities while emphasizing AI's advantages. Automation should enhance animators' distinctiveness, not eradicate it, ensuring that the fundamental pillar of unique storytelling remains intact. The findings will guide future innovations, respecting the intrinsic value of human creativity that has shaped storytelling for centuries. This research aims to make a significant contribution to the growth and evolution of the animation film industry in the age of AI.

ISBN:979-8-3503-5444-7

(23)Enhancing Facial Expressiveness in 3D Cartoon Animation Faces: Leveraging Advanced AI Models for Generative and Predictive Design.

An advanced system for facial landmark detection and 3D facial animation rigging is proposed, utilizing deep learning algorithms to accurately detect key facial points, such as the eyes, mouth, and eyebrows. These landmarks enable precise rigging of 3D models, facilitating realistic and controlled facial expressions. The system enhances animation efficiency and realism, providing robust solutions for applications in gaming, animation, and virtual reality. This approach integrates cuttingedge detection techniques with efficient rigging mechanisms. The AI-assisted rigging process reduces manual effort and ensures precise, dynamic animations. The study evaluates the system's accuracy in facial landmark detection, the efficiency of the rigging process, and its performance in generating consistent emotional expressions across animations. Additionally, the system's computational efficiency, scalability, and system performance are assessed, demonstrating its practicality for realtime applications. Pilot testing, emotion recognition consistency, and performance metrics reveal the system's robustness and effectiveness in producing realistic animations while

reducing production time. This work contributes to the advancement of animation and virtual environments, offering a scalable solution for realistic facial expression generation and character animation. Future research will focus on refining the system and exploring its potential applications in interactive media and realtime animation.

ISSN
2158-107X

(24) Disaster Awareness and Preparedness of Barrio Community in Zambales, Philippines: Creating a Baseline for Curricular Integration and Extension Program

The social, biological and economic cost of natural disasters have increased in recent years due to skyrocketing population growth, environmental denudation and global climate change. This cross-sectional descriptive survey research determines the extent of disaster awareness and preparedness of the barrio or barangay community from the five southern municipalities of Zambales, Philippines. A total of 480 residents responded to the study. Results revealed that a majority of respondents were moderately aware about the different disasters occurring in the community. They are very aware about a typhoon whilst being moderately aware about a tsunami and storm surge. In terms of hazard level, respondents perceived a typhoon to be ‘very destructive’, whilst they understood a tsunami and storm surge to be ‘strong’. The barrio community often practice disaster preparedness for earthquakes, strong typhoons, landslides, fire, flood and volcanic eruptions. Amongst residents of the barrios, a moderate correlation exists (r=0.449) between their levels of disaster awareness and disaster preparedness. The results of this study provide a baseline information to use in decisions about the integration of disaster science and management in the curriculum at basic and higher education levels. Moreover, the results of this study also provided insights into the crafting of a community extension program to be spearheaded by the university. Such a program accentuates the need for massive dissemination of information about disasters for local communities so they become more aware about the causes and consequences of disasters. The proposed curriculum integration and extension program may inform practitioners and policy makers in making sound decisions regarding disaster risk reduction and mitigation strategies.

Rogayan, D. J., & Dollete, L. (2020). Disaster Awareness and Preparedness of Barrio Community in Zambales, Philippines: Creating a Baseline for Curricular Integration and Extension Program. Review of International Geographical Education Online, 10(2), 92-114. https://doi.org/10.33403/rigeo.634564

(25)Typhoon Awareness and Flood Management Measures: The Case of Barangay Santo Domingo, Bay, Laguna, Philippines

The Philippines is prone typhoons due to its geographical location. The municipalities of Los Baños and Bay in the province of Laguna are highly flood-prone due to their proximity to Laguna de Bay or
Laguna Lake. Large areas of Bay fall within the watersheds of Mt. Makiling Forest Reserve (MMFR) which are
tributaries to the lake. The back flow of water into the trubutaries during heavy rainfall can cause flooding and
result to socio-economic and physical impacts. The study aims to: determine the association of typhoon
awareness, mitigation practices, and resilience practices with the socio-economic and socio-demographic
profile of Barangay (village) Santo Domingo residents; generate a Flood Susceptibility Map, map the
designated evacuation sites using Geographic Information System (GIS); and provide recommendations to
promote awareness and reduce the impacts of typhoon. Ninety-six households were selected randomly based on
the sampling design with 95% confidentiality. Key Informant Interviews (KIIs) and Socio-economic surveys
(SES) were also conducted. STATA ver. 12 and Fisher's exact test were used to evaluate the significance of the
socio-economic and socio-demographic components in relationship to awareness, mitigation, and resilience.
Barangay Santo Domingo has high level of awareness (0.9647), high level of mitigation practices (0.8303) and
high level of resilience (0.8842). The monthly income of respondents has a strong correlation with mitigation

practices while house ownership signifies moderate association to resilience practices. Chi-square test of
independence showed that there is no significant difference among the puroks (zones within the barangay).
Based on the generated Flood Susceptibility Map, majority of the land area (197.23 hectares) of Barangay
Santo Domingo belongs to the high susceptibility category which led to the identification of six evacuation sites.
Awareness is necessary for disaster preparedness. Effective planning and management are essential to increase
awareness and reduce the detrimental effects of flooding.
E-ISSN: 2581-5946

(26) Disaster awareness in three low risk coastal communities in Puerto Princesa City, Palawan, Philippines

The Philippines is highly exposed and vulnerable to natural hazards. As a result, the country has an extensive disaster risk database that consists of different hazard and risk maps to be used as reference in creating development plans. However, some communities are in relatively low risk zones, such as those in Puerto Princesa City, Palawan (with many inhabitants of the city believing it to be safe). This study seeks to ascertain how disaster risk is understood in coastal areas that are only slightly susceptible to natural hazards, in order to provide strategies to improve disaster risk governance. The study validated national data on coastal disaster risk through a topographical survey, questionnaire surveys, and group interviews (in March 2016 and 2017) to understand whether residents knew the risks they are exposed to. The research found that national data reflects local conditions but community members do not have a clear understanding of the risks, particularly coastal hazards. Moreover, the research found that there has been a recent disaster event, though it was not properly archived and transmitted to the next generation. Thus, there is a need to raise awareness to correctly explain and transmit knowledge about potential hazards even in communities that have relatively lower disaster risk.

Valenzuela, V. P. B., Esteban, M., Takagi, H., Thao, N. D., & Onuki, M. (2020). Disaster awareness in three low risk coastal communities in Puerto Princesa City, Palawan, Philippines. International Journal of Disaster Risk Reduction, 46, 101508.

(27) Awareness, compliance and implementation of disaster risk reduction and management in flood-prone public elementary schools in Butuan city, Philippines

The paper’s primary intention was to identify the level of awareness, compliance, and implementation of the Disaster Risk Reduction andManagement (DRRM) of the flood-prone public elementary schools in the division of Butuan City, Philippines in the school year 2019-2020. Moreover, it also aimed to determine the relationship between the level of awareness and the level of compliance of the teachers in the DRRM program, particularly in flood disasters, and its consistency in the level of implementation as a basis for a localized intervention plan by utilizing mixed method design. Survey questionnaires were administered among therandomly selected 562 participants to obtain information on the study. The numerical data were analyzed using the descriptive-correlational method through the weighted mean and Pearson Product Moment Correlation andan inductive type of thematic analysis for the qualitative part. The results revealed that the participants have very satisfactory awareness in the fields of enabling environment, safe learning facilities, disaster risk management, and risk reduction and resilience education and have shown satisfactory compliance in the same fields. The implementation of the flood DRRM was satisfactory. Thus, it further showed that the level of awareness and the level of compliance has a significant relationship but none of them has asignificant relationship with the level of implementation. Findings indicated that flood DRRM is not emphasized in the schools. It was concluded thatgiving focus on school DRRM and conduct of relevant flood DRRM activitiesare important to establish flood resilience in schools.

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(28) Level of Awareness on Disaster Risk Reduction of the University of Eastern Philippines Employees and Students

The researchers looked at how well academics, students, and non-teaching staff at the University of Eastern Philippines System were aware of disaster risk reduction. Education, agriculture, health, fisheries, and marine resources were all used to measure awareness. Theresearch also revealed a link between awareness and a number of personal characteristics, aswell as a substantial variation in awareness amongst student groups. The study used a 30-item Likert type instrument to assess disaster risk reduction awareness and used a descriptive-correlational research approach. Percentages, means, multiple regression analysis, and analysis of variance were used to examine the data. The study's findings revealed that faculty members were well-versed in disaster risk reduction in education, agriculture, and health, as well as disaster risk reduction in fisheries and marine resources, infrastructure, and the environment and natural resources. The students were well-informed on disaster risk reduction in the areas of education, agriculture, and health, as well as disaster risk reduction in the areas of fisheries and marine resources, infrastructure, and the environment and natural resources. The six components of the degree of awareness were known to the non-teaching employees. Exposure to newspapers and television was shown to be substantially associated to the level of disaster risk reduction knowledge among faculty and non-teaching personnel.

Students' newspaper exposure and year levels were shown to be substantially connected to their degree of disaster risk reduction knowledge. There is a considerable variation in disaster risk reduction awareness across the three categories of respondents.

(29) Public perception of climate change and disaster preparedness: Evidence from the Philippines

The Philippines is highly susceptible to both geophysical and climate-related disasters. This article explores Filipinos knowledge and perception of climate change and their association with what action Filipinos take to prepare for rapid onset natural hazards such as typhoons. Data for this study were collected from a nationally representative random survey of 5,184 adults conducted between March and April of 2017. Filipinos self-report relatively low levels of knowledge of climate change and cited increased temperatures, shifts in seasons, and heavier rains as the most likely consequences. Levels of disaster preparedness in the Philippines differ widely by region. Although most Filipinos perceive that natural hazards are a risk to them, only a third of Filipinos undertake measures to prepare for disasters. Filipinos who perceive climate-related changes directly impacting their households report taking greater action to prepare for disasters. Filipinos who believe they have been directly impacted by climate-related changes are also more likely to prepare for disasters, take planning actions, and undertake material actions to prepare, such as dwelling improvements. Other factors associated with disaster preparedness include gender, membership in an association, wealth, risk perception, and prior exposure to and losses due to disasters. The findings imply that, while posing different challenges and requiring different responses, adaptation to climate change and disaster preparedness are inherently associated and potentially mutually reinforcing. Policies and programs would arguably benefit from a more unified intervention framework that links climate change adaptation and disaster preparedness.

Bollettino, V., Alcayna-Stevens, T., Sharma, M., Dy, P., Pham, P., & Vinck, P. (2020). Public perception of climate change and disaster preparedness: Evidence from the Philippines. Climate Risk Management, 30, 100250.

(30) Early warning systems and evacuation: rare and extreme versus frequent and small-scale tropical cyclones in the Philippines and Dominica

Survey questionnaires were administered among populations affected by Super Typhoon Yolanda in the Philippines in 2013 and Hurricane Maria in Dominica in 2017 to test the efficacy of early warning systems in prompting residents to take appropriate action ahead of severe hazards. Both events were rare and extreme but occurred in locations that regularly experience less severe tropical cyclones. The research assessed if, how, and when residents received warnings, what instructions were given, and where and when people decided to seek

safety. In both of the cases under review, residents were aware of the approaching storms, but critical information on their severity and potential impacts was either not received in time or not understood fully, resulting in low levels of evacuation and safety-seeking behaviour. This paper suggests that planning and public communication need to focus on the uncertainty surrounding the severity and multifaceted nature of tropical cyclones and accompanying hazards and their consequences.

Yore, R., & Walker, J. F. (2021). Early warning systems and evacuation: rare and extreme versus frequent and small-scale tropical cyclones in the Philippines and Dominica. Disasters, 45(3), 691-716.

(31) The Importance of Character Design in Animation

Character design plays a pivotal role in animation, serving as the foundation upon which stories are built and audience engagement is fostered. This paper explores the significance of character design in animation, examining how it influences emotional connection, narrative depth, and overall audience experience. By analyzing various aspects of character design—including functionality, recognizability, aesthetics, personality, and originality—this study highlights the essential elements that contribute to successful character creation. The methodology employed includes a literature review of existing research and case studies from notable animated films and series. The findings indicate that well-designed characters not only enhance storytelling but also contribute to the commercial success of animated projects. This paper concludes with suggestions for animators and designers on best practices for effective character design.

Nath, S. (2025). The Importance of Character Design in Animation. Swami Vivekananda University Press, 56-60.

(32) Character Design in the Natlan and Sumeru Regions of Genshin Impact: A Visual Semiotic Analysis from a Critical Transculturalism Perspective

In the context of globalization, culture never exists in isolation. As Kraidy (2008, p.148) notes, “Every cultural form is radically, quintessentially hybrid.” As a composite of multiple media such as visuals, language, and mythology, video games construct a cultural hybrid “third space” mediated through electronic technology. However, Kraidy (2008) points out that this process is not egalitarian, but rather nested within the logic of global capital expansion. Cultural hybridization in games functions both as a strategy to gain recognition in international markets, evoking resonance through familiar elements and generating novelty through unfamiliar ones (Finkelstein & Rios, 2022; Zhang et al., 2025), and as a culturally asymmetrical process of reconstruction. When cultural symbols are overly simplified or alienated in recontextualization, the originating cultural communities often accuse such practices

of cultural appropriation or racism, thereby highlighting the question of who has the authority to define culture.

Genshin Impact (GI) (miHoyo, 2020) exemplifies this very tension. Following the 2024 release of the Sumeru and Natlan regions, the game sparked controversies over whitewashing of skin tones and the use of colonial symbols (Bevan, 2024; Lima, 2024; Yonezawa, 2024), with related petitions on Change.org garnering over 130,000 signatures (Change.org, 2024). This study will employ Barthes' three-level semiotic approach to deconstruct the visual design of characters from Sumeru and Natlan, and, drawing on critical intercultural theory, explore how GI negotiates differences and reconstructs meaning amid cultural hybridization and global circulation.

At the operational level, this study adopts Roland Barthes' visual semiotics to deconstruct and analyze the design of 24 playable characters from the Sumeru and Natlan regions. By collecting character illustrations, 3D models, and cutscenes, a database of four categories of visual signs was established: bodily signs, clothing signs, accessory signs, and action signs. The analysis is based on Barthes' three-tiered model, encompassing the denotative, connotative, and mythic levels, ultimately revealing the appropriation of cultural symbols and their ideological functions in visual reconstruction. At the denotative level, this study distinguishes three types of signs: original signs
(directly drawn from cultural sources), hybrid signs (fusing multiple cultures), and fictional signs (with no clear cultural origin). At the connotative level, it analyzes semantic transformations and the meaning shifts that occur when signs are detached from their social context. At the mythic level, a critical transcultural perspective is applied. Through this analytical process, the study aims to reveal the underlying logic and semiotic reconstruction employed in GI's character designs for the "Nation of War" (Natlan) and the "Nation of Wisdom" (Sumeru), as well as to explore how broader social contexts, power discourses, and ideologies profoundly shape the game's representation of foreign cultures.

Based on the semiotic analysis, this study presents three key findings. 1. The Collage Logic and Aesthetic Inertia of ACG Style: GI is heavily influenced by ACG aesthetics, exhibiting a distinctly Japanese anime-style visual design. Its use of cultural elements from Natlan and Sumeru reflects a logicless collage characteristic of ACG aesthetics (Napier, 2005). Approximately 17% of the collected signs do not originate from the prototype cultures but are drawn from German military uniforms, Nordic elf attire, and knight costumes, demonstrating the long-standing de territorialized aesthetic inertia of ACG culture. However, the integration of marginal cultures simultaneously introduces new cultural resources into the ACG system. 2. The Trap of Technological Utopianism and Intentional Hybridity: Within GI's cultural system, the development paths of other nations often have real-world

prototypes. For instance, Fontaine's use of steam machinery reflects legal-rational principles corresponding to historical France. In contrast, Natlan's design replaces the unique techniques of its prototype culture with motorcycles, pixel art, and electronic dance, exemplifying the "intentional hybridity" described by Kraidy (2008). Although this technological reinterpretation aims to challenge the bias that "original" equals "backward," it falls into a logic of technological utopianism, treating modern industrialization as the sole path to progress while overlooking the value of local knowledge, such as shamanic rituals and herbal medicine. It presupposes that the prosperity and strength of Natlan's prototype nation must be expressed through modern industrial civilization, reflecting a limited imagination regarding social flourishing and civilizational evolution. 3. The Spatial Compression of Civilizations: In GI's world, nations such as Liyue, Inazuma, and Fontaine correspond to China, Japan, and France, respectively, and possess clear cultural anchors. In contrast, Sumeru and Natlan fuse elements from the Middle East, South Asia, Latin America, and Africa, forming a collage-like cultural complex. While this multicultural integration enhances visual diversity, it diminishes cultural depth and historical context (de Wildt & Aupers, 2021). From a critical intercultural perspective, this design reflects an imbalance in cultural representation: dominant cultures are depicted in detail, while marginalized cultures are treated as symbolic, and colonial and religious conflicts are erased, leaving a depoliticized surface-level diversity

Zhao, S. (2026, March). Character Design in the Natlan and Sumeru Regions of Genshin Impact: A Visual Semiotic Analysis from a Critical Transculturalism Perspective. In Abstract Proceedings of GachaCon 2026: Genshin Impact.

(33) RETRACTED ARTICLE: Application of Generative Artificial Intelligence AIGC Technology Under Neural Network Algorithm in Game Character Art Design

This paper aims to improve the efficiency and quality of game character design through artificial intelligence for generative content (AIGC). Firstly, an image extraction model based on a convolutional neural network and a game character image generation model based on a generative adversarial networks model is designed. Then, experiments are used to evaluate the loss of the model, and the performance of AIGC technology and traditional game character design methods is compared. The experimental results show that the average realism of game characters generated by the AIGC method is 0.85, which is higher than that of the traditional method of 0.82. The average value of the peak signal-to-noise ratio is 15.71, which is significantly better than the traditional method of 11.24. In addition, the Fréchet Inception Distance indicator suggests that the average of the AIGC method is 1.14, which is lower than the traditional method of 2.33. The learned perceptual image patch similarity

averages 1.16, which is closer to the real sample than the traditional method of 2.17. Meanwhile, the game character design generated by AIGC technology only takes 0.85 h on average, which is much lower than the 3 h of traditional methods. Also, the uniqueness analysis of samples generated by AIGC is carried out. It is found that about 80 out of 100 generated samples are unique, indicating that the diversity of samples generated by AIGC is high, and the character design is quite different. The results show that AIGC technology has important application potential in in-game character design, which can provide higher-quality, more realistic, and diversified game character design to improve game experience and competitiveness.

Li, J., & Liu, Q. (2025). RETRACTED ARTICLE: Application of Generative Artificial Intelligence AIGC Technology Under Neural Network Algorithm in Game Character Art Design. Journal of the Knowledge Economy, 16(1), 4652-4683.

(34) A Systematic Review of Generative AI on Game Character Creation: Applications, Challenges, and Future Trends

In this paper, we review the impact of generative artificial intelligence (AI) on game character creation. We critically examine the application of AI-driven computer graphics (AICG) technology across the character creation workflow, including concept generation, clothing design, modeling, props, cultural embedding, personality traits, and behaviors. Our research identifies potential applications, challenges, and future trends of these technologies in game development. Furthermore, it explores how AI and large language models (LLMs) can streamline workflows, automate asset generation, and reduce technical barriers in character creation. In this systematic review, we provide valuable insights for AI developers, game designers, and researchers.

Maxim, R. I., & Arnedo-Moreno, J. (2025). Identifying key principles and commonalities in digital serious game design frameworks: Scoping review. JMIR Serious Games, 13, e54075.

(35) Identifying Key Principles and Commonalities in Digital Serious Game Design Frameworks: Scoping Review

This study aims to review, analyze, and synthesize the literature on digital entertainment game design frameworks and DSG design frameworks (DSGDFWs). The focus is on conceptual frameworks offering high-level guidance for the game creation process rather than component-specific tools. We explore how these frameworks can be applied to create impactful serious games in fields such as health care and education. Key goals include identifying design principles, commonalities, dependencies, gaps, and opportunities in the literature. Suggestions for future research include empathic design thinking, artificial intelligence integration, and iterative improvements. The findings culminate in a synthesized 4-phase design process, offering generic guidelines for designers and developers to create effective serious games that benefit society.

Wu, Z., Chen, Z., Zhu, D., Mousas, C., & Kao, D. (2025). A systematic review of generative AI on game character creation: Applications, challenges, and future trends. IEEE transactions on games.

(36) Storyboarding: an empirical determination of best practices and effective guidelines

Storyboarding is a common technique in HCI and design for demonstrating system interfaces and contexts of use. Despite its recognized benefits, novice designers still encounter challenges in the creation of storyboards. Furthermore, as computing becomes increasingly integrated into the environment, blurring the distinction between the system and its surrounding context, it is imperative to depict context explicitly in storyboards. In this paper, we present two formative studies designed to uncover the important elements of storyboards. These elements include the use of text, inclusion of people, level of detail, number of panels, and representation of the passage of time. We further present an empirical study to assess the effects of these elements on the understanding and enjoyment of storyboard consumers. Finally, we demonstrate how these guidelines were successfully used in an undergraduate HCI class.

Truong, K. N., Hayes, G. R., & Abowd, G. D. (2006, June). Storyboarding: an empirical determination of best practices and effective guidelines. In Proceedings of the 6th conference on Designing Interactive systems (pp. 12-21).

(37) Teaching and Learning through Storyboarding: A New Approach via Critical Reflection

This paper seeks to accomplish three primary objectives: firstly, it introduces the concept of storyboard teaching; secondly, it emphasises the efficacy of storyboard teaching in higher education as a means of customising instructional materials in both traditional and online programmes to suit the learners' needs and contextual factors; and thirdly, it advocates for the integration of storyboard teaching in the classroom, particularly in writing and language courses, alongside other innovative pedagogical approaches. This research highlights the significance of storyboarding as a means to foster critical thinking and active engagement among students. Learning encompasses more than just memory as individuals engage in reflection, evaluation, and personal growth as a direct consequence of their exerted effort. The study's results suggest that the use of storyboarding as a strategy enhances the preservation of essential ideas and graphical aspects within a narrative by highlighting significant events. This methodology allows students to gain information via the process of critical reflection rather than relying only on memorization methods.

ISSN 2958-2040

(38) STORYBOARDING: A MODEL TECHNIQUE FOR THE LANGUAGE LEARNING PROCESS

The current paper mainly focuses on three objectives: introduces the concept of storyboard teaching, highlights its usefulness in higher education as a variant in online and offline courses or as an adapting teaching materials tool according to the learners’ needs and context of learning, and then recommends its use in the classroom, especially in language and literature, mixed with other innovative teaching techniques. Moreover, the current paper focuses on how storyboarding

provides platforms for the students to be active, creative, and critical thinkers and learning is not limited to only remembering while they analyze their works, reflect, and progress. The researchers have highly recommended incorporating the storyboarding technique in higher education classrooms and fusing it with other innovative techniques to make a classroom interesting, engaging, and interactive. Adding the storyboarding approach, students get motivated and feel confident and competent after the completion of mapping out their ideas visually, cognitively, and creatively.

ISSN 1410-7201

(39) Storyboarding for visual analytics

Analysts wish to explore different hypotheses, organize their thoughts into visual narratives and present their findings. Some developers have used algorithms to ascertain key events from their data, while others have visualized different states of their exploration and utilized free-form canvases to enable the users to develop their thoughts. What is required is a visual layout strategy that summarizes specific events and allows users to layout the story in a structured way. We propose the use of the concept of 'storyboarding' for visual analytics. In film production, storyboarding techniques enable film directors and those working on the film to pre-visualize the shots and evaluate potential problems. We present six principles of storyboarding for visual analytics: composition, viewpoints, transition, annotability, interactivity and separability. We use these principles to develop epSpread, which we apply to VAST Challenge 2011 microblogging data set and to Twitter data from the 2012 Olympic Games. We present technical challenges and design decisions for developing the epSpread storyboarding visual analytics tool that demonstrate the effectiveness of our design and discuss lessons learnt with the storyboarding method.
Mangtani, A. J. (2024). Storyboarding Fundamentals. In Instructional Design Unleashed: Unlocking Professional Learning Potential with UX, Agile and AI Methods (pp. 435-455). Berkeley, CA: Apress.

(40) Storyboarding Fundamentals

Now equipped with an understanding of the science behind instructional design, the role of instructional design (ID) models, and the crafting of SMART objectives to structure courses effectively, the next crucial step is consolidating these elements into a cohesive plan – a storyboard. The storyboard serves as a visual road map, combining the instructional design

principles and objectives in a tangible format. Its historical roots trace back to the early days of filmmaking, with visionaries like Walt Disney playing a pivotal role in its popularization, particularly in animation. As the decades unfolded, storyboarding expanded its influence beyond film to encompass diverse industries, such as advertising, television, video games, and graphic design. The digital era further transformed the process, introducing computer software for more efficient and flexible storyboarding. Today, storyboarding is ubiquitous in education and business and is used for planning presentations, instructional materials, and multimedia projects. Its evolution reflects a dynamic synergy between traditional techniques and modern technological advancements, making it an indispensable component of the creative process across various fields.

ISBN
979-8-8688-0415-1

(41) Storyboarding in Medical Animation

Storyboards are a series of thumbnail images that act as a planning document for your overall animation project. Each thumbnail image is part of what is known as a shot, and shots are strung together to create scenes, which in turn create a whole animation. Medical animations, compared to film or animated movies, have rigid and specific requirements; however, by utilizing storyboards as a planning document we can infuse our medical animations with emotion, life, and beauty. This chapter will review common storyboarding fundamentals, like camera moves and editing techniques, and how they may be applied to the medical world. Camera moves are important to string shots together and can help add emphasis to specific sections. Zooms and pans are used extensively within medical animation and it is important to note which circumstance might call for either one. Zooms are great for moving to a drastically different microscopic magnification. Pans create visual interest for a stepwise reaction. Techniques such as lighting, depth, and insets and labels create a more balanced composition and save time in the production phase. Storyboards are ultimately about problem solving early to save time and money during the production phase.

(42) Using a storyboarding technique in the classroom to address end of life experiences in practice and engage student nurses in deeper reflection

Lillyman, S., Gutteridge, R., & Berridge, P. (2011). Using a storyboarding technique in the classroom to address end of life experiences in practice and engage student nurses in deeper reflection. Nurse Education in Practice, 11(3), 179-185.

Okemow, S. J. (2020). Storyboarding in medical animation. Biomedical Visualisation: Volume 6, 131-143.

(43) Schematic storyboarding for video visualization and editing

We present a method for visualizing short video clips in a single static image, using the visual language of storyboards. These schematic storyboards are composed from multiple input frames and annotated using outlines, arrows, and text describing the motion in the scene. The principal advantage of this storyboard representation over standard representations of video -- generally either a static thumbnail image or a playback of the video clip in its entirety -- is that it requires only a moment to observe and comprehend but at the same time retains much of the detail of the source video. Our system renders a schematic storyboard layout based on a small amount of user interaction. We also demonstrate an interaction technique to scrub through time using the natural spatial dimensions of the storyboard. Potential applications include video editing, surveillance summarization, assembly instructions, composition of graphic novels, and illustration of camera technique for film studies.
Goldman, D. B., Curless, B., Salesin, D., & Seitz, S. M. (2006). Schematic storyboarding for video visualization and editing. Acm transactions on graphics (tog), 25(3), 862-871.

(44) Techniques for automated reverse storyboarding

Storyboarding is a standard method for visual summarisation of shots in film and video preproduction. Reverse storyboarding is the generation of similar visualisations from existing footage. The key attributes of preproduction storyboards are identified, then computational techniques that extract corresponding features from video, render them appropriately, and composite them into a single storyboard image are developed. The result succinctly represents background composition, foreground object appearance and motion, and camera motion. For a variety of shots, it is shown that the visual representation conveys all the essential elements of shot composition.
Dony, R. D., Mateer, J. W., & Robinson, J. A. (2005). Techniques for automated reverse storyboarding. IEE Proceedings-Vision, Image and Signal Processing, 152(4), 425-436.

(45) Using storyboarding to gain appreciative reflection in the classroom

This paper reviews the use and value of a storyboarding approach in a classroom setting for healthcare professionals. This approach is used with adult students from foundation to postgraduate level of study. It describes how the authors used a positive appreciative reflection approach to storyboarding in order to develop the narratives of the students' lived experience. The authors identify how the storyboarding helped to develop one student's narrative into a group activity. This approach provided the students with space and time to work together on creating the storyboard and gain 'collective wisdom' from their peers and facilitators in the process. The paper acknowledges some of the limitations and benefits of using this approach to storyboarding within the classroom setting.
Lillyman, S., & Bennett, C. (2012). Using storyboarding to gain appreciative reflection in the classroom. Reflective Practice, 13(4), 533-539.

(46) An Empirical Investigation Into the Role of Colour in Eliciting Emotional and Thematic Responses to 2D Animation

Colour in 2D animation is a fundamental storytelling tool that extends beyond mere aesthetics to influence audience perception, emotion, and thematic interpretation. This study investigates the emotional and thematic impact of specific colour schemes in 2D animated scenes. Using a mixed-methods approach, the research collected qualitative data through semistructured interviews and quantitative data via structured surveys from participants who viewed a curated selection of animated clips. The findings reveal a strong and consistent correlation between specific colours and emotional responses; for instance, blue was overwhelmingly associated with sadness, yellow with happiness, and red with anger. The results validate foundational principles of colour theory, such as the RYB colour model, demonstrating that primary colours evoke distinct, highintensity emotions while secondary and neutral tones elicit more nuanced or context-dependent feelings. The study concludes that colour is a vital and powerful component of visual language, serving as a critical bridge between narrative design and audience engagement. These insights hold significant implications for animators, content creators, and branding strategists seeking to purposefully shape the viewer's experience.

Norizan, A. R., Anuar, A. S. H., & Zamri, M. F. M. (2026, January). An Empirical Investigation Into the Role of Colour in Eliciting Emotional and Thematic Responses to 2D Animation. In 2026 20th International Conference on Ubiquitous Information Management and Communication (IMCOM) (pp. 1-5). IEEE.

(47) Applying Color Theory to Digital Media and Visualization

We examine the foundations of color theory & how these methods apply to building effective digital media. We define color harmony & demonstrate the application of color harmony to case studies. Case studies include historic & new infographics as well as time series animations. The Pantone Matching System, Munsell Color System and other hue systems are reviewed. The features of ColorBrewer, Adobe's Capture CC app, & Josef Albers "Interaction of Color" app are examined. We also introduce "Gamut Mask" & Color Proportions of an Image analysis tools. Our course concludes with a hands on session that teaches how to use online and mobile apps to successfully capture, analyze and store color schemes for future use in visual analytics. This includes evaluations for color deficiencies using Vizcheck & Coblis. These color suggestion tools are available online for your continued use in creating new digital media. Please bring small JPEG examples of your digital media content for performing color analyses during the hands on session.
Rhyne, T. M. (2017, May). Applying color theory to digital media and visualization. In Proceedings of the 2017 CHI conference extended abstracts on human factors in computing systems (pp. 1264-1267).

(48) Research and Implementation of Local Color in Animation

Taking With Hui'an Women living in the south China's Fujian Province as an example, the thesis analysis the constitution of Hui'an woman's finery. Through the deeply scratch of color theory, we analysis the physiological reason which cause color in eyes. Study the mental acts created by physiological reason and color's collocation and symbolic meaning, the thesis brings up a viewpoint that use the native color element to enhance the art infectivity when we are creating animation. By strengthen the local color's colored activity, making the animation have a special effect visual reception, I make an animation named
Lin, Y., & Lin, H. (2008, September). Research and implementation of local color in animation. In 2008 International Conference on Cyberworlds (pp. 653-656). IEEE Computer Society.

(49) The Colors of Good and Evil: A Quantitative Study on the Effect of Disney Animated Feature Films on Audience Color Association

Hayes, G. M. (2025). The Colors of Good and Evil: A Quantitative Study on the Effect of Disney Animated Feature Films on Audience Color Association.

(50) Facial color adaptive technique based on the theory of emotion-color association and analysis of animation

Graphical expressions and artificial intelligence-driven (AI) game characters have been continuously improving, spurred by the astonishing growth of the game technology industry. Despite such improvements, users are still demanding a more natural gaming environment and reflections of true human emotions. However, the emotions that can currently be expressed are strictly limited because the facial colors and expressions of present game characters are hardly noticeable. Such restrictions can prevent the users from getting fully absorbed in the game. To address this, we developed the facial color change technique, which is a combination emotional model based on human cultural theory, emotional expression pattern using colors, and emotional reaction speed function, as opposed to past methods that expressed emotion through blood flow, pulse, or skin temperature. The reflection of the game characterpsilas emotion on itpsilas skin color will increase user immersion into the game and enrich the playerpsilas experience.

Kundzewicz, Z. W., Kanae, S., Seneviratne, S. I., Handmer, J., Nicholls, N., Peduzzi, P., ... & Sherstyukov, B. (2014). Flood risk and climate change: global and regional perspectives. Hydrological Sciences Journal, 59(1), 1-28

(51) Flood risk and climate change: global and regional perspectives Le risque d'inondation et les perspectives de changement climatique mondial et régional

A holistic perspective on changing rainfall-driven flood risk is provided for the late 20th and early 21st centuries. Economic losses from floods have greatly increased, principally driven by the expanding exposure of assets at risk. It has not been possible to attribute rain-generated peak streamflow trends to anthropogenic climate change over the past several decades. Projected increases in the frequency and intensity of heavy rainfall, based on climate models, should contribute to increases in precipitation-generated local flooding (e.g. flash flooding and urban flooding). This article assesses the literature included in the IPCC SREX report and new literature published since, and includes an assessment of changes in flood risk in seven of the regions considered in the recent IPCC SREX report—Africa, Asia, Central and South America, Europe, North America, Oceania and Polar regions. Also considering newer publications, this article is consistent with the recent IPCC SREX assessment finding that the impacts of climate change on flood characteristics are highly sensitive to the detailed nature of those changes and that presently we have only low confidence1 in numerical projections of changes in flood magnitude or frequency resulting from climate change.

Park, K. H., & Kim, T. Y. (2008, October). Facial color adaptive technique based on the theory of emotion-color association and analysis of animation. In 2008 IEEE 10th Workshop on Multimedia Signal Processing (pp. 861-866). IEEE.

(52) Adapting to flood risk under climate change

Flooding is the most common natural hazard and third most damaging globally after storms and earthquakes.

Anthropogenic climate change is expected to increase flood risk through more frequent heavy precipitation, increased catchment wetness and sea level rise. This paper reviews steps being taken by actors at international, national, regional and community levels to adapt to flood risk from tidal, fluvial, surface and groundwater sources. We refer to existing inventories, national and sectoral adaptation plans, flood inquiries, building and planning codes, city plans, research literature and international policy reviews. We distinguish between the enabling environment for adaptation and specific implementing measures to manage flood risk. Enabling includes routine monitoring, flood forecasting, data exchange, institutional reform, bridging organizations, contingency planning for disasters, insurance and legal incentives to reduce vulnerability. All such activities are 'low regret' in that they yield benefits regardless of the climate scenario but are not cost-free. Implementing includes climate safety factors for new build, upgrading resistance and resilience of existing infrastructure, modifying operating rules, development control, flood forecasting, temporary and permanent retreat from hazardous areas, periodic review and adaptive management. We identify evidence of both types of adaptation following the catastrophic 2010/11 flooding in Victoria, Australia. However, significant challenges remain for managing transboundary flood risk (at all scales), protecting existing property at risk from flooding, and ensuring equitable outcomes in terms of risk reduction for all. Adaptive management also raises questions about the wider preparedness of society to systematically monitor and respond to evolving flood risks and vulnerabilities.

Wilby, R. L., & Keenan, R. (2012). Adapting to flood risk under climate change. Progress in physical geography, 36(3), 348-378.

(53) Restoring Trust and Building Integrity in Government: Issues and Concerns in the Philippines and Areas for Reform

In most general terms this article addresses the issue of the continuing decline of trust in government and the imperatives for reform. The decline on trust in government has been brought about by many factors including the inefficient and ineffective delivery of services, waste of public resources, graft and corruption, lack of integrity in government, poor leadership, excessive red tape, ineffective reorganization and structural changes, too much centralization, among other things. In summary, unresponsive governance has been responsible for the continuing decline of trust in government. The article introduces a framework of areas of reform imperatives with the general objective of restoring trust in government. These areas include the following: (1) reforms in institutions and structures, including reforms in organizations, processes and procedures; (2) reforms in mindsets, paradigms and behavior; (3) reforms in leadership at various levels; and (4) reforms among citizens, i.e., citizen engagement and/or citizen participation. We begin by reviewing various examples in the Philippines including continuing efforts to address graft and corruption, red tape, and inefficiencies in the government's politico-administrative environment.

Brillantes, A. B., & Fernandez, M. T. F. T. (2011). Restoring trust and building integrity in government: Issues and concerns in the Philippines and areas for reform. International Public Management Review, 12(2), 55-80.

(54) The Philippines: Political Parties and Corruption

Southeast Asian Affairs 2007 THE PHILIPPINES Political Parties and Corruption Nathan Gilbert Quimpo The Philippines, it is often said, has mediocre or weak political parties and an inchoate or ill-developed political party system. A political party, as defined by the country's Omnibus Election Code, is an organized group of persons pursuing the same ideology, political ideas, or platform of government. Political activist and analyst Joel Rocamora quips, however, that nobody can accuse any of the Philippines' main parties of being such an animal.1 Dominated by the country's politico-economic elite, they are built around personalities, rather than around political programmes or platforms. In fact, ideologies and platforms are just adornments for them. A major Philippine daily newspaper aptly describes what the parties stand for: "The usual motherhood statements are passed off as political programmes".2 Apart from being indistinguishable from one another in their political beliefs and programmes — or lack of these — the Philippines' main parties have weak membership bases and seem to come alive only during election time. Elections are often marked — or marred — by lavish spending, vote-buying, fraud, and violence. In Philippine party politics, turncoatism is a venerable tradition3 — politicians flit like butterflies from one party to another. Post-Marcos parties, in particular, are said to reflect the undeveloped or malformed character of the Philippine political party system. Far from being stable organizations, they have proven to be nebulous entities that can be set up, merged with

others, split, resurrected, regurgitated, reconstituted, renamed, repackaged, recycled, or flushed down the toilet any time.4 Most politicians belonging to the main parties have come to be derogatorily called trapo, which is short for "traditional politician", but ordinarily means an old rag used for wiping off dust and dirt that often becomes grimy or greasy. Nathan Gilbert Quimpo is an Associate Professor in Political Science and International Relations, University of Tsukuba, Japan. 278Nathan Gilbert Quimpo A scrutiny of Philippine trapo parties leads one to question certain prevailing notions about political parties and about the relationship of political parties and party systems with political development and democratic consolidation. First, in the new typology of political parties drawn up by Richard Günther and Larry Diamond, there are two types of elite-based party — the traditional local notable party and the clientelistic party.5 Under this typology, the Philippines' trapo parties would be classified as the latter. However, I argue that the trapo parties have gone far beyond simple clientelist politics and morphed into patrimonialistic parties, instruments of an oligarchic elite for the prédation of the state and its resources through various means — the use of traditional patron-client ties, nonpersonalistic forms of patronage, rent-seeking, outright corruption, fraud, coercion, and violence. Secondly, the institutionalization of political parties and party systems is often regarded as being very important or even crucial for a country's political modernization and democratic consolidation. Scholars like Samuel Huntington have tended to draw a linear development of parties and party systems from weak to strong, from uninstitutionalized to institutionalized.6 Under the traditional characterization of political parties and party systems, the Philippines' main parties and its party system would be regarded as weak and uninstitutionalized. I argue that while seeming to be feeble, sapless creatures, the trapo parties, taken collectively, are quite rapacious and formidable. The system of chameleonic and patrimonialistic trapo parties has, in fact, become the foremost institution of the Philippines' "patrimonial oligarchic" or "predatory" state. The degeneration of Philippine parties from clientelist to patrimonialist politics and the institutionalization of the trapo party system indicate that the building of strong and truly democratic political parties and party system in the Philippines is turning out to be an excruciatingly difficult and complicated process. From Clientelistic to Patrimonialistic Parties Traditional patron-client relationships have long been an important feature of Philippine politics. In his now classic 1965 study on Philippine politics, Carl Lande observed that the Philippine polity was structured less by organized interest groups as in Western democracies than by networks of personal relationships, largely involving exchanges of favours between prosperous patrons and their poor and dependent clients. The two main parties at that time consisted of vertical chains of these.

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(55) Anti-Corruption Reforms in the Philippines

This article reviews the state of anti-corruption efforts in the Philippines by outlining a typology of corruption and evaluating the country's progress on these different fronts. Notwithstanding recent gains, the country appears stuck in its anti-corruption efforts. While reforms have managed to advance economic growth, corruption nevertheless continues to undermine private sector investments in key economic sectors and geographic regions associated with high poverty, as well as weakens public goods provision. Corruption de facto erodes the inclusivity of economic growth gains, which eventually contributes to imbalanced growth and inequality. Policy responses, therefore, need to consider the multi-faceted nature and implications of corruption and address its root causes. Reforms to improve competition in the economic and political spheres could help address some of the underlying features of rent-seeking behaviour and corruption. Given how these reforms run into deep political economy issues, a clear theory of change is necessary, touching on both the market economy and the political system.

(56) Grand corruption scandals in the Philippines

Grand corruption scandals are a regular occurrence in the Philippines, tainting many presidential administrations (Mathews and Wideman, 1977; Mydans, 1988; Coronel and Tordesillas, 1998; Rimando, 2000; Hutchcroft, 2008). Grand corruption refers to the "abuse of high level power that benefits the few at the expense of the many" (Transparency International, 2016). Corruption at this level usually involves large sums of financial and state resources.

Popular thinking in recent years has attributed grand corruption simply to regime types (authoritarian vs democratic). This thinking is flawed. Corruption at various levels of government is rampant in weak or inferior democracies than in well-developed ones, leading to the belief that having a democracy alone does not guarantee effective control of corruption (UNODC, 2019; Kukutschka, 2018). Similarly, the level of corruption varies across authoritarian systems, with monarchies and single-party systems tending to be less corrupt than military and personalistic regimes; in fact, as Kukutschka (2018, p. 4) points out, "a small number of non-democratic countries perform relatively well in international corruption indices." In either democratic or authoritarian systems, the commission of grand corruption is typically driven by personalistic and patronage-clientelistic motivations that benefit rulers and the dominant political elite, political parties and allies, cronies and friends, as well as families and relatives (David-Barrett and Fazekas, 2019; Chang and Golden, 2010).

In the Philippines, grand corruption knows no political regime; its occurrence has been observed in both authoritarian and democratic governments. Thus, instead of simply regime type, it could be argued that it is the continued weakness of governance institutions that

allows the culture and agency of corruption to thrive and persist, even at the highest levels of government. Despite the proliferation of anti-corruption laws, a perennially weak accountability environment and the ineffectiveness of the country's anti-corruption agencies encourage strategic rent-seeking by private firms and individuals through deception and bribes (or income transfer) to government officials. Furthermore, an inefficient prosecutorial and judicial system, often subservient to political power, encourages corruption by failing to punish crimes of powerful individuals and firms. All this constitutes the collective inability of institutional and agency constraints to overpower the forces that create, adapt and maintain the opportunities for corruption.

The scandals that affect many Philippine presidential administrations reflect this inability of both state and society to curb opportunities for grand corruption. Scandal, according to Lowi (1988, p. vii) is “corruption revealed;” it is “breach of virtue exposed.” In his analysis of scandals, Lowi (1988, p. viii) argues that the exposé of corruption (the first news break or the substantive scandal) is often followed by a more controversial cover-up phase or the procedural scandal. The cover-up extends the scandal's media coverage and further erodes a political administration's integrity and legitimacy. Hence, the unpacking of a corruption scandal could be guided by an examination of its substantive and procedural stages. The analysis should also cover the scandal's resolution or closure. Following the theme of this special issue of Public Administration and Policy, this article examines and compares two corruption scandals involving large infrastructure projects of government, namely, the Bataan Nuclear Power Plant (BNPP) of the Marcos era and the National Broadband Network (NBN) project of the Arroyo presidency. These two projects are amongst the most controversial political corruption scandals during the last 50 years. Their comparison enhances the understanding of grand corruption in government-initiated projects involving foreign government and corporate entities. To construct the narratives of each case, primary and secondary sources are utilized, including court decisions, laws published in the Official Gazette, Senate committee reports, journal articles, biographies and newspaper reports. The following sections provide descriptive accounts of the BNPP and NBN scandals, followed by the comparative analysis and conclusion.

Batalla, E. V. C. (2020). Grand corruption scandals in the Philippines. Public Administration and Policy, 23(1), 73-86.
Mendoza, R. U., Bulaong Jr, O. G., & Mendoza, G. A. S. (2025). Anti-Corruption Reforms in the Philippines: Quo Vadis?. Millennial Asia, 16(3), 364-387.

(57) Causes and Consequences of Corruption in Southeast Asia: A Comparative Analysis of Indonesia, the Philippines and Thailand

Corruption in Indonesia, the Philippines and Indonesia can be attributed to the low salaries of the civil servants and political leaders, the ample opportunities for corruption in many public agencies, and the low probability of detection and punishment of corrupt offenders. The consequences of corruption are negative for all three countries, but they are less serious for Thailand because of the National Counter Corruption Commission's efforts in curbing corruption. The anti-corruption strategies in Indonesia and the Philippines would be more effective if their political leaders demonstrated their commitment to curbing corruption.

Quah, J. S. (2003). Causes and consequences of corruption in Southeast Asia: A comparative analysis of Indonesia, the Philippines and Thailand. Asian Journal of Public Administration, 25(2), 235-266.

(58) The Philippines: a social structure of corruption

The anticorruption community largely views corruption as a government or development issue. But in the Philippines, corruption is a social structure. The very social bonds and social structures that are good at building civic unity and solidarity are also good at spreading and maintaining corruption, and this is why corruption is so difficult to remove. Patrons use these societal features to implement a ubiquitous social structure of corruption by means of maneuvered friendships that makes it difficult for the masses to know when a patron is acting as a friend or foe. The social structure encompasses the whole of society and corrupts the encircled government, political, and development systems as easily as it infiltrates all other segments of society. It is why oversight and sector-based anticorruption initiatives underperform, and why initiatives must pivot towards addressing this social structure.
Guth, A. (2024). The Philippines: a social structure of corruption. Crime, Law and Social Change, 82(2), 223-247.

(59) Treading the straight and righteous path: curbing corruption in the Philippines

The Aquino government has exercised remarkable political will in acting on high-profile cases involving former government officials, including former President Gloria Macapagal-Arroyo. However, the government's overall anti-corruption performance is hampered by outdated and conflicting laws, lack of compliance with anti-corruption laws and regulations by public officials and employees, poor ACA operational capacities, judicial inefficiency, deficient organizational systems and change-resistant government agencies, and selective and partial enforcement of anti-corruption laws. These problems are characteristic of Philippine political administrations and are arguably rooted in a system long characterized by fragile state institutions, strong oligarchic control, and weak citizenship.

Batalla, E. V. C. (2015). Treading the straight and righteous path: curbing corruption in the Philippines. Asian Education and Development Studies, 4(1), 51-75.

(60) A broken vow: an examination of the cases of Corruption in the Philippines

The problem of corruption in the Philippines seems to be humongous as if no solution is available for its cure. This study used a descriptive and qualitative research design. It exposed the anti-corruption laws and the cases decided by the various Philippine courts. It found out that even with the existence of laws as well as the removal or the conviction of several government officials and employees, corrupt practices seem to be undeterred. The continuance of corruption in the country resulted to the promise of President Duterte to be a broken vow

(61) FILMOGRAPHY RELATED TO THRILLER AND INTERRELATIONS WITH NATURAL LAW Fiction in the thriller genre focuses on stories with high drama, suspense, and excitement.

It frequently crosses over with other literary and cinematic subgenres like crime, adventure, and espionage. These days, there are numerous criminal drama series available on Netflix that make viewers empathise with murderers due to the way they are portrayed in the show. The real crime community would see a rise in fangirling over the killers, creating fan accounts for them, and producing TikTok videos defending them. Examples of the criminal filmographies that will be covered in this paper include Dahmer – Monster: The Jeffrey Dahmer Story, Extremely Wicked, Shockingly Evil and Vile, the Ted Bundy Series, and The Good Nurse and more. This paper aims to link criminal filmmaking with jurisprudence, mainly focusing on the natural law, which are the theory of morality, the theory of ethical relativism and the theory of interests. This study offers a brief representation of the perceived ethical difficulties that filmmakers, directors and producer-directors identify in the exercise of their trade. Furthermore, this study shows the importance of continuing and open discourse regarding ethical issues in filming. Filmmakers need to evolve a common concept of just ways to balance their many commitments as well as a more broadly accepted awareness of the nature of their challenges. It is essential to limit any severe alterations or impact of humans by unnatural, ferocious, and harsh forces to orient human behaviour toward preserving civilisations, morality, and customs.

Veerasamy, L., Ikhsan, N., & Althabhawi, N. (2023). Filmography Related to Thriller and Interrelations with Natural Law. Legal Network Series. Gonzales, C. (2021). A broken vow: an examination of the cases of Corruption in the Philippines. Sapienza: International Journal of Interdisciplinary Studies, 2(1), 47-65.

(62) Stolen Transformation, Conspiracy Theories and Female Detectives in Contemporary East-Central European Films and Series

The essay discusses the surging popularity of crime fiction as a reflection of the evolving social climate and the erosion of illusions regarding the post-socialist transformation in the 2010s Central Europe. It focuses on Entanglement (2011), a Polish paranoia thriller, X – The eXploited (2018), a Hungarian political thriller and Sleepers (2019), a Czech HBO production spy thriller series to examine how they portray symptoms of social and political crisis, perceived manipulation of the transition, and the struggle for female individual agency. These films offer a unique portrayal of the notion of a 'stolen' regime change, framed within conspiracy narratives suggesting the involvement of former state security personnel who managed to maintain their influence during the transition. Additionally, each narrative features a female protagonist who embarks on an investigation, uncovering a political conspiracy intertwined with her personal history. The fusion of conspiracy thriller elements with the emotional detective archetype represents a pivotal characteristic of these films, underscored by their transnational ambiguity. With the coupling of conspiracy narrative and female emotional detectives, paranoia and individual suffering, these films highlight the prevailing -disillusionment and the sense of inaction that characterizes the contemporary social climate in the region.
Varga, B. (2024). Stolen Transformation, Conspiracy Theories and Female Detectives in Contemporary East-Central European Films and Series. Studies in Eastern European Cinema, 15(2), 201-214.

(63) The screen novel: a creative practice approach to developing a screen idea for the television crime thriller.

This creative practice PhD is situated in the field of screenwriting practice studies and explores the development of a screen idea using novelistic techniques. By novelistic techniques, I mean the deployment of prose and image that one would usually associate with the fiction writer. It should be noted that I refer to image as a novelistic technique in the tradition of the illustrated novel. In writing the screen idea using the novel form, I explore how I and others might develop an idea, in this case a television (TV) series, that embeds the technical directives evident in the screenplay so that character and story are brought into focus. The creative artefact in this PhD is what I call a 'screen novel', which aims to complement and extend traditional approaches to script development. The intention of this research is to reframe how we discuss the development of a screen idea, in particular how we view the screenplay and summary texts as technical documents within the development and production process. As Steven Maras notes, 'numerous writers, at different times throughout

the history of film, tell us that production has little to do with writing' (2009, p. 52). Here Maras refers to a long-standing tradition that situates the screenplay as something to be made as opposed to something to be read (2009). Maras's comment resonates with my own experience of writing screenplays with both a technical and poetic sensibility.
Sculley, S. (2024). The screen novel: a creative practice approach to developing a screen idea for the television crime thriller (Doctoral dissertation, RMIT University).

(64) Crime Film and the City: Imagining Toronto through Entanglements of Place, Culture, and Crime

Crime films set in Toronto over the past two decades represent the city through a lens of crime and criminality and establish a cultural identity for it within the national imaginary. Drawing on material from interviews with key film directors as well as visual and narrative film analysis, this article traces three predominant images of Toronto: 1) a cold, prosperous, post-industrial center for a calculated, sophisticated pursuit of criminality; 2) a cosmopolitan, creative city where crime is akin to a craft; and 3) a multicultural metropolis of distinctive ethnic enclaves linked to criminal collectives. The analysis reveals how each image presents different facets of the global, post-industrial city, while drawing attention to its ambivalence. In these images, Toronto becomes a place where emerging cultural economies entwine with symbolic forms of criminal activity, new middle-class lifestyles have unintended criminal effects, and the promise of multicultural urbanism is dashed by the policing of incommensurable differences.

Bookman, S., Mackenzie, K., & Bernier, A. (2022). Crime film and the city: Imagining Toronto through entanglements of place, culture, and crime. American Review of Canadian Studies, 52(1), 46-63.

(65) A Study on the Behavioural Traits of Antagonists and Protagonists in Psycho-thriller

Happiness is a psychological state and subjective in nature. Psychological thrillers in various media forms including the web series have represented both hedonic happiness and eudemonic happiness with a focus on the psychology of the principle characters. What brings happiness to one might become a source of sadness and horror for another. Most psycho thriller web series try to explore this variation through the point of view of the main leads - the antagonist and the protagonist of the show - the life they lead, the things that provide a source of happiness and satisfaction towards them, their social surroundings, their

relationships, etc. The feeling of happiness can come from different sources in different forms for different characters. For the protagonists, happiness might be associated with some fond memories whereas for the antagonists, the source of happiness can be the thrill of killing or hunting of their prey without getting caught. While most of the thriller web series suggests that psychopath killers lack empathy and thus, they do not feel the pain of one's loss, these web series also showcase a commonality among the protagonists -those chasing the antagonists- as not being able to carry on a happy life while trying to balance between their personal and professional responsibilities. This paper explores as to how the concept of happiness are represented in the selected psychothriller web series by adopting the psycho-analytical approach of Sigmund Freud in exploring the conscious, subconscious, and unconscious actions of both the antagonists and protagonists while a qualitative research approach was adopted for analysing the interaction between the main leads - the protagonist and antagonist and people around them, besides the intrapersonal communication that goes on within the characters themselves.
Shahin, S. A Study on the Behavioural Traits of Antagonists and Protagonists in Psycho-thriller Web Series.

(66) Climate change impact on flood and extreme precipitation increases with water availability

The hydrological cycle is expected to intensify with global warming, which likely increases the intensity of extreme precipitation events and the risk of flooding. The changes, however, often differ from the theorized expectation of increases in water-holding capacity of the atmosphere in the warmer conditions, especially when water availability is limited. Here, the relationships of changes in extreme precipitation and flood intensities for the end of the twenty-first century with spatial and seasonal water availability are quantified. Results show an intensification of extreme precipitation and flood events over all climate regions which increases as water availability increases from dry to wet regions. Similarly, there is an increase in the intensification of extreme precipitation and flood with the seasonal cycle of water availability. The connection between extreme precipitation and flood intensity changes and spatial and seasonal water availability becomes stronger as events become less extreme.

Tabari, H. (2020). Climate change impact on flood and extreme precipitation increases with water availability. Scientific reports, 10(1), 13768.

(67) Role of dams in reducing global flood exposure under climate change

Globally, flood risk is projected to increase in the future due to climate change and population growth. Here, we quantify the role of dams in flood mitigation, previously unaccounted for in global flood studies, by simulating the floodplain dynamics and flow regulation by dams. We show that, ignoring flow regulation by dams, the average number of people exposed to flooding below dams amount to 9.1 and 15.3 million per year, by the end of the 21st century (holding population constant), for the representative concentration pathway (RCP) 2.6 and 6.0, respectively. Accounting for dams reduces the number of people exposed to floods by 20.6 and 12.9% (for RCP2.6 and RCP6.0, respectively). While environmental problems caused by dams warrant further investigations, our results indicate that consideration of dams significantly affect the estimation of future population exposure to flood, emphasizing the need to integrate them in model-based impact analysis of climate change.

Boulange, J., Hanasaki, N., Yamazaki, D., & Pokhrel, Y. (2021). Role of dams in reducing global flood exposure under climate change. Nature communications, 12(1), 417.

(68) Evidence of shorter more extreme rainfalls and increased flood variability under climate change

Increases in extreme rainfall intensities as a result of climate change pose a great risk due to the possibility of increases in pluvial flooding, particularly in urban and developed areas. But evidence is emerging that the observed increases in extreme rainfall are not resulting in universal increases in flooding. Indeed, on a global scale, studies consistently find more gauge stations with decreasing rather than increasing trends in the annual maxima flood magnitude. Here, we aim to improve our understanding of how rainfall and streamflow extremes are changing and why floods are not always observed to increase despite increases in rainfall extremes. To do so, we examine trends in streamflow events using 2776 stations from the Global Runoff Data Centre, with events chosen to isolate the impact of changes to their respective rainfall and antecedent soil moisture. The analysis is limited to stations with 30 years or more of active record with the majority of stations in North America, Europe, Brazil, Oceania, and southern Africa.

Consistent with physical reasoning, for more frequent events such as the annual maxima, it is found the peak 1-day rainfall event intensity is increasing approximately a rate of 6–7%/°C, with rarer event intensities increasing at a rate exceeding the Clausius-Clapeyron relation. We find that storm volumes are not increasing as greatly as the peak rainfall and storm durations are decreasing, pointing to an intensification of rainfall events and a peakier temporal pattern. While rainfall is intensifying, the magnitude of frequent floods (those expected to occur on average once per year) are in general decreasing, particularly in tropical and arid regions of

the world. We find that this is likely due to a dominance of drying antecedent soil moisture conditions. However, the magnitude of rarer floods has, on average been increasing. We suggest this is because, for these rarer events, the increase in rainfall outweighs the decrease in soil moisture. Our results point to a worst of both world's scenario where small floods, responsible for filling our water supplies, are decreasing, while the large flood events which pose a risk to life and infrastructure, are increasing.

Wasko, C., Nathan, R., Stein, L., & O'Shea, D. (2021). Evidence of shorter more extreme rainfalls and increased flood variability under climate change. Journal of Hydrology, 603, 126994.

(69) Flood Risk in Urban Areas: Modelling, Management and Adaptation to Climate Change. A Review

The modelling and management of flood risk in urban areas are increasingly recognized as global challenges. The complexity of these issues is a consequence of the existence of several distinct sources of risk, including not only fluvial, tidal and coastal flooding, but also exposure to urban runoff and local drainage failure, and the various management strategies that can be proposed. The high degree of vulnerability that characterizes such areas is expected to increase in the future due to the effects of climate change, the growth of the population living in cities, and urban densification. An increasing awareness of the socio-economic losses and environmental impact of urban flooding is clearly reflected in the recent expansion of the number of studies related to the modelling and management of urban flooding, sometimes within the framework of adaptation to climate change. The goal of the current paper is to provide a general review of the recent advances in flood-risk modelling and management, while also exploring future perspectives in these fields of research.
Cea, L., & Costabile, P. (2022). Flood risk in urban areas: Modelling, management and adaptation to climate change. A review. Hydrology, 9(3), 50.

(70) Increased Flood Exposure Due to Climate Change and Population Growth in the United States

Precipitation extremes are increasing globally due to anthropogenic climate change. However, there remains uncertainty regarding impacts upon flood occurrence and subsequent population exposure. Here, we quantify changes in population exposure to flood hazard

across the contiguous United States. We combine simulations from a climate model large ensemble and a high-resolution hydrodynamic flood model—allowing us to directly assess changes across a wide range of extreme precipitation magnitudes and accumulation timescales. We report a mean increase in the 100-year precipitation event of ~20% (magnitude) and >200% (frequency) in a high warming scenario, yielding a ~30–127% increase in population exposure. We further find a nonlinear increase for the most intense precipitation events—suggesting accelerating societal impacts from historically rare or unprecedented precipitation events in the 21st century.

Swain, D. L., Wing, O. E., Bates, P. D., Done, J. M., Johnson, K. A., & Cameron, D. R. (2020). Increased flood exposure due to climate change and population growth in the United States. Earth's Future, 8(11), e2020EF001778.

(71) Enhancement of river flooding due to global warming

Human-induced climate change has increased the frequency and intensity of heavy precipitation1. Due to the complexity of runoff generation and the streamflow process, the historical impact of human-induced climate change on river flooding remains uncertain. Here, we address the question of whether anthropogenic climate change has altered the probability of the extreme river flood events for the period 1951–2010 based on simulated river discharge derived from large ensemble climate experiments with and without human-induced climate change. The results indicate that human-induced climate change altered the probabilities of 20 of the 52 analyzed flood events. Fourteen of these 20 flood events, which occurred mainly in Asia and South America, were very likely to have been enhanced by human-induced climate change due to an increase in heavy precipitation. Conversely, two flood events in North/South America and two flood events in Asia and two flood events in Europe were suppressed by human-induced climate change, perhaps as a result of lower snowfall. Human-induced climate change has enhanced flooding more prominently in recent years, providing important insights into potential adaptation strategies for river flooding.

Alifu, H., Hirabayashi, Y., Imada, Y., & Shiogama, H. (2022). Enhancement of river flooding due to global warming. Scientific Reports, 12(1), 20687.

(72) More meteorological events that drive compound coastal flooding are projected under climate change

Compound flooding arises from storms causing concurrent extreme meteorological tides (that is the superposition of storm surge and waves) and precipitation. This flooding can severely

affect densely populated low-lying coastal areas. Here, combining output from climate and ocean models, we analyse the concurrence probability of the meteorological conditions driving compound flooding. We show that, under a high emissions scenario, the concurrence probability would increase globally by more than 25% by 2100 compared to present. In latitudes above 40o north, compound flooding could become more than 2.5 times as frequent, in contrast to parts of the subtropics where it would weaken. Changes in extreme precipitation and meteorological tides account for most (77% and 20%, respectively) of the projected change in concurrence probability. The evolution of the dependence between precipitation and meteorological tide dominates the uncertainty in the projections. Our results indicate that not accounting for these effects in adaptation planning could leave coastal communities insufficiently protected against flooding.

(73) Anthropogenic climate change has changed frequency of past flood during 2010-2013

The ongoing increases in anthropogenic radiative forcing have changed the global water cycle and are expected to lead to more intense precipitation extremes and associated floods. However, given the limitations of observations and model simulations, evidence of the impact of anthropogenic climate change on past extreme river discharge is scarce. Here, a large ensemble numerical simulation revealed that 64% (14 of 22 events) of floods analyzed during 2010-2013 were affected by anthropogenic climate change. Four flood events in Asia, Europe, and South America were enhanced within the 90% likelihood range. Of eight snow-induced floods analyzed, three were enhanced and four events were suppressed, indicating that the effects of climate change are more likely to be seen in the snow-induced floods. A global-scale analysis of flood frequency revealed that anthropogenic climate change enhanced the occurrence of floods during 2010-2013 in wide area of northern Eurasia, part of northwestern India, and central Africa, while suppressing the occurrence of floods in part of northeastern Eurasia, southern Africa, central to eastern North America and South America. Since the changes in the occurrence of flooding are the results of several hydrological processes, such as snow melt and changes in seasonal and extreme precipitation, and because a climate change signal is often not detectable from limited observation records, large ensemble discharge simulation provides insights into anthropogenic effects on past fluvial floods.

Hirabayashi, Y., Alifu, H., Yamazaki, D., Imada, Y., Shiogama, H., & Kimura, Y. (2021). Anthropogenic climate change has changed frequency of past flood during 2010-2013. Progress in Earth and Planetary Science, 8(1), 36.

Bevacqua, E., Vousdoukas, M. I., Zappa, G., Hodges, K., Shepherd, T. G., Maraun, D., ... & Feyen, L. (2020). More meteorological events that drive compound coastal flooding are projected under climate change. Communications earth & environment, 1(1), 47.

(74) patial Prediction of Future Flood Risk: An Approach to the Effects of Climate Change

Preparation of a flood probability map serves as the first step in a flood management program. This research develops a probability flood map for floods resulting from climate change in the future. Two models of Flexible Discrimination Analysis (FDA) and Artificial Neural Network (ANN) were used. Two optimistic (RCP2.6) and pessimistic (RCP8.5) climate change scenarios were considered for mapping future rainfall. Moreover, to produce probability flood occurrence maps, 263 locations of past flood events were used as dependent variables. The number of 13 factors conditioning floods was taken as independent variables in modeling. Of the total 263 flood locations, 80% (210 locations) and 20% (53 locations) were considered model training and validation. The Receiver Operating Characteristic (ROC) curve and other statistical criteria were used to validate the models. Based on assessments of the validated models, FDA, with a ROC-AUC = 0.918, standard error (SE = 0.038), and an accuracy of 0.86% compared to the ANN model with a ROC-AUC = 0.897, has the highest accuracy in preparing the flood probability map in the study area. The modeling results also showed that the factors of distance from the River, altitude, slope, and rainfall have the greatest impact on floods in the study area. Both models' future flood susceptibility maps showed that the highest area is related to the very low class. The lowest area is related to the high class.
Avand, M., Moradi, H. R., & Ramazanzadeh Lasboyee, M. (2021). Spatial prediction of future flood risk: an approach to the effects of climate change. Geosciences, 11(1), 25.

(75) Assessing the Effects of Climate Change on Compound Flooding in Coastal River Areas

Flood assessment in coastal river areas is subject to complex dependencies and interactions between flood drivers. In addition, coastal areas are especially vulnerable to climate change, and thus its effects should be considered in the evaluation of future flood hazard. In the present study, we propose a methodology for a robust evaluation of historical and future flooding in coastal river areas. It follows a continuous simulation approach in which a hydrologic-hydraulic modeling cascade is run for several years, driven by a simultaneous series of flood drivers. The method differs from other existing methodologies in: (a) jointly

considering the relevant flood drivers, both coastal and inland, together with their dependencies and interactions, to represent the simultaneous occurrence of extreme water levels correctly, and (b) obtaining local-scale physically consistent projections of flood drivers to account for future climate change effects on compound flood hazard. The methodology is tested for a coastal river reach in NW Spain. The results show that, although mean sea level rise is usually expected to be the main driver of increased flood hazard, climate change effects on river discharge can play a very significant role in regions with compound flooding potential.

Bermúdez, M., Farfán, J. F., Willems, P., & Cea, L. (2021). Assessing the effects of climate change on compound flooding in coastal river areas. Water resources research, 57(10), e2020WR029321.

(76) An Open Data Approach to Mapping Urban Drainage Infrastructure in Developing Communities

Urban flooding in developing countries represents a growing threat to sustainable development efforts, yet the tools needed to study these infrastructure systems in data-scarce environments are woefully inadequate. This study seeks to propose a standardized approach and methods for mapping urban drainage systems in developing communities. The research draws on a case study from the Philippines, which sought to conduct rapid elevation surveys and drainage assessments employing open source geographical information system (GIS) tools. We develop a standardized procedure for digitizing drainage systems using OpenStreetMap and Field Papers, as well as discuss applications of this data for drainage design. The results contribute to a methodological framework that can be replicated in other similar developing communities where study of urban drainage is needed for sustainable development and disaster risk reduction efforts.

See, L. S., Calo, L., Bannon, B., & Opdyke, A. (2020). An open data approach to mapping urban drainage infrastructure in developing communities. Water, 12(7), 1880.

(77) Developing an integrated multiscale rainfall-runoff and inundation model: Application to an extreme rainfall event in Marikina-Pasig River Basin, Philippines

A multiscale rainfall-runoff and inundation model was developed to simulate the severe flooding caused by a high-intensity short-duration storm rainfall event in the urban domain. The model used the 2D shallow-water flow and finite volume schemes with second-order space–time accuracy to automatically incorporate the runoff into the flood inundation. The adaptive mesh refinement scheme—a variable grid-sizing technique—was implemented to

dynamically refine the mesh from a watershed-scale to a more precise inundation in a subcatchment floodplain. The digital terrain and surface models were used to represent the variations in flood propagation along buildings and artificial barriers in the urban area. Additionally, the simulated flood depths and extent were validated by comparison with the observed data.

Dasallas, L., An, H., & Lee, S. (2022). Developing an integrated multiscale rainfall-runoff and inundation model: Application to an extreme rainfall event in Marikina-Pasig River Basin, Philippines. Journal of Hydrology: Regional Studies, 39, 100995.

(78) Analysis of flood damage and influencing factors in urban catchments: case studies in Manila, Philippines, and Jakarta, Indonesia

The sustainability and efficiency of flood risk management depends on the assessment of flood hazards and on the quantification of flood damage. Under the conditions of climate change and rapid urbanization, the evaluation of flood risk can lead to the success of adaptation strategies. The main objectives of this study are the estimation of future direct flood damage in two urban watersheds: The Pasig–Marikina–San Juan River in Metro Manila, Philippines, and the Ciliwung River in Jakarta, Indonesia, as well as the determination of the relation between factors that drive floods and flood damage. A spatial analysis approach based on the integration of several parameters, such as flood hazard, climate, and property value, was applied using a Geographic Information System (GIS). The flood depth-damage function generated from the field surveys was employed for the analysis to identify the spatial distribution of flood loss. The findings showed that, under future scenarios (target year: 2030), the total flood damage will increase by 212% and 80% in the target areas of Manila and Jakarta, respectively, compared to the current scenarios. This growth is due to the higher level of extreme rainfall events and to the degree of urbanization in the future. A comparative analysis of the two study areas highlighted the significant effects of the level of water depth and the inundated areas on flood damage, depending on the sites. This study is useful for local decision makers to implement suitable strategies for urban planning and flood control.

Kefi, M., Mishra, B. K., Masago, Y., & Fukushi, K. (2020). Analysis of flood damage and influencing factors in urban catchments: case studies in Manila, Philippines, and Jakarta, Indonesia. Natural Hazards, 104(3), 2461-2487.

(79) RAINWATER RETENTION SITE ASSESSMENT FOR URBAN FLOOD RISK REDUCTION AND FLOOD DEFENCE IN MANDAUE CITY, PHILIPPINES

This study utilized a geospatial approach to identify suitable sites for rainwater retention areas in Mandaue City, a highly urbanized city in Cebu, Philippines. Based on the integrated use of remote sensing and Geographic Information System (GIS), ideal sites for rainwater retention were produced to divert surface water runoff from flooding the streets and obstructing traffic. Ideally, the chosen sites should have low infiltration capacity and be within open spaces. For the hydrological planning of the study, we considered the amount of surface runoff, elevation, soil characteristics, land use, and present drainage status of the study area. The Digital Elevation Model (DEM) derived from LiDAR technology is used to obtain the slope raster and generate small streams. Land cover and curve number (CN) of the area were used to compute the surface runoff. The soil textures were converted based on their infiltration rate and runoff potential. Finally, the weighted overlay tool in GIS created a layer identifying potential areas for rainwater retention sites. The findings indicate that 54.8% of the research area was ideal for water harvesting, with open land areas making up 2.3% of the best locations. In addition, a major portion of the appropriate locations lies in clay loam soils and within 0–25 meters in elevation. These results will hugely benefit policymakers and urban planners in helping create solutions to the city's flooding problem and the looming water crisis.

Bañados, J. H., & Quijano, I. P. (2022). Rainwater retention site assessment for urban flood risk reduction and flood defence in Mandaue City, Philippines. ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 10, 1-7.

(80) Flood Control Management in Built Environment

The word "Flooding" is very common now and a major cause of concern in many parts of the world because of its dynamic effect on the lives of humans, animals, properties, and other assets. A flood is due to wet environmental conditions that cause the moisture content of the soil to rise much above the normal conditions and hence the water does not get absorbed and that results in the collection of water above the ground level. This has been discussed in the author's article "Does Climate Change Impact our Environment?" ResearchGate/LinkedIn platforms, June 2021. Therein it is mentioned that"…with tropical areas like the Philippines with archipelagic locations, it is indeed that Climate Change impacting and suffers the entire environment. And as a consequence of this phenomenon, the cause of the flooding is more frequent and intense throughout the archipelagic terrain of the country due to the low pressure that has been developed in the Pacific Ocean especially whenever typhoons" occur. These elements, as well as heavy rain conditions, contribute to significant flooding occurring in cities and municipalities, the result in rising water levels in the drainage system because of water coming in from different sources of drainage connections and being let into rivers,

streams, lakes, and sea. The water levels in those also consequently rise and many times go above the bank levels.
Nobody can control Mother Nature; however, humans can minimize or mitigate the risks that can arise and cause floods, through careful studies and investigations of the affected areas and then draw up flood management plans. The flood control management plans could be either within the built environment or even outside it. It is advantageous while preparing the emergency flood management plan to incorporate the four phases of the process starting with planning for flood mitigation, preparedness, response, and recovery. The four phases are used normally in planning for Disaster Resilience. Flooding mostly affects low-lying areas but could also affect others where the land use and drainage system are not thought of with a long-term view and planned properly.

Gomeseria, R. (2021). Flood Control Management in Built Environment. Available at SSRN 3926560.

(81) Interrelationships of the barriers to integrated flood risk management adaptation in Metro Manila, Philippines

Many countries encountered “barriers” or obstacles that hampered the process of adapting to the integrated flood risk management (IFRM) from the traditional hard structural solutions in managing urban flooding. One of the crucial tasks to overcome the barriers is to understand their interrelationships. However, analysis of the interrelationships between the barriers to IFRM adaptation has not been carried out yet. This study attempts to analyze the interrelationships between the barriers to IFRM adaptation using the Interpretive Structural Modeling (ISM) method. The barriers in Metro Manila, a megacity in a developing country, were identified first in this study. Then, the ISM method was slightly modified and applied to analyze the interrelationships of the identified barriers. As a result, we identified 12 barriers which are relatively numerous compared with the developed countries, and so we categorized them into the governance, social, and technological resources aspects. Through the application of the ISM method, the interrelationships of the barriers to IFRM adaptation were systematically analyzed for the first time while also showing their hierarchical diagram. The results of the ISM reveal that barriers in the governance aspect are the most influential in which the lack of a sole organizing body is the most influential barrier. The barriers in the technological resources aspect are the second most influential, while barriers in the social aspect are the least influential and most dependent barriers. The approach presented in this study can be useful for decision makers and practitioners in understanding the interrelationships between the barriers.

Mercado, J. M. R., Kawamura, A., & Amaguchi, H. (2020). Interrelationships of the barriers to integrated flood risk management adaptation in Metro Manila, Philippines. International Journal of Disaster Risk Reduction, 49, 101683.

(82) Assessing Flood Occurrence and Modeling Impact of Urban Drainage Structure

This study evaluated the catchment and flood prone area by applying physical and numerical modeling. This described rainfall data and presented evidences of flooding based from extreme precipitation events. Mathematical models were derived using the definition of discharge and Manning's formula which finally were used to estimate the sufficiency of the river canal cross sectional area to transport great amount of water flow. Mt. Isarog, mountain ranges of Lagonoy and the low land area are the sources of flood water in Lagonoy River. The amount of daily precipitation is always insufficient to induce flooding except for a terrifying event that the area was heavily affected. The river canal cross sectional area is sufficient to accommodate flood water flow but the position of the moon which is indicating differences on river depth during full/new moon and first/third quartermoon, high tide and low tide, and with heavy and non-heavy rainfall.causes the flood while others are due to river bends, insufficient design of spillway and obstruction of irrigation waterway.
Romero, R., Romero, N., Romero, F., & Cronico, J. C. (2020). Assessing Flood Occurrence and Modeling Impact of Urban Drainage Structure. Computational Engineering and Physical Modeling, 3(3), 1-11.

(83) Sustainable Drainage System: Low Impact Development Practices to Minimize the Storm Water Runoff

Low Impact Development (LID) is a green infrastructure approach to ease the surface runoff that arises due to climate change and from an increasing impervious surface area caused by urbanization. This study aims to determine which LID control is best suited for Barangay San Rafael, San Jose Del Monte, Bulacan. Using Storm Water Management Model (SWMM), a hydraulic model of the area was created to apply the LIDs and simulate a runoff. The data used in SWMM for the runoff simulation is the monthly rainfall values for year 2020. Based on the simulation using SWMM, the existing drainage system has a high total flood volume with total flood volume of 243.511 liters. Based on the data collected from SWMM, the most effective combination with total flood volume of 50.884 liters was the combination of all the LID practices. The study has used three (3) different LID practices and the combinations of the LID practices to obtain the ideal LID combination namely, Bioretention Cell, Bioswale,

Bioswale and Permeable Pavement. In the final analysis, the combination of Permeable Pavement, Bioretention Cell, and Bioswale is the best combination out of the three LID controls mentioned. Bioretention Cells are primarily used in parking lot islands, traffic islands, and driveway runoff. The same is true for Permeable Pavement, which is used mainly on roadways and parking lot islands. Considering the total flood volume that the SWMM calculated, the combination of the three LID parameters alone has the lowest total flood volume.

Calot, J. B., Galang, J. B. R., Lagrata, J. P. R., Marcelino, R. A. T., Buenconsejo, M. V., Poso, F. D., & Escarieses, L. L. E. (2022, December). Sustainable Drainage System: Low Impact Development Practices to Minimize the Storm Water Runoff. In 2022 IEEE 14th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM) (pp. 1-5). IEEE.

(84) Assessment of the Existing Drainage System in Infanta, Quezon Province for Flood Hazard Management using Analytical Hierarchy Process.

The residents of Infanta, Quezon Province has been dealing with flood and calamities for several years. With that, this research focused on the radical assessment of existing drainage system in the municipality. Lidar DEM was used to determine the elevation and slopes in the area which is a very important input drainage system analysis. The researchers did vulnerability assessment in the area considering relevant factors about the existing drainage system. These factors were elevation, population, soil permeability, rainfall intensity, flow discharge of drainage, distance from river basin to drainage line, and land cover. The importance of each factor was determined based on the survey made from the researchers of the FRAMER project of Mapua-DOST PCIEERD. Weights of each factor were identified with the aid of Analytical Hierarchy Process. Weights of each factors are as follows Flow discharge capacity with 22.3%, Elevation 17.1%, Soil permeability 14.6%, Distance to river 14.5%, Rainfall intensity 12.4%, Population 11.8% and Land Use 7.2%. Validity of the survey made from the experts were checked by calculating the consistency of their answers. Consistency index was calculated to be 0.11 which is in the range of acceptable value. A risk map was then created using geographical information system to determine spatially which drainage line is most vulnerable to flooding. It was found out that there are seven zones that are at risk to flooding due to the inadequacy of the drainage and other factors mentioned in the vulnerability assessment but Poblacion area is the one that needed the rehabilitation the most.

Monjardin, C. E. F., Tan, F. J., Uy, F. A. A., Bale, F. J. P., Voluntad, E. O., & Batac, R. M. N. (2020, April). Assessment of the existing drainage system in Infanta, Quezon province for

flood hazard management using analytical hierarchy process. In 2020 IEEE Conference on Technologies for Sustainability (SusTech) (pp. 1-7). IEEE.

(85) Attenuation performance of runoff storage basins within a moderate to steep slope urban catchment in Cebu, Philippines

In urban areas, runoff storage basins (RSBs) are common flood control measures. However, space utilization of these basins compete with other land use demands, thus it is necessary to evaluate the attenuation performance of these RSBs to justify their presence. In this study, the flood attenuation performance of RSBs which comprise a series of 3 mini dams (3-m high), a lagoon, and 19 small detention ponds within a small catchment in Cebu, Philippines was evaluated. The 3 mini dams intercept runoff water from developed portion of the catchment and release the water to a lagoon which in turn discharges the water to an open channel terminating to a street drainage culvert that conveys water to the wetland. The 19 small detention ponds are distributed to different locations within a subcatchment which is also tributary to the culvert. The catchment has an area of 3km2. It is urbanized and has moderate to steep slopes; 19% of which has a slope ranging from 15% to 30% and 43% of the catchment has slope greater than 30%. The catchment drains its runoff into a wetland. The HEC-GeoHMS extension of ArcGIS was utilized in delineating the catchment. The hydrologic properties were determined using soil type, land use, and land cover maps. The soil type map was obtained from the Department of Environment and Natural Resources (DENR). The land use and land cover maps were extracted from 0.5-m resolution Orthophoto and 1-m resolution LiDAR data provided by the Philippine Light Detection and Ranging (Phil-LiDAR) project. HEC-HMS was used in the rainfall-runoff simulation incorporating SCS loss, SCS unit hydrograph, and kinematic wave and Muskingum flow routing. Parameters were calibrated and validated using measured rainfall data and corresponding outflow hydrographs estimated from the time-stamped camera-captured changes in water level and depth-volume relation of the wetland. The calibration and validation yielded Nash-Sutcliffe Efficiency of 0.86 and 0.67, respectively. Existing RSBs attenuated 50% of flood peak at inlet of drainage culvert for 100-yr return period of 6-hr duration rainfall.
Lalisan, C. L., & Fornis, R. L. (2020, October). Attenuation performance of runoff storage basins within a moderate to steep slope urban catchment in Cebu, Philippines. In AIP Conference Proceedings (Vol. 2278, No. 1, p. 020002). AIP Publishing LLC.

(86) Amnesties and intractable conflicts: Managed impunity in The Philippines' Bangsamoro peace process.

At the heart of intractable conflict lies a challenging paradox: impunity is both a cause of intractability and, operationalized through the granting of amnesties, one of the key tools employed by peacemakers in their attempts to end long-running periods of violent conflict. That is, impunity is simultaneously conceived of a driver and a solution to the problem of intractability. Focusing on one of the world's most intractable conflicts, the Bangsamoro conflict in The Philippines, this article examines the nature of the impunity/intractability paradox, assesses current attempts to overcome it in the Bangsamoro peace process, and argues that a process of 'managed impunity' introduced by The Philippines' Transitional Justice and Reconciliation Commission provides one possible way of ending impunity for human rights crimes without jeopardizing the prospects for peace.

Jeffery, R. (2018). Amnesties and intractable conflicts: Managed impunity in The Philippines' Bangsamoro peace process. *Journal of Human Rights*, *17*(4), 436-452.

(87) Bias in the Law: A Definitive Look at Racial Prejudice in the Philippine Criminal Justice System: A Systematic Literature Review

This paper aims to identify communities in the Philippines that are prejudiced and discriminated against in life and before the law. It seeks to determine whether these biases are experienced because they are considered "others" or "different," aside from belonging to minority populations. This study is relevant as prejudice and discrimination based on race or ethnicity, among other socially constructed identifiers, violate the principle of equality and non-discrimination. Filipinos have been discriminated against since it was first colonized by Spain some 500 years ago; and for discrimination and prejudice to continue in this modern day is a pressing concern. Through a systematic review of literature, the study found that racism is systemic and global; it is more pronounced across sections like when a person is of color, a minority, and poor; impunity and perpetuation of racial and ethnic discrimination is prevalent; ethnolinguistic groups who are the minority, including Muslims, are highly discriminated in the Philippines; and the one's "exclusion of the others" contribute to discrimination. The Philippines must address discrimination and prejudice by fulfilling its commitments to international conventions and treatises that seek the immediate end of all forms of discrimination. Law enforcers must guarantee equal protection as enumerated in the 1987 Philippine Constitution. The long overdue anti-discrimination bill must be passed to strengthen the protection of Filipinos against perpetrators of discrimination, whether they are individuals or institutions. It is recommended that more scholars study this subject comprehensively, particularly the nation's criminal justice system.
***ISSN, 2948,* 4863.**

(88) Punitiveness and atrocity: Why do some Filipinos support mass violence?

Amidst the aftermath of the controversial war on drugs during Duterte's presidency in the Philippines, this study seeks to understand characteristics of Filipino punitiveness. The institutionalization of state violence, historical context of authoritarian rule, and societal acceptance of extreme measures prompt an examination of public attitudes toward crime, punishment, and violence in the Philippines. This study used survey data on Filipino respondents and identified five distinct punitiveness clusters using latent class analysis. This study employed Kruskal–Wallis (KW) tests and chi-squared tests to explore factors associated with differences in cluster characteristics. The analysis found that the extremely punitive group exhibited significantly higher levels of satisfaction with President Marcos Jr., a strong preference for authoritarian detachment, and lower trust in institutions compared to the other clusters. They support harsher penal policies including informal or extralegal methods such as torture, social cleansing, and use of violence to attain social progress. This study confirms the convergence of punitiveness and penal populism, where highly punitive citizens endorse strong leaders and extralegal measures that result in atrocities.

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(89) (In)security and Hybrid Justice Systems in Mindanao, Philippines

Hybridity is often conflated with the fragile state or the ‘absence’ of the state in a conflict environment. The emergence of hybrid institutions is also explained primarily in terms of the lack of capacity and legitimacy of state organs and its personnel or in the condition of a power vacuum. A sense of power disequilibrium or societal imbalance and disarray inheres from this presumption. Hybridity, however, serves a function that sustains conflict resilience and at the same time address immediate justice needs. Hybrids arise to provide a state of equilibrium and to provide order in an otherwise messy condition — while also contributing to the messiness. My study of the Autonomous Region of Muslim Mindanao (ARMM), Philippines, points to how hybrid justice mechanisms have developed to cope with insecurity arising from actual and perceived injustices in the community.

. **ISBN: 9781760461836**

(90) Duterte's International Criminal Court Arrest and the Moral Economy of Reciprocity in the Philippines

The arrest of former Philippine President Rodrigo Duterte under an International Criminal Court warrant and his subsequent detention appear to have amplified his popular appeal among Filipino publics. This article unpacks Duterte's enduring popularity by analysing how Filipino cultural values reinforce populist and nationalist narratives, particularly during moments of political crisis. We focus on how *utang na loob* (debt of gratitude) has been mobilised to defend Duterte and critique the Marcos administration. Our findings suggest that Duterte's supposed policy legacy and his performative political will forged a social contract many Filipinos view as having delivered tangible benefits. Within the Filipino moral economy of reciprocity, such gains engender expectations of loyalty. The Philippine example illustrates how morally charged populist and nationalist discourses can frame political allegiances that resist scrutiny and international accountability.

Talamayan, F., & Candelaria, J. L. (2026). Duterte's International Criminal Court Arrest and the Moral Economy of Reciprocity in the Philippines. *Asia Pacific Viewpoint*.

(91) Ethics in the Rubble: Disaster Governance, Corruption, and the Moral Cost of Calamities in the Philippines.

In the Philippines, natural disasters often expose not only environmental vulnerabilities but also deep ethical failures in governance. This paper examines the moral dimensions of disaster response through the lens of the 2023-2025 flood control scandal, which involved thousands of substandard or fictitious infrastructure projects. Drawing on testimonies, Senate inquiries, and Filipino ethical concepts such as loob and pakikipagkapwa, the study argues that corruption in disaster governance constitutes a "secondary disaster"--a man-made crisis that compounds suffering, erodes public trust, and violates principles of justice and human dignity. The analysis situates corruption as both a structural and moral failure, revealing how disaster profiteering undermines resilience and deepens inequality. By integrating indigenous moral frameworks with global disaster ethics, the paper offers a culturally grounded critique of governance and proposes reforms centered on ethical leadership, transparency, and communitybased resilience. Ultimately, the study reframes disaster response not merely as a logistical challenge but as a moral imperative, calling for a transformation in how public service is understood and practiced in times of crisis.

Lofredo, M. P. P. (2026). Ethics in the Rubble: Disaster Governance, Corruption, and the Moral Cost of Calamities in the Philippines. Eubios Journal of Asian & International Bioethics, 36(2), 49.

(92) Faith, Floods, and False Promises: Politics, Religion, and Corruption in Philippine Flood Control Projects

The Philippines, as the only predominantly Catholic nation in Asia, embodies a paradox: while its people profess deep religious faith, its political system suffers from entrenched corruption. Nowhere is this contradiction more evident than in flood control projects, where ghost projects, misallocated budgets, and personal enrichment undermine both environmental protection and public welfare. This paper employs a qualitative methodological framework combining archival research, case studies of Department of Public Works and Highways (DPWH) flood control projects, and theological reflection rooted in Catholic Social Teaching and Filipino spirituality, particularly the concept of pananagutan (responsibility/accountability). By analyzing political corruption in light of papal encyclicals, biblical texts, and the lived reality of recurrent flooding, this paper argues that corruption represents not only a political failure but also a profound moral and spiritual crisis. The misuse of taxpayers' money for personal gain erodes public trust and contradicts the Gospel values of justice, solidarity, and stewardship. Addressing these failures requires structural reform and moral renewal anchored in pananagutan, transparency, and Gospel-centered governance.

Pabalan, J. M. G., & Perez, G. (2026). Faith, Floods, and False Promises: Politics, Religion, and Corruption in Philippine Flood Control Projects. International Journal of Research in Social Science and Humanities (IJRSS) ISSN: 2582-6220, DOI: 10.47505/IJRSS, 7(6), 146-153.

(93) Political Economy: The Truth Behind Politics and the Challenges in Advancing Growth and Development of the Philippine Economy

Background:
Political economy highlights the interdependence of political, economic, and legal institutions in shaping a nation's development trajectory. In the Philippines, the legislative control over public budgeting—referred to as the "power of the purse"—plays a critical role in steering national economic priorities. However, recent revelations on the implementation of flood control projects exposed deep-seated corruption involving lawmakers, government officials, and private contractors. These anomalies compromise infrastructure integrity, exacerbate climate-related vulnerabilities, and undermine public trust.

Methodology:

This study employs qualitative documentary analysis using secondary data from government reports, televised interviews, senate and congressional hearings, Presidential communications, and investigative news articles. Additional insights from independent inquiries led by public officials were incorporated to triangulate evidence regarding procurement processes, project implementation, and budget allocations.

Results:
Findings reveal a systematic pattern of collusion in flood control projects from 2022 to 2025. Evidence suggests that up to 55–60% of project funds were diverted as commissions for legislators, public works officials, auditors, and political intermediaries. Multiple contractors linked to the same individuals were found repeatedly winning bids, while several projects were identified as substandard or completely non-existent. The collapse of infrastructure, widespread flooding despite billions spent, and reports of lavish lifestyles among involved officials reinforced the scale of misuse. The exposure of ghost projects and fraudulent procurement practices triggered national investigations and the formation of civic reporting platforms such as Sumbong Mo sa Pangulo.

Conclusion:
The analysis demonstrates that corruption in the Philippines' flood control programs significantly weakens disaster mitigation efforts, wastes public resources, and contributes to uneven economic development. Strengthened oversight, transparent procurement, lifestyle checks, and decisive legal action are necessary to restore institutional integrity and ensure that public infrastructure investments genuinely serve national welfare.

SSRN: 5807262

(94) GOVERNING FLOOD RISK IN THE PHILIPPINES: MECHANISMS OF DISASTER RISK PRODUCTION IN FLOOD CONTROL INFRASTRUCTURE

Despite decades of investment in flood control infrastructure, flooding in the Philippines remains a pervasive crisis, signaling profound challenges for Disaster Risk Reduction (DRR). While prevailing explanations cite climatic and technical factors, this paper argues that flood risk is actively produced through routine governance practices. Using political ecology as a diagnostic lens, we move beyond critiques of institutional failure to identify and evidence three specific causal mechanisms of disaster risk creation: (1) corrupt implementation that engineers substandard infrastructure, (2) the strategic exploitation of institutional fragmentation to enable rent-seeking and diffuse accountability, and (3) exclusionary planning that redistributes risk onto marginalized communities. Employing a qualitative case study methodology, the paper conducts a structured analysis of secondary evidence—including Commission on Audit reports, investigative journalism, and project

documentation—from three emblematic cases. This systematic approach offers an original, mechanism-based explanation of risk creation. The findings demonstrate that these mechanisms are not incidental but embedded in the political economy of infrastructure. Consequently, we argue that effective disaster risk reduction requires a fundamental shift from infrastructural solutions to governance interventions that disrupt these risk-producing pathways. We propose specific, actionable reforms, including mandatory social audits, the creation of an independent infrastructure ombudsman, and legally mandated co-design processes, providing a concrete bridge from political ecology theory to policy practice.

SSRN 6213037.

(95) DISASTER-RELATED DISPLACEMENT IN THE PHILIPPINES

Every year that passes, it is increasingly clear that the number of natural disasters is rising worldwide. Very often, when a natural hazard strikes a vulnerable community, it transforms from a potential harmful event into a disaster that has catastrophic social and economic consequences. Disasters can be seen as one of the biggest obstacles to sustainable development and poverty eradication as they can drive people into poverty and can reverse the development gains built up over decades. Together with the devastation of lives, livelihoods and economic loss, displacement is one of the major outcomes of disasters. In the aftermath of a disaster, in most cases, part of the affected communities are forced to leave their homes and livelihoods to search for better living conditions. Among all the possible countries that could have been studied for their high propensity to suffer a disaster, the Philippines has been chosen to be the case study of this thesis. The Philippines is regarded to be one of the most disaster prone countries in the world, in fact every year the country suffers multiple sorts of disasters including storms, floods, earthquakes, drought, and volcanic eruptions that cause a large number of causalities and social setbacks. Because of the high number of hazards that hits the archipelago, very often, the inhabitants also suffers extensive displacement. This thesis sets out to investigate how displacement risk is constructed in the country and in particular this research is focused on giving an answer to “Why are the Filipinos still not able to adequately withstand natural disasters and avoid frequent displacements?”. As Socio-economic aspects and the incidence of poverty and inequalities are determinant factors that increase the chances of displacement, the reasons behind a high number of people forced to flee is to be understood in the context of political and economic systems. Therefore to give an answer to this question, the inquiry focuses on two of the possible factors that can increase the level of vulnerability of Filipinos, namely the free-market-oriented economic policies implemented following the Structural Adjustment Programs (SAPs) and the influence of the political landscape in the process of deforestation. The focus on deforestation, and in particular on illegal logging, is very relevant because of

the strong correlation between the misuse of the environment and disaster risk. By employing two theories, Neoliberalism and Elite Theory, it has been possible to analyze the consequences of the economic reforms and the power relations in the logging industry. It was found that SAPs, negatively affected the vulnerability of the poorest strata of the Philippine by reducing their resources and assets, and thus preventing them to build efficient coping capacities to withstand disasters. Moreover, the political landscape of the country has increased the vulnerability of the population by promoting deforestation and by implementing logging bans II without providing alternative livelihoods. All in all, a large part of the Philippine population is unable to adequately withstand natural disasters and avoid frequent displacement because SAPs and the political landscape contributed in several ways to reduce their coping capacities and to increase their vulnerability to hazards.

Zorat, P. I. (2015). DISASTER-RELATED DISPLACEMENT IN THE PHILIPPINES.

(96) Social Systems, Ecological Networks And Disasters: Toward a socio-political ecology of disasters

For residents of southern sections of the Miami metropolitan area, the morning of 24 August 1992 represented a dramatic discontinuity in the world as they knew it. At least temporarily, there were no daily routines or normal lives. Everyday activities, such as getting a drink of water, fixing a meal, or taking a bath, became daunting tasks. The information upon which the routines of daily life had been structured was virtually useless because the "stuff" needed to put it to use was no longer available – no electricity, water, gas, telephone, refrigerator, stove, bed, bathtub, room, or for many, even house. There were no stores or businesses to buy food, supplies, or services. Schools, hospitals, recreation facilities, entire neighborhoods, were virtually gone. The world was now full of uncertainty; new information was needed – where to find things, how to fix things, and how to get things done.

Peacock, W. G., & Ragsdale, A. K. (2012). Social systems, ecological networks and disasters: Toward a socio-political ecology of disasters. In Hurricane Andrew (pp. 20-35). Routledge.

(97) Natural hazards as political events: framing and politicisation of floods in the United Kingdom

This study explores how disasters are framed and politicised in the media to provide a systematic assessment of discursive dynamics and external political contexts of natural hazards. Utilising an actor-focused approach, it contributes with knowledge on how politicisation of disaster discourses unfolds. Two similar natural hazard events, the United

Kingdom floods of 2005 and 2015, are investigated by means of a content analysis and a political claims analysis. The study finds that a tension between the national government and its contestants following the 2015 floods led to a framing contest which was heavily affected by the external political context at the time. The opposition and journalists constructed a narrative of government failure, not least by intertwining the event with the politically tense situation in the United Kingdom to further populist claims about government spending and EU policies. In 2005, the lack of a comparable external context and polarisation between actors in the media prevented a politicisation of the floods in the immediate aftermath of the disaster. These results illustrate the importance of broader political contexts, even those essentially unrelated to the natural hazard, for the politicisation of a disaster.

Albrecht, F. (2022). Natural hazards as political events: framing and politicisation of floods in the United Kingdom. Environmental Hazards, 21(1), 17-35.

(98) Beyond "Socially Constructed" Disasters: Re-politicizing the Debate on Large Dams through a Political Ecology of Risk

Questions of dam safety and hazard potential most often do not take center-stage in contestations and articulations concerning large dams. Through a comparative study of two of Europe's most emblematic dam disasters – Vajont (Italy) and Ribadelago (Spain) – and the ongoing conflict over the safety of the Lower Subansiri Hydroelectric Project in Northeast India, this article argues that the damage caused by dam disasters is often not unavoidable or unforeseen but instead allowed to happen. Our cases show that power relations, economic pressures and profit influence "risky" dam management decisions, often disregarding the vernacular knowledge of concerned communities and silencing critical voices that do not fit dominant narratives of modernization and progress. We posit that an essential requirement for re-politicizing the question of dam safety is to unpack the apolitical notion of "socially constructed disasters," thinking instead about "capital-driven destructions." By emphasizing resistance against dam projects and against dominant risk discourses across space and time, this article seeks to underline the legitimacy of past and ongoing struggles surrounding the construction of large dams.

Huber, A., Gorostiza, S., Kotsila, P., Beltrán, M. J., & Armiero, M. (2017). Beyond "socially constructed" disasters: Re-politicizing the debate on large dams through a political ecology of risk. Capitalism Nature Socialism, 28(3), 48-68.

(99) Advancing "no natural disasters" with care: risks and strategies to address disasters as political phenomena in conflict zones

The understanding that disasters are not natural but rather shaped by a complex interplay of socio-political factors has existed for more than a century (Dynes, 2000; Kelman, 2020), and it has been at the core of disaster studies as a field since its beginnings (Bell, 1975; Blaikie et al., 1994; Dynes and Quarantelli, 1971; O'Keefe et al., 1976). As a result, and whilst the concept "natural disaster" is growingly recognised as a misnomer (Chmutina and von Meding, 2019), it is still the main term in the public discourse to refer to these phenomena.

Whilst the understanding that disasters are not natural has significant social benefits (revised in more detail below), and there has been a growing body of academic literature dedicated to this notion (Chmutina et al., 2017; e.g. Chmutina and von Meding, 2019; Kelman, 2020; Lizarralde, 2021), there is a lack of critical perspectives on the process of des-naturalised disasters (or in better words, of preventing the use of the misnomer by advancing the understanding that disasters are socio-political phenomena). The unreflected promotion that disasters are not natural can generate multiple unintended and even negative impacts, especially in violent conflict-affected settings, including the possibility of exacerbating social conflicts.

It is crucial to cover the impact of disasters and disaster-related processes in conflict-affected areas due to disasters recurring nature and high impact in those settings. Recent research indicates that up to 81% of countries affected by violent conflicts also face or have faced natural hazards and disasters during the conflict (Caso, Hilhorst, Mena and Papyrakis, 2023). Furthermore, over half of all disaster-related deaths in recent decades occurred in conflict-affected nations (Caso, Hilhorst and Mena, 2023).

In this article, drawing on insights and discussions from a decade of research on the interrelationship between disasters and conflict and a critical literature review, I analyse the significance of advancing with care the understanding that disasters are not natural, especially in areas affected by violent conflict. The approach included a literature review of publications directly addressing the idea that disasters are socio-political rather than purely natural events and a review of the literature concerning the disaster-conflict relationship. Subsequently, I examined my field notes documenting the process of advancing, explaining and disseminating the concept that disasters are not natural during research conducted in conflict-affected areas. This review identified challenges and associated risks for multiple actors. The analysis of the systematisation of these notes yielded essential elements for this reflective exercise, including the (de)politicisation and instrumentalisation of disasters, the potential role of humanitarian principles in the process and perspectives on conflict sensitivity and "do no harm." A literature review on these elements further informed the reflective exercise systematised in this article.

(100) Disasters in conflict areas: finding the politics

Despite some 50 years of research, relatively little is known about how disasters in conflict areas are created and discursively framed, and how information on them is publicly consumed. The emphasis in disaster studies has remained on establishing causal linkages, demonstrating the way in which natural hazard-related disasters result in deepening conflict, or ushering in peace. Furthermore, it has been accepted that disaster risk reduction is the state's responsibility. The strengths and limitations of these approaches are examined prior to a political reimagining of disasters in conflict areas. The absence of 'politics' from the wider debate on disasters in conflict areas is not just a benign oversight, but is in fact the politics of disasters in conflict areas. A politics that does not engage with the processes and outcomes of pursuing dominant agendas, such as neoliberal orthodoxy and state imperial control, in areas and communities vulnerable to natural hazards and political conflict needs to be recognised and challenged.

Siddiqi, A. (2018). Disasters in conflict areas: finding the politics. Disasters, 42, S161-S172.

Mena, R. (2023). Advancing "no natural disasters" with care: risks and strategies to address disasters as political phenomena in conflict zones. Disaster Prevention and Management: An International Journal, 32(6), 14-28.

(101) Governance matters: climate change, corruption, and livelihoods in Bangladesh
In world climate science, Bangladesh is considered a poster child of vulnerability. The primary stressors that affect the vulnerability of the Bangladeshi population include sea-level rise, biodiversity degradation, saltwater intrusion, desertification, social exclusion, unstable political conditions, and weak governance systems. Governance is an important non-climatic stressor that has not received sufficient attention. Within this framework, this paper explores the impacts of bribery and extortion on livelihoods and adaptive capacity in the face of climate change and argues that corruption significantly reduces the ability to respond to climatic stressors. Findings draw attention to this critical issue in climate change adaptation and international development in general, particularly for developing countries.

Rahman, M. A. (2018). Governance matters: climate change, corruption, and livelihoods in Bangladesh. Climatic change, 147(1), 313-326.

(102) Corruption Kills: Global Evidence From Natural Disasters

This study examines the relationship between corruption and natural disaster-related fatalities using panel data from 135 countries spanning the period 1980 to 2020. The empirical analysis reveals robust evidence that elevated levels of corruption in developing countries are significantly associated with disproportionately high fatality rates from natural disasters. These outcomes are indicative of systemic weaknesses in infrastructure, risk management, and emergency response mechanisms. By controlling for a comprehensive set of economic, demographic, healthcare, and institutional variables, the study offers a clearer understanding of how governance quality shapes disaster outcomes. Importantly, the analysis identifies nonlinear effects, wherein the adverse impact of corruption on disaster mortality intensifies at higher levels of corruption. These findings underscore the critical importance of institutional integrity and effective governance in mitigating the human toll of natural disasters, especially in low- and middle-income countries.

Cevik, S., & Jalles, J. T. (2025). Corruption kills: Global evidence from natural disasters. Scottish Journal of Political Economy, 72(5), e70024.

(103) The Impact of Corruption on Climate Change-Induced Vulnerability and Inequality

This chapter examines how corruption affects climate change-induced vulnerability and income inequality in Africa. Corruption diverts resources away from crucial environmental and resilience projects, undermining climate adaptation efforts, increasing communities' susceptibility to climate disasters, and exacerbating income inequality by benefiting the elite while marginalizing the poor, who are most vulnerable to climate impacts. Using a dynamic panel data framework, the chapter captures the nonlinear interactions of corruption on climate vulnerability and income disparity. The results indicate that corruption significantly heightens climate vulnerability and income inequality, emphasizing the necessity for robust governance and anti-corruption measures. This chapter provides empirical evidence on the adverse role of corruption in climate change adaptation and social equity, highlighting the importance of transparent institutions for sustainable development in Africa.

Kuttu, S., Soku, M. G., Amidu, M., & Addo, D. A. A. (2026). The Impact of Corruption on Climate Change-Induced Vulnerability and Inequality. In Climate Change in Africa: Financing and Management (pp. 621-658). Cham: Springer Nature Switzerland.

(104) Disasters and corruption: An Empirical Analysis of 16 countries from Asia and the Middle East

Disasters and disaster risks may induce corruption in a country, aggravating the number of deaths and damages while reducing a country's adaptive capacity. Within this framework, our study explores the impact of natural and man-made disasters on corruption. Utilising information from the Political Risk Services (PRS) group and other nationally representative sources, we construct a panel dataset to study the effect of disasters and disaster risk on corruption in 16 Asian and Middle Eastern countries from 1984 to 2018. We estimate the Panel OLS Model, as well as the 2SLS Model. For robustness checks, we run the Panel Ordered Logistic Regression (POLR) and Panel Ordered Probit Regression (POPR) models on the dataset. Our study finds that natural and man-made disasters and disaster risks increase the propensity for corruption in the sample countries. Other key findings of the study reveal that business activities, government stability, national income, and education reduce the likelihood of corruption in our sample countries. Lastly, our findings suggest a higher likelihood of corruption in developing countries. Results from our study support the notion that disasters generate a series of resource windfalls that governments and elites may misuse while increasing petty corruption through bribery and extortion.

Zafar, S., Rahman, I. U., & Ammara, S. (2023). Disasters and corruption: An empirical analysis of 16 countries from Asia and the Middle East. International journal of disaster risk reduction, 90, 103678.

(105) Alexander, D. E. (2017). Corruption and the governance of natural disaster risk. Oxford University Press.

This article considers how corruption affects the management of disaster mitigation, relief, and recovery. Corruption is a very serious and pervasive issue that affects all countries and many operations related to disasters, yet it has not been studied to the degree that it merits. This is because it is difficult to define, hard to measure and difficult to separate from other issues, such as excessive political influence and economic mismanagement. Not all corruption is illegal, and not all of that which is against the law is vigorously pursued by law enforcement. In essence, corruption subverts public resources for private gain, to the damage of the body politic and people at large. It is often associated with political violence and authoritarianism and is a highly exploitative phenomenon. Corruption knows no boundaries of social class or economic status. It tends to be greatest where there are strong juxtapositions of extreme wealth and poverty. Corruption is intimately bound up with the armaments trade. The relationship between arms supply and humanitarian assistance and support for democracy is complex and difficult to decipher. So is the relationship between disasters and organized crime. In both cases, disasters are seen as opportunities for corruption and potentially massive gains, achieved amid the fear, suffering, and disruption of the aftermath. In humanitarian emergencies, black markets can thrive, which, although they support people

by providing basic incomes, do nothing to reduce disaster risk. In counties in which the informal sector is very large, there are few, and perhaps insufficient, controls on corruption in business and economic affairs. Corruption is a major factor in weakening efforts to bring the problem of disasters under control. The solution is to reduce its impact by ensuring that transactions connected with disasters are transparent, ethically justifiable, and in line with what the affected population wants and needs. In this respect, the phenomenon is bound up with fundamental human rights. Denial or restriction of such rights can reduce a person's access to information and freedom to act in favor of disaster reduction. Corruption can exacerbate such situations. Yet disasters often reveal the effects of corruption, for example, in the collapse of buildings that were not built to established safety codes.

Alexander, D. E. (2017). Corruption and the governance of natural disaster risk. Oxford University Press.

(106) Impact of natural disaster on public sector corruption

This paper uses inter-country panel data from 1990 through 2010 to examine how the occurrence of natural disasters affects corruption within the public sector. For a closer analysis, disaster is classified into various categories, including general floods, other floods, tropical storms, other storms, earthquakes, volcanic eruptions, and landslides. Furthermore, this paper explores whether natural disasters have different impacts on corruption levels in developed and developing countries. The study reveals a number of novel findings. (1) Natural disasters that cause substantial damage increase public sector corruption in both developing and developed countries. (2) Natural disasters have a greater impact on public sector corruption in developed countries than in developing countries. (3) In developed countries, natural disaster frequency has a significant impact on the level of corruption. Hence, foreseeable disasters increase corruption in general. In developed countries, an incentive may exist to live within disaster-prone areas because of the potential for disaster compensation payments.

Yamamura, E. (2014). Impact of natural disaster on public sector corruption. Public Choice, 161(3), 385-405.

(107) Strategic Disaster Risk Management in Asia

This book presents strategies for managing disasters and reducing risks in Asian countries. Given the dynamic changes in the natural environment as well as the patterns of land use and management, the growing populations of the developing nations in Asia, migration patterns,

and other social-cultural aspects, the impacts of disasters have increased manifold in Asian countries.

Against this backdrop, the book examines disaster management issues such as disaster preparedness, post-disaster reconstruction, peace, development and corruption. The views of different groups of stakeholders are incorporated in the discussion to ensure a comprehensive analysis of and findings on the governance process, as well as best practices in pre- and post-disaster management. The book also includes chapters focusing on aspects often overlooked in the context of disaster management, such as the need to invest in public education to improve public awareness, and approaches to supporting the disabled, the vulnerable and the elderly from disaster risks. In closing, the book presents research on disaster management methods employed by different countries in the Asian region.

Saharan, V. (2015). Disaster management and corruption: Issues, interventions and strategies. In Strategic disaster risk management in Asia (pp. 193-206). New Delhi: Springer India.

(108) Public Sector Corruption and Natural Hazards

Public sector corruption has been shown to increase death rates and damages from natural disasters. We consider whether natural hazards can lead to rising levels of public sector corruption. This might seem unlikely since natural hazards are predetermined, naturally occurring events. However, when the distinction between hazards and disasters is considered, it becomes clear that corruption may well increase the likelihood that any new hazard will become a disaster, increasing the existing level of corruption within a given country. Based on standard estimation techniques, we find a statistically significant, positive relation between predetermined natural hazards and public sector corruption.

Escaleras, M., & Register, C. (2016). Public sector corruption and natural hazards. Public Finance Review, 44(6), 746-768.

(109) Corruption and disasters in the built environment: a literature review

This paper presents the findings of a review of academic literature concerning the degree to which corruption worsens disasters triggered by natural hazards in the built environment. The research employed a ‘systematic literature review’ methodology to analyse leading academic databases, resulting in a detailed analysis of 59 peer-reviewed, published papers. It found that while much of the literature focuses on earthquakes (relating to building and infrastructure collapse), the quality of governance, and the drivers of corruption, there is presently very

limited scholarship on the general scope, reach, and scale with respect to how disasters are worsened by corruption. It is notable that the Sendai Framework for Disaster Risk Reduction 2015–2030 and a number of other high-level reports fail to mention corruption. The paper argues that this serious gap in understanding and expressing how corruption increases vulnerability in the built environment within disaster studies perversely supports the furtherance of corruption in worsening disasters

Sanderson, D., Patel, S. S., Loosemore, M., Sharma, A., Gleason, K., & Patel, R. (2022). Corruption and disasters in the built environment: a literature review. Disasters, 46(4), 928-945.

(110) The unexpected outcomes: how does climate finance affect corruption in developing countries

Climate finance has become an essential tool for meeting the twin challenges of environmental sustainability and human development. However, in countries with weak governance structures, a large influx of external funds can distort incentives, encourage rent-seeking behaviour, and divert resources from planned projects. The purpose of this paper is therefore to conduct a preliminary analysis of the effect of climate finance on corruption in a panel of 130 developing countries over the period 2000 to 2021. The empirical evidence is based on the instrumental variable two-stage least squares (IV-2SLS) to account for endogeneity. Findings provide evidence that climate finance increases corruption in developing countries. This result is robust to additional controls, to the use of alternative corruption measures, and to alternative identification strategies. The heterogeneity analysis reveals that the effect of climate finance on corruption is not uniform across contexts. First, using a quantile regression approach, we find that climate finance has a less significant effect in countries with already high levels of corruption. In economic terms, the evidence suggests that the marginal impact of additional climate funds diminishes as corruption approaches a saturation point. Second, a regional disaggregation highlights substantial geographic variation: the effect of climate finance on corruption is strongest and statistically significant in Latin America and the Caribbean, moderate in Africa, and insignificant in Asia. Policy recommendations are formulated.
389, 126065.

(111) A Conceptual Governance Framework for Climate Change Adaptation and Disaster Risk Reduction Integration

Climate change adaptation (CCA) and disaster risk reduction (DRR) have similar targets and goals in relation to climate change and related risks. The integration of CCA in core DRR

operations is crucial to provide simultaneous benefits for social systems coping with challenges posed by climate extremes and climate change. Although state actors are generally responsible for governing a public issue such as CCA and DRR integration, the reform of top-down governing modes in neoliberal societies has enlarged the range of potential actors to include non state actors from economic and social communities. These new intervening actors require in-depth investigation. To achieve this goal, the article investigates the set of actors and their bridging arrangements that create and shape governance in CCA and DRR integration. The article conducts a comprehensive literature review in order to retrieve main actors and arrangements. The article summarizes actors and arrangements into a conceptual governance framework that can be used as a backdrop for future research on the topic. However, this framework has an explorative form, which must be refined according to site- and context-specific variables, norms, or networks. Accordingly, this article promotes an initial application of the framework to different contexts. Scholars may adopt the framework as a roadmap with which to corroborate the existence of a theoretical and empirical body of knowledge on governance of CCA and DRR integration.

Forino, G., von Meding, J., & Brewer, G. J. (2015). A conceptual governance framework for climate change adaptation and disaster risk reduction integration. International Journal of Disaster Risk Science, 6(4), 372-384.

(112) Climate change, natural disasters, and institutional integrity

We empirically investigate the effect of climate-induced natural disasters on the quality of institutions in 92 countries using data from 1984 to 2016. An instrumental variable approach is used to account for the reverse causality from the number of people affected by natural disasters to the quality of institutions. We then employ the Hausman Taylor approach to account for the bias in estimating panels with endogenous variables. Estimation results reveal a negative impact of natural disasters on the quality of national institutions. Furthermore, disasters negatively affect the quality of institutions in low-income, non-developed countries, whereas the effect is non-significant for high-income and developed nations. At the regional level, institutions in East Asia, Pacific, and South Asia, Sub-Saharan Africa, Eastern Europe and Central Asia, and Latin America and the Caribbean tend to deteriorate after natural disasters. In contrast, natural disasters in the Middle East and North Africa are often followed by improved institutions. Of the 12 components of the institutional integrity index, metrics mostly affected by disasters include government stability, internal conflicts, law and order, ethnic fragmentation, and democratic accountability. Our results suggest that disaster risk reduction policies and international assistance programs to help climate change adaptation in various economies need a combined “top-down” and “bottom-up” approach. Additionally,

institutional strengthening should be an integral component of disaster preparedness and adaptation efforts.

Khurana, R., Mugabe, D., & Etienne, X. L. (2022). Climate change, natural disasters, and institutional integrity. World Development, 157, 105931.

(113) Corruption : the hidden perpetrator of under-development and vulnerability to natural hazards and disasters : the Pat Reid lecture 2010

"Formerly Leverhulme Senior Research Fellow and co-founder of the Disaster Research Unit, University of Bradford (1973 - 77), James Lewis relocated to the University of Bath and became Visiting Fellow at the Centre for Development Studies. Datum International was created in1980 for consultancy to the United Nations (UNCTAD, UNEP, UNESCO, UN Habitat & WHO), the Commonwealth Secretariat / Commonwealth Fund for Technical Cooperation, and the European Commission. James Lewis is author of Development in Disaster-prone Places: Studies of Vulnerability (1999: IT / Practical Action, London) and of numerous chapters and papers on vulnerability to natural hazards, corruption, island vulnerability, climate change, and interconnections with socio-economic capacity, development and disaster risk reduction. As a commissioned author to Transparency International Berlin, he has contributed to Global Corruption Reports of 2005, Corruption in Construction, and 2011, Corruption and Climate Change. Trained and practiced as an architect and member of the Royal Institute of British Architects (RIBA), James Lewis has been involved in applications, implications, management and inspection of construction, destruction and reconstruction in Algeria, Bangladesh, the Caribbean, China (Hong Kong), the South Pacific, the United Kingdom and the USA. www.datum-international.eu "

Lewis, J. (2011). Corruption: the hidden perpetrator of under-development and vulnerability to natural hazards and disasters: the Pat Reid lecture 2010. Jàmbá: Journal of Disaster Risk Studies, 3(2), 464-475.

(114) Weathering Corruption

Could bad weather be responsible for U.S. corruption? Natural disasters create resource windfalls in the states they strike by triggering federally provided natural-disaster relief. By increasing the benefit of fraudulent appropriation and creating new opportunities for such theft, disaster-relief windfalls may also increase corruption. We investigate this hypothesis by exploring the effect of disaster relief provided by the Federal Emergency Management Agency (FEMA) on public corruption. The results support our hypothesis. Each additional $100 per capita in FEMA relief increases the average state’s corruption by nearly 102

percent. Our findings suggest notoriously corrupt regions of the United States, such as the Gulf Coast, are in part notoriously corrupt because natural disasters frequently strike them. They attract more disaster relief, which makes them more corrupt.

Leeson, P. T., & Sobel, R. S. (2008). Weathering corruption. The Journal of Law and Economics, 51(4), 667-681.

(115) Corruption and SDG 13: Climate Action

In this chapter, we set out the links between corruption and the capacity to take actions to combat climate change and mitigate its impacts. After noting that corruption is linked with increased greenhouse gas emissions and reduced stringency of environmental regulation, we focus on three themes based on the other goals and indicators of SDG 13. We begin by discussing the links between corruption and the damage from natural disasters. Corruption increases death and destruction from natural disasters and can also increase the likelihood of the occurrence of certain types of disasters. We also explore the ways in which corruption undermines planning for and responses to natural disasters, noting the dangers that arise when corrupt officials are tasked with preventing disasters and disbursing resources that are intended to aid in relief and reconstruction. We then discuss the effects that corruption has on the extent and effectiveness of climate financing, noting that while current targets have been met, the dramatic increase projected to be needed speaks to an urgent need to fight corruption in donor and recipient countries. Finally, we provide preliminary evidence on how corruption undermines a fundamental requirement for climate action-citizen understanding of climate change.

Aja-Eke, D., Gillanders, R., & Jha, C. K. (2025). Corruption and SDG 13: Climate Action. Available at SSRN 5086913.

Implementation of 2D and 3D in animation

In terms of the techniques used, a combination of 2D and 3D elements were used (2D animation and 3D background). This combination is said to showcase the artistic visions of two techniques without compromising the visual clarity of the other.

This is according to a thesis by Kristin C. Au named Animation: 2D Versus 3D and Their Combined Effect for Massachusetts Institute of Technology.

In the thesis, they highlight that; “Modern day animated films have been seen to mix the two forms for the sake of artistic effect, expedited production, and general convenience. Though some modern animations combine the two in the exploratory quest to discover new animation forms, few films directly compare the forms to visualize the differences in their perceived qualities. Noticeably, the two animation methods differ in level of detail, dimension, realism, and artistic expression.”

Chapter III

In this section, we’ll explain how our research directly helps us build "Sa Likod ng Baha." We want to show how every part of our study from the 2D/3D art style to the story of Tinoy—actually serves the bigger goal of our capstone. By connecting our research to the final film, we make sure that our work isn't just a school requirement, but a meaningful project that tells a real, high-quality story.

Research Description

For Sa Likod ng Baha, we chose to explore 2D animation by placing hand-drawn 2D characters into a 3D-modeled world. We experimented with 3D rain and shadows to make the environment feel heavy and overwhelming, making our characters look small and fragile by comparison. By telling Tinoy's story this way, we want the audience to truly feel his sadness and the weight of his betrayal through the atmosphere itself.

This blend of 2D and 3D is more than a visual choice—it’s a storytelling tool that reflects the contrast between the human spirit and the harsh, unforgiving realities around him. The delicate, hand-drawn characters symbolize vulnerability and resilience, while the imposing 3D environment represents the powerful, often indifferent forces that shape their lives.

What makes this approach especially meaningful is how it mirrors the lived experience of many who face overwhelming challenges. Just like Tinoy, real people often feel small against systems and circumstances beyond their control. Through this visual metaphor, we hope to

give a face and a voice to those quiet struggles, reminding viewers that behind every statistic or headline is a human story full of pain, hope, and courage.

We also focused heavily on mood and sound design, using dark tones, muted colors, and gritty audio textures to deepen the emotional impact. Every drop of rain, every shifting shadow, is designed to echo Tinoy's internal struggle and the systemic neglect he faces. This immersive atmosphere invites viewers not just to watch, but to experience the story on a visceral level.

At its core, Sa Likod ng Baha is about connection—between the audience and the characters, between art and reality, and between individual grief and a shared social conscience. We want the film to linger in the hearts of viewers, stirring empathy and reflection about the unseen battles many endure every day.

By grounding the animation in both artistic expression and emotional truth, Sa Likod ng Baha aims to connect with audiences in a way that lingers—prompting reflection on issues of loss, injustice, and the quiet strength of those who fight on despite it all.Description:

"Sa Likod ng Baha" is an emotional short film that explores the pain of betrayal during a disaster. Our study looks at how mixing hand-drawn 2D characters with 3D-modeled worlds can make a story feel more heavy and real. By using 3D rain and shadows, we show the contrast between "fragile" people and a cold, uncaring system. Ultimately, we want to show how animation can capture the true "human cost" of corruption through Tinoy's personal journey.

Research Methodology:
These paragraphs dive deeper into the sensory and emotional experience of the digital creation process, illustrating how technical steps like texturing, rendering, and sound design were humanized to match the raw atmosphere of the story. Continuation of Research Methodology (Sensory World-Building and Digital Craftsmanship)Beyond the initial layering of 2D and 3D assets, the methodology required a meticulous process of digital texturing and weathering to ensure the world felt lived-in and honest. In standard 3D animation, surfaces are often mathematically perfect, clean, and smooth. To strip away this artificial sheen, the creative team deliberately introduced human imperfections into the software. We gathered digital textures from real-world references—photographing water stains on concrete, rust eating away at galvanized iron roofs, and the thick, suffocating layers of mud left behind by actual floods. Manually mapping these textures onto our 3D models became an emotional exercise in world-building. It ensured that the environment of Sa Likod

ng Baha didn’t look like a computer-generated backdrop, but rather a visceral, bruised setting that carries the physical scars of continuous institutional neglect.

This painstaking attention to detail extended heavily into the lighting and color grading phases of the pipeline. The methodology moved away from vibrant, saturated digital palettes, opting instead for a restricted, atmospheric color scheme that mirrors the psychological weight of the characters. The team spent long nights fine-tuning the virtual volumetric fog and rain particles in the 3D space, aiming to recreate that specific, terrifying dimness that swallows a community just before a major typhoon makes landfall. This technical manipulation of light—contrasting the cold, clinical glow of the Mayor's office with the warm, fragile candle flicker inside Tinoy’s home—was handled with immense care. Every adjustment was a deliberate attempt to capture the subtext of the scene, making sure the technical environment actively participated in the storytelling by suffocating the viewer with tension.

Ultimately, this production methodology underscores the philosophy that behind every technical step—every line of code, polygon mesh, and keyframe—must be a heartbeat. Combining traditional sketching with 3D technology wasn't treated as a mere marriage of convenience, but as an act of artistic devotion. It bridged the gap between the digital canvas and the raw, unpolished realities of human struggle. By structuring our workflow to preserve the emotional weight of a hand-drawn stroke while leveraging the vast, claustrophobic depth of 3D spaces, this methodology successfully created a visual language that refuses to let the audience look away. It transforms a standard technical production pipeline into a deeply empathetic medium, ensuring that when the final frames of Sa Likod ng Baha fade to black, the audience is left not just admiring the technology, but deeply moved by the truth of the human experience it captured..

Qualitative Research:
Qualitative research played a crucial role in the development of Sa Likod ng Baha. Unlike quantitative approaches that focus on numerical measurements and statistical patterns, qualitative research seeks to understand human experiences, emotions, beliefs, and perspectives in greater depth. Since the central objective of the film is to portray the human consequences of flooding, corruption, and systemic neglect through the story of Tinoy, it was important for the researchers to gather insights that extended beyond factual information and into the lived realities of affected individuals.

Throughout the research process, participants shared personal stories, reflections, and memories related to disasters and community experiences. These narratives provided

valuable perspectives that helped shape the emotional and thematic direction of the project. More importantly, the interviews and discussions revealed the complex ways in which environmental disasters intersect with social, economic, and political issues. The experiences shared by participants highlighted that the effects of flooding cannot be understood solely through physical destruction; rather, they must be examined through the emotional, psychological, and social consequences that continue long after floodwaters have receded.

The qualitative approach enabled the researchers to listen closely to the voices of individuals who have directly experienced hardship and uncertainty. Through these interactions, the team gained a deeper appreciation for the resilience of communities as well as the frustrations and disappointments that often accompany repeated exposure to disaster. These findings became essential in informing both the narrative structure and visual design of Sa Likod ng Baha, ensuring that the story reflects authentic human experiences rather than generalized assumptions.

Understanding Lived Experiences

As we continued engaging with participants, it became increasingly evident that flooding was only one aspect of a much larger story. While participants described the immediate dangers posed by rising waters, damaged homes, and disrupted livelihoods, they frequently returned to broader concerns regarding governance, public accountability, and community support systems. Their experiences demonstrated that disasters are rarely isolated incidents. Instead, they are deeply connected to long-standing social conditions and institutional structures that shape how communities prepare for, respond to, and recover from crises.

Many participants expressed frustration over recurring problems that seemed to remain unresolved despite repeated appeals for assistance. Some described situations in which infrastructure improvements had been promised but never completed. Others recalled instances where aid distribution appeared uneven or delayed. These accounts revealed a growing sense of distrust toward institutions that were expected to provide protection and support during times of crisis.

The repeated nature of these experiences contributed to feelings of hopelessness among some participants. Rather than viewing disasters as extraordinary events, many had come to regard them as predictable parts of everyday life. This normalization of hardship was particularly striking because it reflected how prolonged exposure to adversity can influence perceptions of what is possible or expected. For some individuals, repeated disappointment had diminished confidence in the likelihood of meaningful change.

These insights significantly influenced the conceptual development of Tinoy's character. Rather than portraying him simply as a victim of flooding, the film presents him as someone navigating a broader system of challenges that extend beyond the immediate disaster. His

experiences reflect not only physical vulnerability but also emotional struggles related to trust,

disappointment, and resilience.

The Emotional Dimensions of Memory

One of the most valuable aspects of qualitative research is its ability to reveal emotional realities that may not be captured through surveys or statistical analysis. During interviews, participants often shared memories that carried deep emotional significance. In many cases, these memories were accompanied by subtle expressions that communicated feelings more powerfully than words alone.

Researchers observed moments when participants paused before answering questions about particularly painful experiences. These pauses often suggested ongoing emotional processing, indicating that the memories remained significant even years after the events had occurred. Similarly, changes in tone, volume, and pacing frequently provided insight into the emotional weight of specific recollections.

For example, some participants spoke calmly when describing property damage but became noticeably emotional when discussing family members, neighbors, or community relationships affected by disaster. This distinction highlighted the importance of human connection in shaping experiences of loss and recovery. Material possessions could often be replaced, but damaged relationships, lost opportunities, and emotional trauma left lasting impressions.

These observations reinforced the importance of portraying emotions authentically within the film. Rather than relying on exaggerated dramatic moments, the narrative seeks to capture quieter emotional experiences that resonate with real-life struggles. Moments of silence, hesitation, reflection, and uncertainty became important storytelling elements inspired directly by participant experiences.

Trust and Vulnerability in the Research Process

The success of qualitative research depends heavily on the development of trust between researchers and participants. Throughout the data collection process, efforts were made to create a respectful and supportive environment in which individuals felt comfortable sharing personal experiences.

Building trust required patience, empathy, and active listening. Participants were not merely sources of information; they were individuals sharing deeply personal stories that often involved pain, frustration, and vulnerability. Recognizing this, researchers approached each conversation with sensitivity and genuine interest.

As trust developed, participants frequently shared more detailed accounts of their experiences. These conversations often extended beyond initial questions and evolved into broader discussions about community life, family relationships, and personal aspirations. Such moments provided valuable context that enriched understanding of the challenges faced by affected communities.

Importantly, trust also created opportunities for participants to express perspectives that might otherwise remain unheard. Many individuals appreciated having a space where their experiences were acknowledged and valued. This mutual exchange underscored the collaborative nature of qualitative research and highlighted the importance of ethical engagement throughout the project.

Recognizing Patterns Across Individual Stories

Although each participant's experience was unique, recurring themes began to emerge across multiple conversations. These patterns provided insight into shared challenges and concerns within affected communities.

One prominent theme involved uncertainty about the future. Participants frequently expressed anxiety regarding future disasters and their ability to protect their families. This uncertainty extended beyond environmental concerns and included worries about employment, education, housing, and financial stability.

Another recurring theme involved resilience. Despite significant hardships, many participants demonstrated remarkable determination and adaptability. Stories of neighbors helping one another, families rebuilding homes, and communities organizing relief efforts illustrated the strength of collective action during difficult times.

At the same time, resilience was often accompanied by exhaustion. Participants described the emotional and physical toll of repeatedly recovering from disasters without adequate support. This duality—strength alongside fatigue—became an important thematic element within the film.

Translating Research into Storytelling

A primary objective of the research was to ensure that the narrative of Sa Likod ng Baha reflected authentic experiences. Rather than directly reproducing specific participant stories, the project synthesized common themes and emotional truths into a fictional narrative. Tinoy's journey serves as a representation of broader social realities identified through qualitative research. His challenges reflect experiences shared by many participants, including feelings of vulnerability, disappointment, hope, and perseverance.

The storytelling process involved careful consideration of how emotional experiences could be translated into visual and narrative form. Researchers worked to identify moments that captured universal aspects of human experience while remaining grounded in the specific social context explored during the study.

Empathy as a Research Outcome

One of the most significant outcomes of the qualitative research process was the development of empathy. Through direct engagement with participants, researchers gained a deeper understanding of perspectives that might otherwise remain abstract or distant.

Empathy emerged not only as a personal outcome for researchers but also as a central objective of the film itself. By presenting experiences through Tinoy's perspective, the project seeks to encourage audiences to connect emotionally with issues that are often discussed primarily through statistics and policy debates.

The film aims to remind viewers that behind every disaster-related headline are individuals, families, and communities whose lives have been profoundly affected. By emphasizing personal stories and emotional experiences, the narrative encourages reflection on the human dimensions of social and environmental challenges.

Ethical Responsibility in Representation

Representing real experiences through fictional storytelling carries significant ethical responsibilities. Researchers recognized the importance of avoiding sensationalism, stereotyping, or oversimplification.

The goal was not to exploit suffering for emotional impact but to create a respectful portrayal that honors the experiences shared by participants. This required ongoing reflection regarding narrative choices, character development, and visual representation.

Authenticity became a guiding principle throughout production. Decisions regarding dialogue, character behavior, environmental design, and emotional tone were informed by research findings and grounded in participant experiences whenever possible.

Reflection and Conclusion

The qualitative research process became much more than a method for gathering information. It evolved into an opportunity for meaningful engagement, learning, and reflection. Through conversations with participants, researchers gained valuable insights into the emotional realities of disaster experiences and the broader social conditions that shape them.

These lived experiences became the emotional foundation of Sa Likod ng Baha. They informed the development of characters, themes, environments, and narrative structure while reinforcing the project's commitment to authenticity and social relevance.
Ultimately, the research demonstrated that storytelling has the potential to bridge the gap between individual experiences and broader public understanding. By listening carefully to participants and translating their insights into animation, the project seeks to create a film that not only tells a compelling story but also encourages empathy, reflection, and awareness regarding the human consequences of disaster, corruption, and systemic neglect.
This qualitative foundation ensures that Tinoy's journey is not merely fictional—it is rooted in the emotional truths, resilience, and humanity shared by the people whose voices helped shape the story.

Quantitative Research:
Moreover, the survey data goes beyond individual habits to map the invisible community infrastructure created by these enthusiasts. When we quantify variables such as membership in retro gaming online forums, attendance at local arcade conventions, or participation in global speedrunning leaderboards, the statistics reveal a profound social truth: retro gaming is a deeply communal anchor. In a modern era where online multiplayer games can often feel toxic or overwhelming, the data demonstrates that retro spaces offer a sanctuary of mutual respect and shared preservation. By translating these community touchpoints into percentages and frequency charts, the research mathematically captures the warmth of a subculture that actively welcomes newcomers, preserves dying hardware through crowdsourced knowledge, and celebrates the collective triumph of conquering decades-old software bugs. The resulting charts do not just display data; they prove that the shared love for a vintage aesthetic creates a resilient social fabric capable of bridging geographical and cultural divides.

This quantitative approach also uncovers the unique cognitive relationship players have with the limitations of early technology. When the questionnaires measure factors like preferred display outputs (such as replicating scanlines on modern monitors) or the specific choice to play on original hardware versus emulation, the data points reveal an intentional rejection of modern, over-stimulating entertainment. The metrics capture a deliberate human preference for the tactile feedback of a physical button click, the warmth of a cathode-ray tube screen, and the rewarding difficulty of games that didn't hand-hold the player. By tracking these aesthetic choices numerically, the study provides a fascinating look into human psychology—proving that the modern mind actively craves the focused, uncluttered challenge of retro

design. It shows that the data is not just tracking a consumer trend, but rather a collective yearning for an era of gaming where imagination filled the gaps left by the pixels.

Ultimately, integrating these deeply humanized metrics ensures that the quantitative chapter of this study does not feel like a sterile corporate market report. Instead, it elevates the numbers to serve as an empathetic mirror for the community itself. By demonstrating the undeniable scale of this subculture—not just in the money spent, but in the passion invested, the generations connected, and the mental well-being preserved—this design gives the research true cultural weight. It concludes that every percentage point yielded by our surveys is rooted in an authentic human smile, a nostalgic sigh, and an unyielding declaration that these classic digital worlds are an enduring, invaluable treasure in our shared technological history.

Data Analysis:

The collected data undergoes rigorous analysis, integrating both qualitative coding and statistical techniques. Qualitative data is systematically coded to identify recurring themes and patterns, while quantitative data is subjected to statistical analysis to derive meaningful insights and trends. The analysis ensures that the interpretation of findings is grounded in both qualitative depth and quantitative breadth. Beyond just numbers and codes, this process is about uncovering the stories and emotions behind the data. By carefully examining the words and experiences shared by participants, we gain a deeper understanding of their realities—their struggles, hopes, and frustrations. These insights bring richness and context to the statistical trends, making the findings more human and relatable. This careful blending of methods allows us to honor the voices of those who shared their stories, ensuring that their experiences are faithfully represented and not lost in abstraction. It reminds us that behind every data point is a person with a story, feelings, and a unique perspective. The combined analysis also allows us to cross-validate results, ensuring that what we observe in the data reflects genuine experiences and perceptions. This balance between depth and scale strengthens the reliability of our conclusions and helps us craft a narrative that is both impactful and truthful. Ultimately, this thorough and thoughtful approach to data analysis transforms raw information into a meaningful story—one that fosters empathy and understanding. It ensures that the final work we create not only stands on solid evidence but also touches the hearts of viewers, honoring the lived experiences and resilience of the people at the center of the project.

Ethical Considerations:

Ethical considerations serve as the foundation of this research project. More than a set of academic requirements, ethics represents our commitment to treating every participant with dignity, fairness, and respect throughout the entire research process. Because this study explores experiences, perspectives, and personal realities that may involve sensitive topics, maintaining high ethical standards is essential to ensuring the credibility, integrity, and humanity of the research.

At the heart of this study is the recognition that every participant is a person first and a source of information second. Behind every response, interview, and shared experience is an individual with unique emotions, beliefs, values, and life circumstances. For this reason, ethical principles guide every stage of the research process—from planning and data collection to analysis, interpretation, and dissemination of findings. These principles help ensure that participants feel safe, respected, and empowered while contributing to the study.

One of the most important ethical principles observed in this research is informed consent. Before participating, individuals are provided with clear and comprehensive information regarding the purpose of the study, its objectives, procedures, expected duration, potential benefits, and any possible risks associated with participation. The information is communicated using language that is simple, understandable, and free from technical jargon to ensure that participants can make informed decisions about their involvement.

Participation in the study is entirely voluntary. No individual is pressured, manipulated, or coerced into taking part. Participants are given sufficient time to review the information presented to them and are encouraged to ask questions if any aspect of the study is unclear. Only after they fully understand the nature of the research and agree to participate are they asked to provide their consent. This process ensures that participation is based on genuine willingness rather than obligation.

Respect for participant autonomy is another key ethical consideration. Every individual has the right to decide whether to participate in the study and to determine the extent of their involvement. Participants are informed that they may decline to answer any question that makes them uncomfortable and may withdraw from the study at any time without facing penalties, negative consequences, or the need to provide an explanation. This respect for personal choice acknowledges the participant's right to control their own experiences and information.

Privacy and confidentiality are also central components of this research. Participants entrust researchers with personal information, experiences, and perspectives that may be sensitive in nature. Protecting this information is therefore a critical responsibility. To uphold

confidentiality, all data collected during the research process are handled with strict care and professionalism.

Personal identifiers such as names, addresses, contact details, and other identifying information are removed or replaced with codes whenever possible. This anonymization process helps ensure that individual participants cannot be identified in research reports, presentations, or publications. Any quotations used in the study are carefully reviewed to prevent the disclosure of information that could reveal a participant's identity.

In addition to anonymization, secure data management practices are implemented throughout the project. Digital files are stored using password-protected systems, while physical documents are kept in secure locations accessible only to authorized researchers. Data are handled responsibly and retained only for the period necessary to fulfill the objectives of the study and comply with academic requirements. These measures help minimize the risk of unauthorized access, misuse, or disclosure of sensitive information.

The principle of beneficence also guides this research. Beneficence refers to the responsibility of researchers to maximize potential benefits while minimizing possible harm. Every effort is made to design research activities that contribute meaningful knowledge while reducing physical, psychological, emotional, social, or professional risks to participants. Because some discussions may involve personal experiences or difficult memories, researchers remain attentive to participants' emotional well-being throughout data collection. Questions are carefully designed to avoid unnecessary distress while still allowing participants to share valuable insights relevant to the study. Researchers maintain a respectful and compassionate approach during interactions, recognizing that certain topics may evoke strong emotions or uncomfortable reflections.

If a participant experiences discomfort during an interview or survey, appropriate measures are taken to support their well-being. Participants are reminded that they may pause the session, skip questions, take breaks, or discontinue participation altogether. This flexibility helps ensure that individuals remain in control of their level of involvement and are not placed in situations that compromise their emotional health.

The principle of nonmaleficence, often summarized as "do no harm," further reinforces the commitment to participant protection. Researchers continuously evaluate research procedures to identify and reduce potential risks. This includes considering how questions are phrased, how data are collected, and how findings are presented. Ethical research requires not only avoiding intentional harm but also anticipating unintended consequences that could negatively affect participants.

Justice is another important ethical principle observed in this study. Justice involves ensuring fairness in participant selection and treatment. Individuals are invited to participate based on criteria relevant to the research objectives rather than personal bias, favoritism, or discrimination. All participants are treated equally and respectfully regardless of age, gender,

socioeconomic status, educational background, ethnicity, religion, or other personal characteristics.

Fair treatment also extends to how participants' contributions are valued. Every response is regarded as important and meaningful. Researchers avoid privileging certain voices while disregarding others. By ensuring that diverse perspectives are acknowledged and represented, the study seeks to produce findings that accurately reflect the experiences of the population being examined.

Transparency plays a crucial role in maintaining ethical integrity. Researchers strive to communicate honestly with participants regarding the purpose, methods, and intended use of the research. Misleading information, deception, and hidden agendas are avoided. Participants have the right to know how their contributions will be used and how the findings may be shared with academic, professional, or public audiences.

Ethical responsibility also extends beyond data collection into the interpretation and reporting of results. Researchers have a duty to represent participant experiences accurately and respectfully. Findings are reported truthfully without fabrication, falsification, manipulation, or selective omission of data. The integrity of research depends on presenting results honestly, even when findings differ from initial expectations or hypotheses.

In analyzing data, researchers remain mindful of personal biases that could influence interpretation. Reflexivity—the process of critically examining one's own assumptions, beliefs, and perspectives—is practiced to promote objectivity and fairness. By acknowledging potential biases, researchers can better ensure that participant voices remain central to the analysis rather than being overshadowed by preconceived notions.

The ethical use of quotations and narratives is another important consideration. Personal stories often carry emotional significance and deserve careful handling. When participant statements are included in research reports, they are presented respectfully and within their proper context. Researchers avoid sensationalizing experiences or using narratives in ways that distort meaning or exploit vulnerability.

Cultural sensitivity is equally important in ethical research. Participants may come from diverse cultural, social, and linguistic backgrounds. Researchers strive to understand and respect these differences throughout the study. Communication methods, interview approaches, and interpretations are adapted when necessary to ensure that participants feel understood and valued. Respect for cultural diversity strengthens both the ethical quality and validity of the research.

Trust forms the foundation of any meaningful research relationship. Participants choose to share their experiences because they believe that researchers will treat their information responsibly and respectfully. This trust is not something that can be assumed; it must be earned through honesty, professionalism, empathy, and consistent ethical conduct. Every interaction with participants provides an opportunity to demonstrate this commitment.

The researcher-participant relationship is characterized by mutual respect. Researchers recognize that participants are not merely subjects of study but active contributors whose insights make the research possible. Their willingness to share experiences represents a valuable contribution to knowledge and understanding. Acknowledging this contribution helps foster a collaborative and respectful research environment.

Ethical considerations are particularly important when dealing with potentially vulnerable populations. Extra precautions are taken to ensure that individuals who may face additional risks receive appropriate protection. Researchers remain sensitive to power dynamics that could influence participation and take steps to ensure that consent remains genuinely voluntary.

Another essential aspect of ethical research involves accountability. Researchers are responsible not only to participants but also to academic institutions, professional communities, and society as a whole. This responsibility requires adherence to established ethical guidelines, research standards, and institutional policies. Accountability helps maintain public confidence in research and supports the production of trustworthy knowledge.

The dissemination of research findings is approached with equal care and responsibility. Results are shared in ways that protect participant confidentiality and preserve the integrity of their experiences. Researchers avoid presenting findings in a manner that could stigmatize, stereotype, or harm individuals or communities represented in the study. Instead, efforts are made to highlight insights constructively and respectfully.

Ethics also involves recognizing the broader social impact of research. Research findings have the potential to influence policies, practices, attitudes, and public understanding. Therefore, researchers consider how their work may affect individuals, communities, and stakeholders beyond the immediate study population. Responsible dissemination helps ensure that findings contribute positively to knowledge and social awareness.

Throughout the research process, empathy remains a guiding value. Ethical research is not simply about following rules or completing procedural requirements. It is about recognizing the human stories behind the data and responding to them with compassion, respect, and integrity. Participants entrust researchers with pieces of their lives, and honoring that trust requires continual ethical reflection and responsibility.

This commitment to ethics shapes every decision made within the study. From obtaining informed consent and protecting confidentiality to ensuring participant well-being and reporting findings honestly, ethical principles remain embedded in every aspect of the research process. These practices help safeguard participants while strengthening the credibility and validity of the research outcomes.

Ultimately, ethical considerations are not merely a section of the research methodology but a core value that informs the entire project. They reflect a commitment to respecting human

dignity, protecting participant welfare, promoting fairness, and advancing knowledge responsibly. By upholding these principles, the study seeks to create a research environment founded on trust, integrity, compassion, and respect.

Through this unwavering dedication to ethical practice, the research not only meets academic and professional standards but also honors the individuals whose voices, experiences, and perspectives make the study possible. Their contributions are treated with the care and respect they deserve, ensuring that the research remains both academically rigorous and deeply human-centered.

Conclusion:

Research is often evaluated through figures, percentages, statistical analyses, and measurable outcomes. These tools are undeniably important because they provide structure, evidence, and credibility to academic work. However, some realities cannot be fully understood through numbers alone. Some truths exist within memories, emotions, lived experiences, and stories that statistics are unable to capture. This study began with a simple yet important question: What truly lies behind the floodwaters that repeatedly return to vulnerable communities? At first glance, the answer seemed straightforward. Flooding is commonly associated with heavy rainfall, overflowing rivers, inadequate drainage systems, and environmental conditions. Yet as the voices of community members emerged throughout this research, it became increasingly clear that the floodwaters carry something much deeper than rain. Beneath every rising river and submerged street lies a story of people whose lives have been disrupted, whose trust has been challenged, and whose resilience has been tested repeatedly over time.

This study set out to explore the human realities surrounding recurring flood disasters and to understand how these experiences are shaped not only by environmental factors but also by systemic corruption, institutional neglect, and failures of governance. Through qualitative methods such as interviews and focus group discussions, participants were given the opportunity to share their experiences in their own words. Rather than speaking through statistics or generalized assumptions, the community spoke directly. Their voices revealed a reality that is often overlooked in official reports and public discussions. What emerged from this process was not merely a collection of observations about flooding but a powerful narrative about survival, injustice, disappointment, and hope.

The findings demonstrate that floods are not experienced solely as natural disasters. For many participants, flooding has become a recurring cycle that reflects broader social and institutional failures. While heavy rainfall may trigger the immediate event, the severity of its consequences is frequently linked to human decisions, policy shortcomings, inadequate infrastructure, and the unequal distribution of resources. Communities repeatedly exposed to flood disasters often find themselves trapped in a cycle where recovery remains incomplete

before the next disaster arrives. This cycle creates not only physical destruction but also emotional exhaustion and long-term socio-economic hardship.

One of the most significant insights from this study is the understanding that disaster vulnerability is not evenly distributed. Certain communities experience greater exposure to risk because of historical inequalities, economic limitations, and insufficient institutional support. Participants described how recurring floods affect every aspect of daily life, from housing and employment to education and family stability. Homes are damaged or destroyed, livelihoods are interrupted, possessions accumulated over years of hard work are lost within hours, and future opportunities become increasingly difficult to pursue. These impacts accumulate over time, creating a burden that extends far beyond the duration of the flood itself.

The stories shared during interviews revealed that many residents no longer view flooding as an isolated emergency. Instead, it has become an expected reality woven into the rhythm of everyday life. Participants spoke of preparing for the next flood even while recovering from the previous one. This constant anticipation creates a state of prolonged uncertainty and stress. Families develop coping mechanisms, emergency routines, and survival strategies not because they feel secure, but because repeated exposure has taught them that assistance may not arrive when it is needed most. Such conditions reveal the profound psychological impact of recurring disasters.

The emotional dimension of flood experiences emerged as a central theme throughout the study. Participants frequently described feelings of fear, frustration, anxiety, helplessness, and exhaustion. These emotions were not limited to the moments when floodwaters entered their homes. Instead, they persisted long after the water receded. The anticipation of future floods often generated chronic stress, particularly among parents concerned about the safety and well-being of their children. Many participants expressed feelings of uncertainty regarding their future, questioning whether meaningful improvements would ever occur.

Equally significant was the theme of trust. Trust is essential for effective disaster management because communities rely on institutions to provide protection, support, and leadership during times of crisis. However, many participants described a growing sense of disillusionment with public institutions. Repeated experiences of unmet promises, delayed responses, inadequate interventions, and perceived corruption contributed to a weakening of public confidence. Participants frequently expressed frustration with systems that appeared disconnected from the realities faced by ordinary citizens.

The issue of systemic corruption emerged as one of the most powerful and recurring concerns within community narratives. Participants described situations in which resources intended for public welfare appeared to be mismanaged, delayed, or distributed inequitably. Whether these perceptions reflected documented incidents or broader frustrations, they significantly influenced how communities understood their experiences. Corruption was not viewed

merely as an administrative problem. Instead, it was seen as a factor that directly contributed to suffering by limiting the effectiveness of disaster prevention, preparedness, and recovery efforts.

These accounts reveal an important truth: disasters are rarely natural in their consequences. While natural hazards may be unavoidable, the extent of their impact is often shaped by human systems and decisions. When governance structures fail to function effectively, communities become more vulnerable. Infrastructure projects remain incomplete, risk reduction measures are insufficiently implemented, and emergency responses may fail to meet actual needs. As a result, environmental events that could have been mitigated become sources of prolonged hardship.

Another important finding concerns the concept of disaster fatigue. Public narratives often celebrate affected communities as resilient and resilient populations certainly deserve recognition. However, the participants in this study challenged simplistic narratives that portray resilience as an endless resource. While individuals demonstrated remarkable strength and determination, many also expressed exhaustion. They spoke of being tired of rebuilding, tired of waiting for solutions, and tired of hearing promises that failed to translate into meaningful change.

This distinction is important because resilience should not be used to excuse institutional shortcomings. Communities should not be expected to bear the burden of adaptation indefinitely while systemic issues remain unresolved. The narratives gathered during this research reveal that resilience and vulnerability can coexist. Individuals may continue to persevere despite experiencing profound hardship, but perseverance should not be mistaken for the absence of suffering.

The study also highlighted the importance of collective memory. Flood experiences are not forgotten when recovery efforts conclude. They become part of community identity, shaping how individuals understand their environment, their institutions, and their future. Stories of previous disasters are passed between generations, creating shared memories that influence perceptions of risk and preparedness. These memories serve both as warnings and as reminders of unresolved challenges.

At the same time, community narratives demonstrated remarkable solidarity. Participants repeatedly described acts of mutual support during and after disasters. Neighbors helped one another evacuate, shared food and supplies, provided temporary shelter, and offered emotional encouragement. In many cases, these informal support networks became essential sources of assistance when formal systems were perceived as inadequate. Such examples illustrate the strength of community bonds and the importance of social cohesion during times of crisis.

The findings of this study carry significant implications for disaster management and public policy. Effective disaster response requires more than technical solutions. While

infrastructure improvements, flood control systems, and emergency preparedness programs are essential, they must be accompanied by transparent governance, meaningful community participation, and accountability mechanisms. Policies designed without genuine engagement with affected communities risk overlooking the realities experienced by those most vulnerable to disaster impacts.

Community members possess valuable knowledge derived from lived experience. They understand local risks, historical patterns, and practical challenges that may not be visible within official reports. Therefore, their voices should play a central role in decision-making processes. Including community perspectives in planning and implementation can help create solutions that are both effective and responsive to local needs.

The creation of Sa Likod ng Baha represents an effort to transform these research findings into a form that can reach broader audiences. While academic research provides valuable insights, its impact can be limited if findings remain confined to written reports. Film offers a powerful medium for communicating human experiences because it engages audiences emotionally as well as intellectually. Through storytelling, visual imagery, and authentic narratives, the film seeks to bridge the gap between research and public understanding. The purpose of Sa Likod ng Baha extends beyond documenting flood experiences. It aims to reveal the deeper realities that exist beneath public narratives and official statistics. The title itself reflects the central message of the study: behind every flood lies a complex story involving people, institutions, decisions, and consequences. The film invites audiences to look beyond the immediate disaster and consider the broader systems that shape vulnerability and resilience. Importantly, the research process demonstrated that storytelling can serve as a form of advocacy. When individuals share their experiences, they challenge silence and invisibility. Their narratives create opportunities for empathy, understanding, and social reflection. Stories have the power to humanize issues that might otherwise be reduced to abstract discussions or policy debates. Throughout this study, participants entrusted researchers with deeply personal experiences. Many spoke about painful memories, difficult losses, and ongoing frustrations. Their willingness to share these stories reflects courage and generosity. As researchers and storytellers, we carry a responsibility to honor that trust by representing their experiences accurately, respectfully, and ethically. This responsibility extends to how findings are interpreted and presented. The goal is not to exploit suffering for emotional impact but to create greater awareness of the realities faced by affected communities. Ethical storytelling requires balancing honesty with respect, ensuring that participants are portrayed as complete individuals rather than symbols of tragedy.

One of the most powerful lessons learned through this research is that listening is itself an important act. In many cases, participants expressed appreciation simply for having an opportunity to share their experiences. Their comments suggest that affected communities often feel overlooked or unheard within broader public discussions. Creating spaces where

people can speak openly about their experiences can foster recognition, validation, and empowerment.

The study also underscores the interconnected nature of environmental, social, economic, and political issues. Flood disasters cannot be understood solely through meteorological explanations. They are shaped by patterns of development, governance decisions, resource allocation, social inequalities, and historical circumstances. Addressing flood vulnerability therefore requires comprehensive approaches that recognize these interconnected dimensions.

As this project concludes, it becomes clear that the central question guiding the research has led to a deeper understanding of both disaster and humanity. Behind the floodwaters are families striving to protect their loved ones. Behind damaged homes are years of effort, memories, and dreams. Behind statistics are individuals whose lives cannot be fully represented through numerical data. Behind public policies are decisions that carry real consequences for real people.

Most importantly, behind every story shared during this research is a voice asking to be heard. These voices challenge us to reconsider how disasters are understood and addressed. They remind us that effective solutions require more than technical expertise. They require empathy, accountability, transparency, and a genuine commitment to justice.

Ultimately, this study affirms that research is more than the collection and analysis of information. It is an act of witnessing. It involves listening carefully, documenting responsibly, and presenting findings in ways that honor the experiences of those who contribute to knowledge creation. The purpose of this research was not simply to identify problems but to illuminate realities that might otherwise remain unseen. The conclusion of this study is therefore not an ending but an invitation. It invites policymakers to listen more closely to affected communities. It invites institutions to strengthen accountability and transparency. It invites citizens to engage critically with the systems that shape disaster vulnerability. And it invites audiences of Sa Likod ng Baha to look beyond the water itself and consider what lies beneath.

In the end, the floodwaters recede, but the stories remain. The voices gathered through this research continue to speak long after interviews have concluded and data analysis has ended. Their stories remind us that every disaster has a human face, every statistic represents a life, and every call for change deserves attention. By listening to these voices and carrying their stories forward, this project seeks not only to document reality but also to contribute to a future where communities are safer, more empowered, and more fully heard.

This is the enduring message of Sa Likod ng Baha: that understanding begins with listening, accountability begins with truth, and meaningful change begins when the experiences of ordinary people are no longer ignored. Their voices are the heart of this research, and their stories remain its most important finding.

Final Summary

This research has shown us that behind every flood, there is a story. Behind every statistic, there is a person. Behind every problem, there is a cause. Sa Likod ng Baha is the result of this understanding. It is built entirely on the voices of those who live with this reality every day, and it carries their message with honesty, respect, and responsibility.

We have learned that the true cost of these disasters is not measured in pesos or lost property—it is measured in broken trust, lost opportunities, and the quiet erosion of hope. We have learned that resilience without justice is only a burden, and that dignity means having the right to live without fear. By looking closely at what lies behind the waters, we do not just tell a story—we give a voice to those who have been silenced, and we remind everyone that safety, security, and fair governance are not privileges reserved for a few, but rights that belong to all.

In the end, this work confirms that research and creativity are powerful partners. Together, they can bring truth into the light, turn suffering into awareness, and make sure that what lies behind the flood is no longer hidden, but finally seen.

Research Design

Every study begins with a question, but the way you answer that question determines what you will find. If you look only for numbers, you will find only numbers. If you look only for technical explanations, you will find only technical answers. But if you look for the truth of what people experience, you must be willing to meet them where they are, and to see the world through their eyes. For this research, our goal was not just to understand why flooding happens, but to understand what it does—to lives, to homes, to trust, and to hope. This required a research design that is as much about presence as it is about procedure.

Research design is often presented as a set of rules and steps: define the population, select the sample, choose the instruments, set the schedule, and analyze the results. These steps are important, but if followed without deeper purpose, they can create a study that is technically correct but emotionally empty. For Sa Likod ng Baha, the design was built to be both rigorous and respectful, systematic and sensitive. It was planned to capture not just information, but reality in its full form—including the sights, sounds, smells, feelings, and memories that make up daily life in flood-prone communities.

This chapter explains how this design was developed and implemented. It describes the choice of location, the approach to data collection, the integration of environmental observation, and the direct link between research findings and the creative process. It shows that this design is more than a method—it is a commitment to honesty, a way of bearing witness, and a tool for ensuring that the voices of the community are heard clearly and without distortion.

To truly capture this phenomenon, the research design demanded that the fieldwork be conducted not in sterilized, artificial meeting rooms, but directly within the geography of the tragedy itself. There is a common belief that research should be done in neutral spaces so that the environment does not influence the responses. But in this case, the environment is not an influence—it is the subject. The flooding, the infrastructure, the layout of the community, and the weather patterns are all inseparable from the experiences of the people living there. To separate them would be to remove the very context that gives meaning to their words.

When we first began planning this study, we considered holding interviews in barangay halls, community centers, or schools—places that are clean, organized, and easy to access. But as we discussed it further, we realized that this would create a barrier. If we invited people to come to our space, we would be asking them to leave their reality behind. We would hear their stories, but we would not understand the setting in which those stories take place. We would learn that the water rises, but we would not see how it moves, where it comes from, or what it leaves behind. We would hear about broken promises, but we would not see the physical evidence of those failures standing right in front of us.

Therefore, the decision was made to conduct all interviews and observations on-site. This meant traveling to the communities, walking through the streets, and sitting in the very places where families live, work, and raise their children. It meant accepting the conditions of those areas—heat, humidity, dust, uneven ground, and sometimes the smell of stagnant water—as part of the work, not as obstacles to be avoided. It meant being present, not as visitors or outsiders, but as people willing to understand what life is really like there.

This meant conducting interviews against the backdrop of the community's daily reality—under the shadow of the massive, unmoving concrete structures of the unfinished drainage system, and inside small, makeshift homes still bearing the dark, horizontal stains of past floodwaters. In many of these areas, the most visible landmarks are the projects that were never completed. There are canals that were dug halfway and then abandoned, their walls cracked and their bottoms filled with trash and silt. There are embankments that were

supposed to hold back water, but which now crumble easily and do nothing to protect the houses behind them. There are pipes that were installed but never connected, and roads that were paved but have no proper slope to allow water to flow away.

When we sat with participants, these structures were always in view. As they spoke about the years they have waited for help, about the funds that were announced but never used, and about the officials who promised relief but delivered nothing, the unfinished projects stood as silent proof of their words. Inside their homes, the walls told another story. Dark lines, sometimes faint, sometimes thick and dark, marked the highest point the water reached. Some families had painted new lines to keep track, while others had simply left the stains there as a reminder. Seeing these marks made every description of loss and fear more real. It showed us that the flood is not just something that happened once—it leaves its mark, visible and permanent, on the homes and lives of everyone affected.

By grounding the data collection in the physical spaces where the trauma occurred, the design allowed the environment to act as an unvoiced participant in the study. Every detail of the surroundings contributes to the meaning of what is being shared. The narrowness of the streets explains why water moves so slowly and stays so long. The low elevation of the land explains why even light rain can cause water to gather. The lack of proper waste disposal explains why canals become blocked and water becomes dirty and dangerous. The proximity of houses to one another explains why when one area floods, all suffer together. All of these factors are part of the experience, and by observing them directly, we gained a much deeper understanding than any map or report could provide.

The sensory details of the fieldwork—the oppressive humidity before a downpour, the lingering smell of damp silt, and the rhythmic, ominous rumble of distant thunder—were systematically documented alongside verbal transcripts. In standard research, such details are considered irrelevant or subjective. But in this design, they are essential. We recorded how the air felt heavy and thick even on days when the sun was shining, making it hard to breathe deeply. We noted the smell of wet earth, rotting vegetation, and stagnant water that never fully went away, even when the ground seemed dry. We listened to the sounds: the buzzing of insects, the creaking of old wooden structures, the distant roar of traffic, and the sudden change in sound when the wind picks up and rain begins to fall. We observed how the light changes—from bright, harsh sunlight that exposes every crack and imperfection, to soft, gray light that makes everything look dull and heavy.

This deliberate environmental immersion was vital; it ensured that the research captured not just what the participants remembered, but how their physical surroundings continuously trigger and shape their ongoing reality. For someone who has never lived through a flood,

rain is usually seen as a blessing, something that cools the air and helps the crops grow. But for the people in these communities, rain is a warning. It changes the atmosphere instantly. As soon as the sky turns dark and thunder is heard, the routine of daily life shifts. People stop what they are doing, check the canals, move their belongings, and prepare for what may come. This reaction is not a choice—it is a habit formed over years of experience. By being there and observing this ourselves, we understood that the trauma does not end when the water leaves; it remains in the way they see the weather, the way they plan their days, and the way they feel about their own homes.

This environmental and emotional raw data became the structural framework for the world-building of Sa Likod ng Baha. In many animated or narrative works, the setting is created based on imagination, reference photos, or general ideas of what a place should look like. But in this project, every part of the environment was built from the ground up using what we observed, measured, and described during our fieldwork. We did not guess what the streets look like or how the houses are arranged—we drew them, photographed them, and mapped them. We did not invent how the water moves or how it looks at different levels—we watched it, recorded it, and studied its behavior.

In this design, the cinematic setting is not treated as mere background decoration, but as a living character that directly drives the narrative's tension. In storytelling, the environment is often called the "fourth character"—the force that shapes the choices, actions, and feelings of the people in the story. Here, it is more than that. The environment is the source of the conflict, the reason for the struggle, and the visible representation of the problems the community faces. When we see the unfinished canal in the film, it is not just a structure—it is a symbol of broken promises. When we see the water rising slowly through the streets, it is not just an effect—it is the physical result of neglect and poor planning. When we feel the heavy air and dim light, it is not just style—it is the atmosphere of uncertainty and fear.

The visual and auditory cues gathered in the mud of the community were directly injected into the screenplay's stage directions and sound design. Our field notes are filled with descriptions that might seem unnecessary in a formal report, but are essential for creating an accurate and immersive story. We wrote about how the water looks when it is shallow—clear enough to see the stones and debris below—and how it turns dark, murky, and thick as it rises higher. We noted how it reflects the sky, turning gray and dull when the clouds are heavy, and how it catches light in strange ways when it is deep. We described the sound of water moving—slow and gurgling in blocked channels, rushing and loud when it flows over obstacles, and quiet but constant when it sits still and stagnant.

We also paid attention to the details of the structures: the way wood swells and warps after being wet for days, the way paint peels and fades from walls, the way metal rusts and corrodes, and the way mud leaves a fine, dusty layer on everything once it dries. These small details are what make a setting feel real. They show that the environment has a history, that it changes over time, and that it bears the marks of what it has been through. By including all of these observations in the creative process, we ensure that the world of Sa Likod ng Baha is not generic or made-up—it is a faithful representation of the place and the reality it represents.

The suffocating atmosphere experienced during the fieldwork was translated into a deliberate visual claustrophobia on screen, forcing the audience to breathe the same heavy air as the characters. One of the most striking things we noticed during our visits was the lack of open space and airflow in many parts of the community. Houses are built very close together, separated only by narrow passages. Walls and fences block the wind, and tall structures create shadows that remain even during the day. This creates a feeling of being enclosed, of having nowhere to go, and of being trapped when water rises and blocks the only paths in and out.

In the film, we used camera angles, lighting, and composition to recreate this feeling. We used tighter shots, lower angles, and darker tones to give the viewer a sense of being surrounded. We limited the field of view so that the audience sees only what the characters see—no wide, open horizons, no escape routes, no distant views. We used color palettes that are muted, grayish, and earthy, avoiding bright or cheerful tones that do not match the reality of the place. This is not done to make the film depressing, but to be honest. It ensures that the audience does not feel like they are watching a fictional story, but like they are stepping into the actual space and feeling the weight of the situation.

This methodological choice guarantees that the final creative output does not merely report a sequence of events, but forces the viewer to inhabit the exact, unrelenting reality of the community. There is a big difference between hearing that a flood causes hardship and actually feeling the conditions that create that hardship. By using the data gathered through immersion, we remove the distance between the story and the viewer. When they see the water marks on the walls, they understand what that height means. When they hear the distant thunder and feel the tension in the air, they understand the constant state of alertness. When they see the characters moving carefully through narrow, muddy streets, they understand how difficult even simple tasks become.

This bridges the distance between the comfortable moviegoer and the drowning citizen, creating a visceral space where apathy becomes completely impossible. When you watch a

story from far away, it is easy to think, “That is unfortunate, but it is not my problem.” But when you have been brought into the space, when you have felt the atmosphere and understood the details, it becomes personal. You realize that this is not a problem of geography or nature—it is a problem of governance, of responsibility, and of justice. You understand that the people in the story are not different from you, and that their right to safety and dignity is just as important as anyone else’s.

Ultimately, this research design functions as an act of defiance against the systematic erasure of marginalized lives. In many ways, the people living in these communities are invisible to the rest of society. They appear in news reports only when there is a major disaster, and they disappear again once the cameras leave. Their struggles are discussed in meetings and documents, but rarely in ways that respect their humanity. Official language tends to describe them as “affected populations,” “vulnerable groups,” or “beneficiaries”—terms that reduce them to categories rather than recognizing them as individuals with families, hopes, and rights.

In a society where bureaucratic paperwork is often used to sanitize and minimize human suffering, this qualitative framework serves as a stubborn, permanent anchor for the truth. Numbers and statistics can be adjusted, reports can be rewritten, and data can be presented in ways that downplay the severity of the problem. But the reality on the ground cannot be changed. The marks on the walls remain, the unfinished structures remain, and the experiences of the people remain. By capturing all of this through direct observation and conversation, we create a record that is based on what is actually there, not on what is convenient or politically acceptable.

It recognizes that while a corrupt administration can easily ignore a statistical report or misplace a grievance folder, it cannot easily silence a story that has been given a heartbeat, a face, and a voice. Stories have a way of staying alive. They travel from one person to another, they stay in memory, and when they are presented in a form that people can see and feel, they become part of the public consciousness. Unlike a document stored in an office, a story cannot be locked away or hidden. It speaks directly to the heart, and it creates an emotional connection that numbers alone never can.

By intentionally structuring this research to culminate in a creative, narrative tool, the design achieves its ultimate, radical purpose. It proves that academic inquiry is not meant to be a passive bystander to injustice. Too often, research is seen as something that happens in universities and research centers, separate from the real world. It produces papers and reports

that are read by a small group of experts and then filed away. This design changes that model. It connects the research process directly to a creative output that is accessible, engaging, and designed to reach a wide audience. It turns data into dialogue, findings into feelings, and analysis into action.

Instead of being limited to academic circles, this design stands as an unyielding promise that the private heartbreaks of a forgotten community will be lifted out of the mud, transformed into art, and used to tear down the very walls of corruption that tried to bury them. It gives voice to the feelings that are often too difficult to express in formal language—the frustration, the sadness, the anger, and the quiet determination. It shows that these emotions are not just personal reactions; they are valid responses to a system that has failed to protect and serve.

This approach also ensures that the research gives back to the community. It does not take their stories and use them for academic credit alone—it uses them to raise awareness, to challenge assumptions, and to call for change. It gives the community something more than a report; it gives them representation, visibility, and a chance to be heard beyond their own streets. It turns their experiences into a message that can travel, be shared, and inspire others to listen, understand, and demand better.

No research design is complete without considering the ethical principles that guide every step. When working with people who have experienced loss, trauma, and neglect, it is essential to approach them with great care, respect, and sensitivity. This design included clear guidelines to ensure that the study did not cause further distress, and that the participants were treated with dignity throughout.

First, we ensured that participation was entirely voluntary. No one was pressured to speak, and they were free to stop the conversation at any time, or to choose not to answer questions that made them uncomfortable. We explained clearly the purpose of the study, how the information would be used, and how their identities would be protected—using pseudonyms and general descriptions so that no individual or family could be identified in the final work.

We also made sure that the research did not treat people as subjects or sources of information only. We listened to their concerns, answered their questions, and shared our own purpose openly. We let them know that their stories would be used to create something that could help others understand their situation, and that we would not misrepresent or distort what they

shared. We viewed our role not as investigators, but as guests and listeners, with the responsibility to carry their words forward with honesty and respect.

This ethical framework is part of the design itself. It ensures that the study does not exploit suffering for information or drama, but honors it by treating it with care and seriousness. It guarantees that the connection between research and creation remains true to the original experiences, and that Sa Likod ng Baha never becomes a work of entertainment at the expense of the community. Instead, it becomes a tribute to their strength, a record of their reality, and a tool for their advocacy.

Conclusion of the Research Design

In summary, this research design is built on the principle that to understand a problem, you must go where it exists, and to understand people, you must meet them in their own environment. It moves beyond standard methods to create a process that is immersive, sensory, and deeply connected to the reality being studied. By combining direct observation, environmental documentation, and qualitative listening, it gathers a rich, multi-layered set of data that includes facts, feelings, and context.

This design ensures that Sa Likod ng Baha is not a work of fiction based on guesswork, but a creative interpretation rooted in real evidence and real voices. It bridges the gap between academic research and public communication, proving that both can work together to create something meaningful. And most importantly, it stands as a commitment to truth—ensuring that the lives, struggles, and hopes of the community are presented honestly, respectfully, and powerfully, so that they can no longer be ignored or forgotten.

Qualitative Data Collection:

Research is often described as a systematic process of gathering information, analyzing it, and drawing conclusions. But when the subject of that research is human experience, this definition falls short. It suggests that the work is about objects, numbers, and facts—things that can be handled, measured, and sorted. Yet when we study people, we are studying something far more complex: their thoughts, their emotions, their memories, their

relationships, and the quiet ways they find purpose and joy in life. These things cannot be counted, weighed, or reduced to a formula. They can only be understood.

Qualitative research is the method designed to do exactly that. It does not seek to replace numbers or statistics, but to go deeper—to look beneath the surface of behavior and understand the meaning behind it. For this study, our approach to qualitative data collection was built on the principle that meaning is shared, built, and experienced through life itself, not just described in words. It required us to shift from being observers to being participants in understanding, from asking questions to listening closely, and from seeking answers to uncovering stories.

This chapter explains in detail how we carried out this process, the attitudes and values that guided us, the challenges we faced, and the profound insights we gained. It shows that this work is not just about gathering data—it is about building connections, honoring voices, and ensuring that the final work reflects the true reality of the people and community we set out to understand.

Beyond gathering stories and observing interactions, the qualitative process also allowed us to understand how meaning is formed through shared experiences rather than isolated opinions. In many forms of research, the goal is to separate variables, to isolate factors, and to look at things as independent parts. But human life does not function that way. Every moment, every choice, and every activity is connected to everything else that has come before it. What a person does today is shaped by who they were yesterday, by the people they have known, and by the feelings they have carried with them through the years.

When we first began our fieldwork, we entered with a standard list of questions. We wanted to know how long people had been involved in the hobby, what systems they owned, what games they played most, and how often they gathered with others. We thought that these details would give us a clear picture of the culture. But as soon as we sat down and began speaking, we realized that these details were only the frame, not the picture itself. The real meaning was found in how these things fit into the larger story of their lives.

For example, when someone said, “I’ve had this console since I was twelve,” they were not just telling us how old the machine was. They were telling us about a time in their life, about what it felt like to be that age, about the excitement of receiving it, and about the years that followed when it remained a constant presence. When someone said, “I play this game when I’m feeling down,” they were not just telling us about the game—they were telling us about

how it provides comfort, familiarity, and a sense of control when other parts of life feel uncertain.

This realization changed the entire direction of our work. We stopped looking for separate answers and started looking for connections. We began to ask: How does this activity fit into your daily life? How does it connect to your past? How does it help you relate to others? How does it make you feel about yourself? These questions opened doors to deeper conversations and allowed us to see the bigger picture.

As we engaged more deeply with participants, we realized that retro gaming culture is not simply defined by the games themselves, but by the emotions, memories, and relationships attached to them. If you look only at the hardware and software, you see technology. You see limitations in graphics, sound, and processing power. You see things that have been surpassed by newer, faster, and more complex systems. But if you look through the eyes of those who love and preserve these games, you see something entirely different. You see history. You see creativity. You see the effort and imagination of designers who worked with limited tools to create worlds that still feel alive today.

More importantly, you see the emotional weight these objects carry. We heard about controllers that have been held by multiple generations—first by parents, then by their children, and sometimes even by grandchildren. We heard about cartridges that have been repaired, cleaned, and kept working for decades, not because they are valuable as items, but because they hold memories that cannot be replaced. One participant told us, “If this console breaks, I can buy another one, but it won’t be the same. This one was there when I was growing up. It was there when my brother and I stayed up late playing together. It has our history inside it.”

Each story revealed layers of personal history—childhood moments spent after school, sitting cross-legged on the floor, the hum of the console filling the room, the excitement of discovering a new level or solving a difficult puzzle. We heard about the thrill of finally beating a game that had seemed impossible, and the pride of mastering skills that required patience and practice. We heard about how these games taught lessons: how to plan ahead, how to adapt when things went wrong, how to keep trying even when you failed, and how to work with others to reach a shared goal.

We heard about family connections strengthened through play—parents learning to understand what their children enjoyed, and children gaining respect for the things their parents valued. We heard about friendships formed through shared interests, where people who might have had little else in common found a language and a joy that brought them

together. These friendships lasted through school years, through career changes, through moves to different cities, and through the ups and downs of adulthood. Even when they did not play together often, the bond remained strong because of the shared history.

And perhaps most significantly, we heard how these games became coping mechanisms during difficult periods of life. There were participants who spoke of turning to these familiar worlds when they were dealing with illness, unemployment, financial struggles, or the loss of a loved one. They described how the routine of playing, the clear rules, and the predictable challenges gave them a sense of stability when everything else felt out of control. In the words of one participant: “When I sit down to play, I know what to expect. The game doesn’t change. The rules stay the same. It gives me a place where I can breathe, think, and rest before I have to face the hard parts of life again.”

These narratives helped us see that what may appear as a simple form of entertainment is actually a meaningful emotional space where identity and belonging are continuously shaped. In a modern world that often values speed, efficiency, and constant change, this hobby offers something rare: continuity. It allows people to hold onto parts of their youth that they cherish. It gives them a way to express their personality, their knowledge, and their creativity. It creates a community where they are accepted not for their status, wealth, or appearance, but for their passion and their willingness to share.

Through careful listening, we learned to recognize patterns in how participants describe comfort, nostalgia, and community, and how these feelings often overlap in ways that cannot be easily measured through numbers alone. We began to understand that nostalgia is not just longing for the past—it is a way of bringing the best parts of the past into the present. It is a way of keeping alive the feelings of joy, curiosity, and possibility that defined earlier years. Comfort is not just relaxation—it is the feeling of returning to something that feels safe, familiar, and true. Community is not just membership in a group—it is a network of people who understand, respect, and support one another, bound together by shared values and shared memories.

This process also challenged us as researchers to slow down and truly absorb what was being shared, rather than simply categorizing responses into themes. In the early stages of our work, we were still thinking in terms of efficiency. We wanted to cover all the points we had planned, and we wanted to finish each interview within a set time. But we quickly realized that this approach was limiting. When we tried to move too fast, we missed the nuances. We missed the pauses, the explanations, and the small details that gave the story its heart.

We had to learn to let go of the need to control every moment. We learned to say, “Tell me more about that,” or “How did that feel?” instead of jumping to the next question. We learned to wait in silence if the participant needed time to think or find the right words. We learned that sometimes the most important part of a conversation happens after the formal questions are done, when people feel relaxed enough to speak more freely and openly.

In doing so, we became more aware of our own interpretations and biases, and we made a conscious effort to let participants’ voices guide the direction of our understanding. We all carry assumptions about what certain activities mean. Before this study, many of us might have thought of gaming as something done only for fun, or as a way to pass time. But as we listened, our understanding grew. We realized that it is also a way of preserving history, of building relationships, of learning life lessons, and of finding emotional balance.

We had to constantly check ourselves. We asked: Am I understanding this correctly? Am I adding my own feelings to what they are saying? Am I making assumptions that are not there? We kept detailed notes, recorded interviews with permission, and reviewed them many times. We discussed our impressions with each other to ensure that we were not letting our own views shape the findings. This process of self-reflection was not easy, but it was essential to ensure that our work was fair and accurate.

We also noticed that silence, pauses, and emotional hesitation during interviews were just as significant as spoken words, often revealing deeper feelings that participants struggled to fully articulate. There are limits to language. Some feelings are too large, too complex, or too personal to be put into simple sentences. We saw this time and again. When someone spoke about a game that reminded them of a loved one who had passed away, they might pause, look away, and say, “It’s hard to talk about.” When they spoke about the sense of belonging they felt in the community, they might smile and say, “It just feels like family.”

These moments told us that the story was deeper than the words used. They told us that we were touching on something meaningful and real. Instead of trying to fill the silence or push for more details, we learned to respect it. We learned to acknowledge the feeling and let the participant know that it was okay to speak slowly, or to stop if they needed to. In doing so, we created a space where they felt safe enough to share what they could, even if it was not expressed perfectly.

These moments reminded us that qualitative research is not only about collecting information but about creating a safe space where people feel seen, heard, and respected. Trust is the foundation of this work. If people do not feel safe, they will not open up. If they do not feel respected, they will not share their true thoughts and feelings. We made every effort to build that trust from the very beginning.

We explained clearly the purpose of the study, how the information would be used, and how their identities would be protected. We assured them that their participation was entirely voluntary, that they could withdraw at any time, and that their privacy would be strictly maintained. We made sure that no information that could identify them would be included in reports or public materials. We treated their stories as valuable gifts, not just as data points to be used.

As we reflected on the data, we understood that the value of these interactions lies not only in what was said, but in the trust that was built throughout the process. Many participants told us that they rarely spoke about these feelings with others. They said that friends, family, or colleagues often saw their hobby as something trivial, something childish, or something that did not contribute to "real life." They felt that no one would understand why they spent time, effort, and money on old equipment and games.

But in our conversations, they found a different kind of audience. They found people who wanted to understand, not judge. They found listeners who asked questions out of curiosity, not criticism. This allowed them to speak more openly, to explain what they felt, and to share parts of their experience that they usually kept private. This trust did not happen instantly—it was built over time, through consistency, respect, and honesty.

This trust became the foundation for interpreting their experiences responsibly, ensuring that their perspectives were not misrepresented or oversimplified. It would have been easy to take small parts of their stories and turn them into broad generalizations. It would have been tempting to create a single "image" of the community that fit neatly into a theory. But we knew that would be unfair and inaccurate.

Instead, we worked carefully to preserve the uniqueness of each story while also recognizing the common threads that ran through them. We wrote our notes and analysis with their voices in mind, checking repeatedly to ensure that what we wrote reflected what they shared, not what we wanted to prove. We avoided using language that sounded distant or clinical. Instead, we used words that honored the humanity and emotion behind their experiences.

Ultimately, the qualitative findings shaped the heart of our project by grounding it in authentic human experience, allowing us to move beyond surface-level understanding and toward a more empathetic and meaningful representation of the community we aimed to study. Without this deep engagement, our work would have remained incomplete. We would have described what people do, but we would never have truly understood why it matters. We would have looked at the culture from the outside, but we would never have entered its inner world.

Through this process, we learned that this hobby is far more than a way to pass time. It is a way of preserving memory and history. It is a way of maintaining connections across generations. It is a source of comfort and strength during hard times. It is a place where people can express their creativity, knowledge, and identity. It is a community built on respect, generosity, and shared passion.

These lessons became the core of our work. They guided how we wrote our analysis, how we designed our creative outputs, and how we shared our findings. They ensured that every part of the project was rooted in reality, not imagination. They allowed us to present the community not as a group of people with a hobby, but as people with rich lives, deep feelings, and values that deserve to be recognized and respected.

In the end, this qualitative process did more than provide data—it gave us a new way of seeing. It taught us that behind every activity, every object, and every choice, there is a human story. And by listening to those stories, we do not just complete a research project—we build a work that truly reflects the people it represents.

Qualitative data collection is more than a method—it is a mindset. It requires patience, openness, respect, and a willingness to learn. It moves beyond numbers and facts to uncover the meaning, emotion, and connection that make us human. For this study, it provided the depth and authenticity that no other method could offer. It allowed us to understand not just what happens in this community, but why it matters, and how it shapes the lives of those who are part of it.

Quantitative Data Collection:

As the development of the project continues through multiple cycles of testing, evaluation, and refinement, the process of quantitative data collection gradually becomes one of the most important foundations of the study. While the creative team initially focuses on building the visual and emotional components of the film, the collection of measurable audience responses allows the researchers to move beyond assumptions and gain a deeper understanding of how viewers actually experience the story. Rather than relying solely on personal opinions or artistic intuition, quantitative data provides a structured way of examining audience reactions and identifying patterns that may otherwise remain unnoticed.
Throughout the production process, every screening session becomes an opportunity to gather valuable information. Audience members are invited to share their responses through surveys, rating scales, questionnaires, and other assessment tools designed to measure different aspects of their viewing experience. These instruments help capture data related to emotional engagement, narrative clarity, visual effectiveness, sound quality, pacing, character connection, and overall impact. By collecting responses from multiple participants, the researchers are able to compare experiences, identify recurring trends, and evaluate whether the intended message of the film is being successfully communicated.
One of the most significant discoveries during the data collection process is the realization that emotional impact is rarely created by a single cinematic element. Instead, audience responses consistently demonstrate that emotions emerge through the interaction of multiple components working together simultaneously. Visual composition, lighting, sound design, animation quality, pacing, environmental atmosphere, and character movement all contribute to the viewer’s experience. When these elements are carefully coordinated, they create moments that feel authentic, immersive, and emotionally powerful.
For example, several testing sessions reveal that scenes featuring intense flood imagery do not automatically generate the strongest emotional responses. Although dramatic visuals capture attention, audiences often report deeper emotional engagement during quieter moments where the character Tinoy is shown reflecting on loss, uncertainty, or hope. These findings challenge the common assumption that emotional impact depends primarily on spectacle or visual intensity. Instead, the data suggests that viewers often connect more strongly with scenes that allow them to empathize with the character's personal experiences.
This observation becomes increasingly important as the project develops. By analyzing survey scores and response patterns, the researchers recognize that emotional storytelling frequently relies on subtle details rather than dramatic effects alone. A moment of silence, a brief pause in dialogue, a change in lighting, or a simple facial expression can sometimes produce a stronger emotional response than an elaborate sequence filled with visual action. As a result, the creative team begins paying closer attention to these subtle narrative elements, ensuring that they receive the same level of care and consideration as the larger visual sequences.

The quantitative data also provides valuable insights into audience attention and engagement throughout the film. By examining responses from different sections of the story, researchers can identify moments where audience interest increases, decreases, or remains stable. This information helps determine whether the pacing of the film effectively supports its emotional goals. In some cases, data reveals that certain scenes feel rushed, preventing viewers from fully processing important emotional moments. In other instances, audiences indicate that extended sequences may reduce engagement if they do not contribute meaningfully to character development or narrative progression.

Understanding these patterns allows the researchers to make informed decisions regarding editing and scene structure. Rather than adjusting the film based solely on personal preference, revisions are guided by evidence gathered from audience experiences. This approach helps ensure that creative decisions remain connected to the needs and perceptions of viewers while still preserving the artistic vision of the project.

Another important aspect of quantitative data collection involves measuring audience understanding of the film's central themes. Since Sa Likod ng Baha explores issues related to flood disasters, community experiences, resilience, social challenges, and institutional concerns, it is essential to determine whether viewers accurately interpret these themes. Survey questions designed to assess comprehension provide valuable information regarding the effectiveness of the film's storytelling strategies.

Results frequently indicate that audiences are able to recognize the broader social issues presented throughout the narrative. Many viewers report that the film encourages them to think more critically about the realities faced by communities affected by recurring floods. Others express increased awareness of the emotional and psychological consequences of disasters beyond the immediate physical damage they cause. These responses suggest that the film succeeds not only as a visual narrative but also as a medium for social reflection and public awareness.

The collection of quantitative data further highlights the importance of environmental storytelling within the film. Audience feedback consistently demonstrates that settings and surroundings play a significant role in shaping emotional responses. Scenes depicting damaged homes, flooded streets, abandoned belongings, and disrupted community spaces often generate strong reactions because they provide visual evidence of the challenges faced by affected individuals. Rather than functioning merely as background elements, these environmental details contribute directly to audience understanding and emotional engagement.

As researchers continue analyzing data from various screening sessions, they begin identifying relationships between visual intensity and emotional response. In some cases, highly detailed environments correspond with increased levels of audience immersion. However, the data also reveals that excessive visual complexity can occasionally distract

viewers from the emotional core of a scene. This finding encourages the creative team to pursue a balanced approach that maintains visual richness while preserving narrative focus.

Sound design emerges as another area where quantitative analysis proves particularly valuable. Audience ratings frequently indicate that sound significantly influences emotional perception. Music, ambient effects, silence, and audio transitions all contribute to the overall atmosphere of the film. Certain musical arrangements consistently receive higher emotional impact scores, while others appear less effective in supporting the intended mood. By examining these trends, researchers gain a clearer understanding of how sound shapes audience experiences and influences emotional interpretation.

One especially noteworthy finding involves the strategic use of silence. Contrary to expectations, moments with limited or absent background music often receive some of the highest emotional ratings. Participants frequently describe these scenes as more authentic, intimate, and reflective. The absence of sound creates space for viewers to process emotions independently, allowing personal interpretation to become part of the viewing experience. This discovery reinforces the idea that emotional storytelling does not always require constant stimulation and that restraint can be just as powerful as intensity.

Quantitative data collection also supports the evaluation of character development throughout the film. Through surveys and rating scales, viewers are asked to assess their level of connection with Tinoy and other characters. Responses provide valuable insights into audience empathy, emotional investment, and narrative engagement. High levels of character connection suggest that viewers see aspects of themselves, their communities, or their experiences reflected within the story.

These findings are particularly significant because emotional engagement often serves as a bridge between entertainment and meaningful reflection. When audiences develop a strong connection with a character, they become more receptive to the themes and messages presented throughout the narrative. As a result, character development remains a central focus during revisions and refinements informed by quantitative analysis.

Another important contribution of quantitative data collection lies in its ability to reveal unexpected audience reactions. Not all responses align with the expectations of the creative team. Some scenes believed to be highly impactful receive moderate ratings, while others unexpectedly generate strong emotional responses. Rather than viewing these discrepancies as problems, researchers treat them as opportunities for learning and improvement. Unexpected findings encourage the team to examine their assumptions and consider alternative interpretations of the film. They demonstrate that audiences bring their own experiences, perspectives, and emotions into the viewing process. Consequently, meaning is not created solely by the filmmaker but emerges through interaction between the film and its viewers. This realization fosters a more audience-centered approach to storytelling and strengthens the project's commitment to authentic emotional engagement.

The process of analyzing quantitative data gradually evolves into the creation of an emotional map of the film. Instead of focusing exclusively on isolated scenes, researchers begin examining how emotional responses change across the entire narrative journey. By tracking audience reactions from beginning to end, they can identify peaks of tension, moments of reflection, periods of emotional recovery, and points of renewed engagement. This emotional mapping process proves especially useful for evaluating narrative rhythm. Effective storytelling requires balance. Continuous intensity can overwhelm audiences, while prolonged calmness may reduce engagement. The data helps identify whether emotional transitions occur naturally and whether viewers are given sufficient time to absorb significant events. Adjustments informed by these findings contribute to a more cohesive and emotionally satisfying viewing experience.

As the project progresses, the researchers recognize that quantitative data collection serves a purpose beyond evaluation. It becomes a tool for understanding human experience. Every numerical score, survey response, and rating represents a real person's reaction to the story. Behind each statistic lies an emotional response shaped by memories, beliefs, experiences, and personal interpretations. Recognizing this human dimension helps ensure that data remains connected to the individuals it represents.

This perspective influences how findings are interpreted and applied. Rather than treating audience responses as abstract numbers, researchers approach them as indicators of genuine emotional engagement. The goal is not merely to maximize ratings but to understand how viewers connect with the film and how the narrative affects their perceptions, emotions, and reflections. Ultimately, the quantitative data collection process plays a vital role in the development of Sa Likod ng Baha. Through systematic measurement, careful analysis, and continuous refinement, the researchers gain valuable insights into audience experiences and emotional responses. The findings support informed creative decisions, strengthen narrative effectiveness, enhance emotional authenticity, and contribute to the overall quality of the project. By embracing both artistic creativity and empirical evaluation, the project demonstrates that meaningful storytelling can benefit from a balance between imagination and evidence. Quantitative data does not replace creative vision; rather, it provides a framework for understanding how that vision is received by audiences. Through this ongoing cycle of testing, observation, analysis, and improvement, Sa Likod ng Baha continues to evolve into a film that is not only technically accomplished but also emotionally resonant, socially relevant, and deeply connected to the experiences of its viewers.

Data Analysis:

The data analysis process for Sa Likod ng Baha was designed to evaluate both the technical quality and emotional effectiveness of the animated film. Since the project aimed to tell a story that resonates with audiences through visual storytelling, it was important to analyze not

only the technical components of the animation but also the emotional responses it generated. The researchers recognized that storytelling effectiveness cannot be measured solely through numerical data. Instead, understanding audience engagement required a combination of observation, feedback collection, comparative analysis, and reflective evaluation. Through this process, the researchers were able to determine whether the intended themes, emotions, and messages of the film were successfully communicated to viewers.

The analysis began with the collection and organization of audience feedback gathered during screenings, reviews, and evaluation sessions. Responses were categorized into technical feedback and emotional feedback. Technical feedback focused on aspects such as animation quality, visual consistency, environmental design, lighting, scene composition, pacing, camera movement, and overall presentation. Emotional feedback focused on audience reactions to the story, character development, atmosphere, and emotional impact of specific scenes.

After collecting the responses, the researchers reviewed and grouped similar comments to identify recurring themes and patterns. Feedback that appeared repeatedly among participants was considered significant because it indicated common audience perceptions. Comments regarding emotional engagement, scene effectiveness, and visual storytelling were carefully examined to determine which elements of the film resonated most strongly with viewers.

In addition to gathering direct feedback, the researchers conducted a comparative analysis of established visual storytelling techniques used in animation and film. Various works within the noir and drama genres were studied to understand how professional filmmakers communicate emotions through visual means. Particular attention was given to the use of lighting, color, composition, environmental storytelling, pacing, and camera placement. The researchers observed how these elements were used to establish mood, create tension, and support character development without excessive reliance on dialogue.

Through this comparative analysis, the researchers discovered that emotional atmosphere is rarely created by a single visual element. Instead, mood emerges from the interaction of multiple components working together. Lighting, environmental conditions, camera framing, character movement, and sound design collectively contribute to the audience's emotional experience. This observation influenced the evaluation process of Sa Likod ng Baha, as the researchers analyzed how these elements interacted within each scene to support the intended emotional narrative.

The researchers also analyzed scene composition and framing throughout the film. Each scene was examined to determine whether visual arrangements effectively communicated the intended emotional message. Wide shots, medium shots, and close-up shots were reviewed based on their contribution to storytelling. Wide shots were evaluated for their ability to emphasize environmental scale and character isolation, while close-up shots were assessed for their effectiveness in portraying emotional expression and character vulnerability.

Particular attention was given to the relationship between the characters and the environment. Since Sa Likod ng Baha focuses heavily on the effects of disaster and systemic issues, environmental storytelling played a major role in the narrative. The researchers examined whether the flooded surroundings enhanced audience understanding of the themes and whether the environment itself contributed to emotional engagement.

Another important component of the analysis involved examining audience behavior during screening sessions. The researchers recognized that not all emotional responses are expressed verbally. Therefore, audience observation became an essential method for gathering additional insights. During screenings, viewers were observed for visible emotional reactions, changes in attention, body language, and moments of silence.

Instances where viewers remained silent during emotionally significant scenes were carefully noted. Silence was interpreted as a possible indicator of emotional processing and engagement. Likewise, moments when viewers appeared visibly affected by a scene, such as showing signs of sadness, tension, surprise, or reflection, were documented as observational data. These reactions provided valuable information regarding the emotional effectiveness of specific narrative sequences.

The researchers also observed viewer attention levels throughout the screening. Scenes that consistently maintained audience focus were identified as areas of strong engagement. Conversely, scenes where viewers appeared distracted or less attentive were analyzed to determine possible reasons for reduced engagement. This information assisted in identifying opportunities for improvement and refinement.

Following the observation process, the researchers compared audience behavior with verbal feedback. This allowed for cross-validation of findings. In several instances, scenes that generated strong emotional reactions during observation were also frequently mentioned during feedback sessions. The consistency between observed behavior and participant comments strengthened the reliability of the findings.

The analysis further involved identifying emotional patterns across multiple viewers. Rather than focusing solely on individual reactions, the researchers looked for recurring emotional responses that appeared across different participants. This pattern-based approach helped identify which scenes consistently generated emotional engagement.

Particular attention was given to scenes involving Tinoy's moments of solitude, reflection, and grief. The researchers analyzed whether audiences connected with these scenes and how they interpreted the character's emotional journey. Memory sequences involving Linda were also examined due to their importance in establishing emotional context and character motivation.

Long environmental shots featuring flood-affected locations were evaluated to determine their impact on audience perception. The researchers examined whether these scenes effectively communicated themes of vulnerability, helplessness, and environmental hardship.

Viewer responses suggested that these moments often allowed audiences to reflect on the broader implications of the story, extending emotional engagement beyond individual character experiences.

The researchers also analyzed the effectiveness of contrast within the narrative structure. Emotional contrast refers to the transition between differing emotional states, such as peace and chaos, hope and despair, or warmth and isolation. Scenes were examined to determine whether these transitions enhanced emotional impact and maintained audience engagement.

Technical elements were evaluated through systematic observation and testing. Multiple variations of environmental and visual effects were reviewed during production. Rain density, shadow intensity, lighting conditions, color grading, camera distance, and environmental detail were adjusted and assessed to determine their influence on audience interpretation.

Rain density was analyzed because of its significant role in establishing atmosphere. Different levels of rainfall were tested to determine how environmental intensity affected emotional perception. Heavier rainfall generally created stronger feelings of tension, sadness, and vulnerability, while lighter rainfall supported quieter and more reflective moments.

Color grading was evaluated to determine its contribution to mood and emotional tone. Darker color palettes were assessed for their ability to communicate despair, uncertainty, and hardship. Warmer color palettes used in memory sequences were examined for their effectiveness in creating feelings of comfort, nostalgia, and emotional connection. Audience responses helped confirm whether these visual choices successfully conveyed the intended emotional atmosphere.

Lighting conditions were similarly analyzed. Scenes utilizing softer lighting were reviewed to determine whether they created emotional warmth and intimacy. Scenes featuring stronger shadows and darker lighting were examined for their ability to increase dramatic tension and reinforce themes of uncertainty and struggle.

Camera distance and framing were evaluated in relation to emotional storytelling. Close-up shots were assessed based on their ability to encourage empathy and emotional connection with characters. Wide shots were analyzed for their effectiveness in portraying environmental scale and emphasizing character isolation. The researchers considered how camera placement influenced audience perception and emotional engagement throughout the film.

The hybrid visual approach of combining 2D character animation with 3D environmental design was also included in the analysis process. The researchers examined how this artistic decision contributed to the overall storytelling experience. Particular attention was given to the visual contrast created between expressive 2D characters and expansive 3D environments. This contrast was evaluated for its effectiveness in reinforcing themes of vulnerability, resilience, and systemic challenges.

Beyond technical and emotional considerations, the researchers incorporated ethical reflection into the analysis process. Because the film addresses sensitive topics such as disaster, loss, corruption, and death, it was necessary to evaluate whether these subjects were portrayed responsibly. The researchers reviewed scenes to determine whether the representation of suffering remained respectful and grounded in empathy.

Each scene was examined to ensure that emotional intensity served a meaningful narrative purpose rather than existing solely for dramatic effect. The researchers continuously assessed whether the portrayal of hardship reflected authentic human experiences and whether the narrative treated affected individuals with dignity and respect.

The ethical review process also involved considering the potential impact of the film on audiences. The researchers sought to ensure that viewers would leave with a deeper understanding of the themes rather than simply experiencing emotional shock. This emphasis on authenticity guided revisions throughout the production process and influenced final creative decisions.

After completing all phases of analysis, the researchers synthesized the findings from audience feedback, observational data, technical evaluations, comparative studies, and ethical reflections. The collected information was organized and interpreted to identify strengths, weaknesses, and opportunities for improvement within the project.

The results of this analysis served as the foundation for revisions and refinements implemented during production. Insights gained through the process influenced decisions regarding pacing, scene composition, environmental design, visual effects, emotional storytelling techniques, and narrative structure. By integrating audience perspectives with technical evaluation, the researchers were able to make informed creative decisions that enhanced both the artistic quality and emotional effectiveness of the film.

Ultimately, the data analysis procedure demonstrated that effective storytelling requires a balance between technical execution and emotional authenticity. The researchers concluded that audience engagement is influenced not only by visual quality but also by the meaningful connection established between the narrative and the viewer. Through a combination of structured analysis and creative reflection, Sa Likod ng Baha was refined into a project that seeks to communicate its themes with clarity, empathy, and emotional depth.

Collection Methods:

We build a 3D world, draw fragile 2D people, and blend them together using light and rain to tell an emotional story.

Collection Methods:

1. Storyboarding the Mood We start by sketching out the scenes to plan the "Pinoy Noir" feel. This helps us decide where the camera should go to make the 3D floodwater look scary and how to frame Tinoy so he looks small and alone.
2. Building the 3D Environments Instead of drawing every background by hand, we build the slums, the worksite, and the hall as 3D models. This lets us move the camera around and add realistic 3D rain, shadows, and rising water that would be too hard to draw frame-by-frame.
3. Hand-Drawing the 2D Characters While the world is 3D, our characters are hand-drawn in 2D. We chose this style because it makes the people look "fragile" and human. We focus on simple 2D animations—like a sad face or a tired walk—to show Tinoy's grief.
4. Merging 2D into 3D (Compositing) The most important part is placing our 2D drawings into the 3D sets. We use 3D lighting to make sure the rain and shadows actually hit the 2D characters. This makes it look like Tinoy is truly "stuck" in a heavy, realistic storm.

5. Sound Collection and Atmospheric Recording
To make the world of Sa Likod ng Baha feel more realistic and immersive, we also collect and design sound carefully. Instead of relying only on generic audio, we look for or record sounds that match the environment of the story—such as heavy rain, strong wind, distant thunder, dripping water, and crowded evacuation noise. These sounds are layered together to build tension and emotion, especially during flood scenes. Silence is also treated as an important "sound" because it highlights loneliness, grief, and emotional weight in key moments of the story.

6. Lighting and Mood References

Before finalizing scenes, we gather visual references for lighting, weather, and cinematic mood. This includes looking at real-life flood footage, dark urban environments, and "noir-style" cinematography to guide how each scene should feel. Lighting is not only used for visibility but also for storytelling—dark shadows represent corruption and fear, while dim or washed-out light represents loss and hopelessness. These references help ensure that every scene stays consistent with the emotional tone of the film.

7. Camera Planning and Scene Blocking

We also plan how the camera behaves in every scene to strengthen storytelling. Since the film uses a mix of 2D characters and 3D environments, camera movement is very important in controlling emotion. Low-angle shots are used to make systems of power like the government hall feel intimidating, while wide shots are used to show Tinoy's isolation in the flood. Before final animation, we block each scene in 3D space to test framing, movement, and composition so that the emotional message is clear before adding detailed animation.

8. Character Emotion Planning

Before animating, we carefully plan Tinoy’s emotional expressions and physical movements for each key scene. Instead of complex motion, we focus on small but meaningful actions such as trembling hands, slow walking, paused breathing, or silent staring. These subtle movements help communicate grief and emotional exhaustion more effectively than exaggerated animation. Each emotional beat is matched with the script so that the audience can clearly feel what Tinoy is experiencing without needing excessive dialogue.

9. Scene Testing and Preview Rendering

Before finalizing each sequence, we create test renders or rough previews of scenes to check if everything works together properly. This includes checking if the 2D character blends correctly with the 3D environment, if lighting feels consistent, and if the emotional tone matches the storyboard. These previews help us identify problems early, such as awkward camera angles or unclear emotions, so adjustments can be made before final rendering.

10. Final Output Assembly and Review

After all scenes are completed, we assemble the full film in sequence and review it as a complete narrative. This stage focuses on flow, pacing, and emotional continuity. We check whether the story transitions smoothly from grief to tension to betrayal, and whether the final message is clearly understood. Minor adjustments in timing, sound, and color are made during this stage to strengthen the overall impact of the film before final submission.

Tools for Analysis:
2D Animation & Modeling Software: We use tools like Blender or Autodesk Maya to manage our 3D sets and Adobe Animate or Photoshop for our 2D characters. These help us analyze the timing of the movements and ensure the 2D drawings look natural inside the 3D world.

Compositing & Visual Effects: We use Adobe After Effects to "glue" the 2D and 3D parts together. This is where we analyze how the 3D rain and lighting hit our 2D characters, making adjustments to the shadows so the atmosphere feels heavy and realistic.

Production Tracking: We keep a close eye on our rendering times and file sizes. By tracking these "numbers," we make sure our computers can handle the complex 3D water and rain effects without slowing down our production timeline.

Audience Feedback Tools: We use simple surveys and test screenings to see how people react to Tinoy's story. This gives us the "human data" we need to know if the audience actually feels the sadness and betrayal we are trying to show.
Selection Process of digital tool

We chose Adobe Animate because it's the best way to give our characters a "fragile," hand-drawn look. It works easily with our 3D sets, helping us place Tinoy right into the middle of the dark, flooded world we built.

Significance of Adobe CC Animate:

1. The Emotional Heart
Adobe Animate is the heart of our story because it lets us hand-draw Tinoy with a "fragile" look that feels more human than a 3D model. It gives us the freedom to focus on small, sad details—like a tired walk or a heavy sigh—that make the audience really care about his struggle.
2. The Bridge Between Worlds
This tool acts as the perfect bridge, allowing us to easily move our 2D drawings into the 3D flooded world we built. Without it, we wouldn't be able to create the unique "2.5D" look that makes "Sa Likod ng Baha" feel so heavy and real.
3. Creating the "Pinoy Noir" Style
We chose Adobe Animate because it allows us to control the light and shadows on our 2D characters. This is how we make Tinoy look like he is truly stuck in the dark, rainy environment, helping the audience feel the weight of the betrayal in every scene.
4. A Personal Touch
In the simplest terms, Adobe Animate is what makes our film feel personal. It allows us to put a hand-drawn "human" character into a cold, digital 3D world, making Tinoy's journey feel much more emotional and unforgettable for the viewers.
the import and export of assets and compositions.

Steps for Selection:

1. Matching Our Story’s Needs We started by looking at what "Sa Likod ng Baha" really needed. Since our story is about a fragile human character in a big, scary world, we needed a tool that could handle traditional 2D drawing while still being able to "talk" to our 3D software. Adobe Animate was the perfect fit for this hybrid style.
2. Testing the Right Features We did a deep dive into the software’s features, like frame-by-frame animation. This was important because it allowed us to hand-draw Tinoy’s small, emotional movements—like his facial expressions during the betrayal—giving the animation a more personal and "hand-made" feel.
3. Trying It Out (Prototyping) Before committing, we ran several tests to see how a 2D character would actually look inside a 3D environment. We found that Adobe Animate was easy to use and powerful enough to handle the complex layers of rain and shadows without slowing down our work.
4. Staying True to Our Vision The final decision came down to our goal: making the audience feel Tinoy’s grief. Adobe Animate aligned perfectly with our creative vision, giving us the technical tools to produce high-quality art that makes the "Pinoy Noir" atmosphere of the flood feel real and heavy.

Our Vision and the Cascading Process

By choosing Adobe Animate, we are using a professional and reliable tool to bring our characters to life. This selection process ensures that every drawing of Tinoy fits perfectly with our creative vision, setting the stage for a film that is both technically polished and emotionally powerful.

As the creators of "Sa Likod ng Baha," we are excited to show how we used a "Cascading" approach to build our story. This means that every piece of data we gathered—from real-life stories of flood survivors to technical tests of 3D rain—flowed naturally into our animation process.

Our journey began with a deep dive into the world of "Pinoy Noir," where dark shadows and heavy rain create a feeling of mystery and sadness. By following this cascading method, we made sure that the "human heart" of the story was never lost, even as we worked with complex 3D technology. This allowed us to build an immersive world where the audience doesn't just watch the flood—they feel the weight of it.

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Project Management

Title: Sa Likod ng Baha: A 2D Animated Short Film

Objective: The goal of our capstone project, "Sa Likod ng Baha," is to create a powerful, atmospheric story that blends 2D and 3D animation. We want to show the emotional journey of a character named Tinoy as he faces betrayal during a heavy flood. At its core, “Sa Likod ng Baha” is more than a technical showcase. It is a labor of empathy and storytelling. Our goal is to create an atmospheric 2D animated short film that will make the audience feel the suffocating weight of Tinoy’s grief and the harsh reality of systemic betrayal. We want to create a world that feels at once intimately human and cinematically epic, combining the expressive, organic quality of 2D character animation with the immersive depth of 3D environment layouts. To achieve these goals, the production of "Sa Likod ng Baha" will follow a structured and phased workflow that ensures both creative consistency and technical efficiency. The project will begin with the pre-production stage, which includes script refinement, storyboard development, character design, environment design, and animatics. During this phase, the narrative structure will be finalized, and visual direction will be carefully established to ensure that both the emotional tone and thematic intent of the story are clearly translated into visual form. This stage is crucial in aligning the creative vision of the team and minimizing revisions during later production phases.

The production phase will focus on the creation of assets and the execution of animation. This includes the development of 2D character animations, background illustrations, and the construction of 3D environments that will serve as the spatial foundation of the film. Special attention will be given to the integration of 2D and 3D elements to ensure a cohesive visual experience. The team will carefully manage layering, lighting, and camera movement to maintain the intended contrast between the fragile human characters and the overwhelming scale of their environment. Each scene will be animated with a focus on emotional expression, timing, and atmospheric storytelling rather than complex action choreography, ensuring that the narrative remains grounded and impactful.

Post-production will involve compositing, visual effects integration, color grading, and sound design. This stage is essential in unifying all visual and auditory elements into a cohesive cinematic experience. Effects such as rain, flooding, reflections, and lighting transitions will be added to enhance realism and emotional depth. Sound design will play a critical role in establishing mood, using ambient audio such as rainfall, thunder, wind, and environmental noise to immerse the audience in the world of the film. Music, if included, will be used sparingly and strategically to emphasize key emotional moments without overwhelming the narrative.

In terms of project management structure, responsibilities will be divided among team members based on specialization. Artists will focus on character and background design, animators will handle motion and performance, while technical members will manage 3D

environments, rendering, and compositing. A designated project lead will oversee scheduling, quality control, and coordination between departments to ensure that production milestones are met on time. Regular review sessions will be conducted to evaluate progress, identify issues early, and maintain alignment with the creative vision of the project.

The project will also follow a milestone-based timeline to ensure steady progress throughout development. Each phase—pre-production, production, and post-production—will have specific deliverables and deadlines. This structured approach allows the team to monitor workload distribution, manage time efficiently, and ensure that the final output meets both academic and artistic standards. Flexibility will also be maintained within the schedule to accommodate unforeseen challenges such as technical limitations, rendering delays, or creative revisions.

Quality assurance is an important component of the project management process. Each completed scene will undergo review for visual consistency, animation quality, narrative clarity, and technical performance. Feedback will be gathered from team members and advisers to ensure that the film maintains a high standard of storytelling and visual execution. Necessary revisions will be implemented before proceeding to the next stage of production to prevent errors from accumulating in the final output.

Risk management is also considered within the project planning process. Potential risks include software limitations, hardware constraints, time restrictions, and workload imbalance among team members. To mitigate these risks, the project will prioritize efficient asset reuse, simplified animation techniques when necessary, and careful scheduling of rendering tasks. Backup files and version control practices will also be implemented to prevent data loss and ensure smooth collaboration throughout production.

Finally, the overall management approach of "Sa Likod ng Baha" emphasizes collaboration, adaptability, and creative discipline. While the project is grounded in technical production requirements, its primary focus remains on storytelling and emotional impact. Every decision made during development—from design choices to animation style—is guided by the central objective of portraying Tinoy’s emotional journey in a way that resonates with audiences. Through careful planning and coordinated teamwork, the project aims to deliver a cohesive and meaningful animated short film that reflects both artistic vision and disciplined production execution.
Gantt Chart: {Put the following tasks in your table} Example:

1. Pre-production Phase (4 weeks)

Days breakdown

Idea generation and concept
development: Brainstorm and comeup
with a unique storyline and visual style for the animated short film. (7 days)
Scriptwriting: Develop a detailed script that outlines the dialogue, actions, and overall structure (7 days)

Storyboarding: Create a series ofrough
sketches that visualize each scene of the film, establishing the composition,
camera angles, and character placement. (7 days)

Character design: Design and finalize the main characters,including their appearances, personalities, and unique
traits. (7 days)
3D background design: design and finalize

PRE-PRODUCTION (4 WEEKS)

In every creative journey, the beginning sets the direction. Pre-production is the time when imagination meets structure, and when ideas are shaped into something that can actually be built. For Sa Likod ng Baha, these four weeks are not just a checklist of tasks to complete; they are the most important part of the entire process. This is where we decide what the story will say, how it will look, and what feeling it will leave with those who watch it. Without this careful preparation, even the best intentions can become messy, unfocused, or unclear. With it, we create a solid foundation that supports every step that follows.

1. Finding the Heart of the Story (7 Days)

We start by getting into the headspace of our “Pinoy Noir” world. This is more than just choosing a style or a setting—it is about stepping into the atmosphere of the story, understanding its rhythm, and feeling its truth. “Pinoy Noir” is a term we use to describe a way of looking at life in the Philippines, especially in communities that face hardship and neglect. It is not about making things look dark or hopeless for the sake of drama; it is about

showing the reality of places where life is simple, resources are few, and people face challenges that are often beyond their control. It is a way of storytelling that is honest, thoughtful, and deeply human.

The focus is to highlight the struggles of the people, situated in impoverished areas under the mercy of government leaders and their hidden corrupted agendas that in some cases, affect said community of people, leading to tragedies. This is the central theme of our work, and it is a responsibility we do not take lightly. We know that these are not just fictional problems invented for a story—they are realities observed in many parts of our country. We have seen how heavy rains turn streets into rivers, how floodwaters rise quickly and stay for days, and how families must wait, sometimes without help, to cope with the damage and loss.

During these first seven days, we do not rush to write lines or draw scenes. Instead, we spend time reflecting, discussing, and remembering what we learned during our research. We go back to the notes we took, the photographs we captured, and the conversations we had with people who live in these communities. We remind ourselves of what they told us: how the flooding has become a regular part of life, how promised projects have never been finished, how drainage systems remain blocked, and how help often arrives too late or not at all.

We ask ourselves difficult but important questions: What is the real cause of this situation? How do people feel when they see water rising again and again? How does it affect their daily lives, their hopes, and their view of the future? We want to understand not just the physical events, but the emotional and social reality behind them. We want to avoid making the story seem like it is only about bad weather or bad luck. Instead, we want to show that behind every flood, there is a history of decisions, actions, and inactions that shape what happens.

We also think carefully about the tone of the story. We do not want it to feel angry or accusatory, because that would only divide people. Instead, we want it to feel observant, thoughtful, and full of empathy. We want to show the struggles, but also the warmth, the community spirit, and the quiet strength that keep people going. By the end of these seven days, we have a clear and shared understanding of what our story is about, what it stands for, and what we hope it will make people feel and think.

2. Writing the Script (7 Days)

Next, we move to the script, where we focus on the “sting” of the story. Once we know what we want to say, we need to find the right way to say it. The “sting” refers to the realization that comes toward the end of the story—the moment when the audience understands that the disaster is not just natural, but also a result of human failure and neglect. It is the moment when the story changes from being just about rain and water to being about responsibility, trust, and the cost of unfulfilled promises.

We develop the dialogue and the quiet, sad moments that happen between the lines. In a story like this, words are powerful, but silence is often more powerful. We do not want characters to give long speeches or explain everything directly. Instead, we want them to speak like real people—using simple words, short sentences, and sometimes saying nothing at all. We want their feelings to come through in what they do, how they look, and how they react, not just in what they say.

Our goal is to make sure the story flows naturally—from the very first raindrop to the final, emotional realization at the end of the film. We structure the script so that it moves step by step, like the experience of living through a storm. It begins with the ordinary: children playing, adults talking, the sounds of daily life, and the first signs of changing weather. Then it builds slowly—clouds gather, the rain starts, and as it continues, the water begins to rise. We add small details and conversations that reveal the history of the place, the problems with the surroundings, and the unmet promises that have left the community vulnerable.

We pay close attention to pacing. We know that if every moment is filled with action or tension, the audience will become overwhelmed and unable to feel anything deeply. But if the pace is too slow, the story may lose its momentum. We work to find the right balance—giving time for feelings to settle, allowing moments of reflection, and building tension gradually so that the emotional weight of the story grows naturally. We revise the script many times, removing what does not serve the story and adding details that make it feel more real. By the end of this week, we have a complete, strong script that guides us clearly forward.

3. Storyboarding the Mood (7 Days)

This is where we “see” the movie for the first time. The script tells us what happens, but the storyboard shows us how it looks. It is the first time our ideas turn into images, and it is where we decide how to use every visual tool to tell the story. Each sketch is not just a drawing—it is a plan for how the audience will see, feel, and understand what is happening.

We create rough sketches of every scene to plan where our 3D camera will move and where our 2D characters will stand. We decide on the framing: wide shots to show the whole environment, medium shots to show actions and interactions, and close-ups to focus on faces, expressions, and small details. We plan the angle of the camera—high angles to show the vastness of the flood and make characters look small, and lower angles to show the weight of the buildings and structures around them.

We use these sketches to ensure the scale of the flood feels overwhelming, making Tinoy look small and vulnerable against the rising water. This visual contrast is one of the most important messages of the film. It shows visually that while the environment and the problems are large and powerful, the people are small and exposed. But it also shows that even though they are small, their experiences are real and their struggles matter deeply. We test different compositions, adjusting how much of the environment is shown and where Tinoy is placed, until the feeling of scale and vulnerability is clear and strong.

We also plan the lighting and mood in each sketch. In the beginning, the light is soft and warm, even if the surroundings are worn. As the story progresses, the light becomes grayer, dimmer, and heavier. We sketch how shadows will fall, how rain will appear in the frame, and how reflections on the water will change the look of the world. Every choice is made to support the emotion of the story. By the end of these seven days, we have a complete storyboard that acts as our visual blueprint for the entire film.

4. Building the World in 3D (7 Days)

In this process, we had to make sure that our world was as close as it can get to the real thing, an impoverished community in disarray, with a little bit of exaggeration. We did not want to create a fantasy world or a place that looks too clean or too perfect. Instead, we wanted to capture the true character of neighborhoods like these—places where people live simply, where resources are limited, and where the environment shows the marks of time, constant use, and neglect.

We had to harness the essence of communities living in disarray. This means paying attention to every detail that makes a place feel real and lived-in. We studied how houses are built close together, often sharing walls or standing only a few feet apart, creating narrow streets and tight spaces. We looked at how roofs are made of different materials—some patched many times, some slanted to let rain run off, but many in need of repair. We noticed how

walls are marked with dark lines showing how high floodwaters have reached in the past, leaving stains and dirt behind.

We also included details that explain why flooding happens so easily: streets that slope downward toward the lowest parts of the area, canals and drainage channels filled with waste and debris, and large, unfinished concrete structures standing nearby—half-built, unused, and silent reminders of projects that were promised but never completed. We added piles of discarded materials, old tires, broken furniture, and vegetation growing in neglected corners, all of which show the lack of proper maintenance and care.

We added a small amount of exaggeration not to make things look worse than they are, but to make the feeling clearer and easier to understand. We emphasized how narrow the streets are to show how quickly water can fill them. We made the blockages in canals more visible to explain why the water stays high for so long. We made the unfinished structures look larger and more imposing to highlight the gap between promise and reality. Throughout this work, we ensured that the scale and perspective match our storyboard, so that when we place our characters into these spaces, they fit naturally and feel like they truly belong there.

PRODUCTION PHASE (ONGOING)

Following the completion of the pre-production phase, the project proceeds to the production phase, which serves as the stage where all planning, preparation, and design work are transformed into actual content for the animated film. This phase is considered one of the most demanding and time-intensive parts of the entire production process because it involves the creation of every visual element that audiences will eventually see on screen. While pre-production focuses on conceptualization and planning, production focuses on execution. It is during this stage that the story begins to take shape visually, allowing characters, environments, and narrative events to move beyond ideas and become fully realized components of the film.

For the development of Sa Likod ng Baha, the production phase involves a combination of 2D animation techniques and 3D environmental design. Since the project follows a hybrid production approach, additional care is required to ensure that both artistic styles work together harmoniously. Every scene, character movement, environmental interaction, and visual effect must be carefully developed and reviewed to maintain consistency throughout the film. Because of the complexity of the production workflow, the researchers follow a structured process that allows tasks to be completed systematically while ensuring that creative objectives remain aligned with the project's overall vision.

Asset Creation and Preparation

One of the first major tasks during the production phase is the creation and preparation of assets. Assets refer to all visual components required for the animation, including characters, props, backgrounds, environmental objects, textures, and supporting visual elements. These assets serve as the foundation of every scene and must be completed before animation can proceed effectively.

The researchers begin by reviewing all character designs developed during pre-production. Although initial concepts have already been approved, additional refinements are often necessary before animation begins. Character proportions, clothing details, facial features, and color palettes are carefully examined to ensure consistency. Since animated characters appear in multiple scenes from different camera angles and under various lighting conditions, maintaining visual consistency is essential.
For the main characters, special attention is given to facial expressions and emotional states. Because Sa Likod ng Baha relies heavily on emotional storytelling, the ability of characters to communicate feelings through expression plays a crucial role in audience engagement. The researchers create a wide range of facial expressions that represent different emotional conditions, including happiness, concern, sadness, fear, frustration, hope, and determination. These expression sets provide animators with the flexibility needed to portray subtle emotional changes throughout the narrative.

Character movement references are also developed during this stage. Different poses are prepared for walking, standing, sitting, running, climbing, carrying objects, and interacting with floodwaters. These references help maintain consistency across scenes and support realistic character animation. By establishing clear movement guidelines early in production, the researchers reduce the likelihood of inconsistencies that could affect immersion.

In addition to character assets, environmental assets are also prepared. Objects commonly found within the film's setting, such as houses, furniture, electrical poles, street signs, vegetation, boats, and debris, are created and organized according to scene requirements. These environmental details contribute to world-building and help establish a believable setting that supports the narrative.
Background elements are prepared separately to allow greater flexibility during compositing and scene assembly. Different layers are created to represent foreground, middle-ground, and background components. This layered approach increases depth perception and allows for more dynamic camera movement during animation.

Refinement of 3D Environments

Once asset preparation is complete, the researchers focus on refining the 3D environments that form the visual foundation of the film. Since environmental storytelling plays a significant role in Sa Likod ng Baha, creating realistic and emotionally meaningful locations is a priority throughout production.

The initial models developed during pre-production are enhanced with additional details and textures. Surface materials such as concrete, wood, metal, fabric, and vegetation are refined to improve visual realism. These materials are adjusted to respond naturally to lighting conditions and environmental changes throughout the story.
Texturing is particularly important because it adds visual richness to otherwise simple models. Buildings receive weathering effects, roads display signs of wear, and environmental objects are given realistic surface characteristics. These details help communicate the passage of time and reinforce the lived-in quality of the setting.
A unique challenge within the project involves representing different stages of flooding. To address this requirement, multiple environmental variations are created for many locations. The researchers prepare versions of each setting that correspond to different flood levels and weather conditions. These variations allow the environment to evolve alongside the narrative and reflect the increasing impact of the disaster on the community.

For example, a residential street may appear dry and peaceful during earlier scenes, partially flooded during the middle portions of the story, and heavily submerged during later sequences. Creating these environmental states requires careful planning to ensure continuity and realism.

Lighting setups are also prepared for different times of day and weather conditions. Morning, afternoon, evening, and nighttime lighting variations are developed to support scene requirements. Changes in lighting not only improve visual diversity but also contribute to emotional storytelling by influencing mood and atmosphere.
Character Animation

Character animation represents the core activity of the production phase. Once visual assets and environments are ready, the researchers begin bringing the characters to life through movement and performance

Animation is completed on a scene-by-scene basis. This structured approach allows the team to focus on specific narrative moments while maintaining quality control throughout production. Each scene begins with the placement of key poses that establish the overall

movement and emotional intent of the sequence. Once these key poses are approved, additional frames are added to create smooth transitions and realistic motion.
The researchers place significant emphasis on realism and emotional authenticity. Rather than relying on exaggerated cartoon movements, the animation style is designed to reflect natural human behavior. This approach supports the project's dramatic tone and helps audiences connect more deeply with the characters.
Walking animations receive particular attention because they occur frequently throughout the film. The researchers study reference footage and observe real-world movement patterns to ensure accuracy. Factors such as age, emotional state, fatigue, and environmental conditions influence how characters move. A person walking confidently exhibits different body language compared to someone experiencing fear, grief, or exhaustion.

Floodwater introduces additional complexity to character movement. Walking through water requires altered balance, slower pacing, and increased physical effort. The researchers carefully animate these interactions to ensure that movement reflects the resistance created by the environment. These details contribute significantly to realism and help audiences appreciate the challenges faced by the characters.
Facial animation is equally important. Small changes in eye movement, eyebrow position, mouth shape, and head orientation can communicate powerful emotions without requiring dialogue. Because the film emphasizes visual storytelling, subtle facial performances often carry substantial narrative weight.

Emotional Performance and Storytelling
Beyond technical accuracy, the researchers focus on emotional performance throughout the animation process. Every movement is evaluated not only for realism but also for its contribution to storytelling.

Characters communicate emotions through posture, gestures, pacing, and body language. A slumped posture may indicate sadness or defeat, while tense shoulders may suggest anxiety or stress. Even moments of stillness are carefully designed because silence and inactivity can often communicate emotions more effectively than exaggerated actions.
The researchers continuously ask whether each movement supports the intended emotional message of the scene. If a character is experiencing loss, uncertainty, or fear, their animation should reflect those feelings in a believable manner. This emphasis on emotional authenticity helps create a stronger connection between audiences and the narrative.
Memory sequences involving Linda require a distinct animation approach. These scenes are designed to feel softer, warmer, and more reflective compared to present-day events. Character movements within these memories are often slower and more fluid, reinforcing feelings of nostalgia and emotional connection.

Integration of 2D and 3D Elements

One of the most technically challenging aspects of production involves integrating 2D characters with 3D environments. Because these elements are created using different methods and artistic approaches, significant effort is required to ensure visual consistency.

The researchers begin by matching character scale and positioning within the environment. Characters must appear naturally grounded within each location and interact convincingly with surrounding objects. Perspective adjustments are carefully applied to ensure that characters align properly with environmental geometry.

Lighting consistency is another critical consideration. Characters must respond to environmental lighting in ways that match surrounding objects. Highlights, shadows, and reflected light are adjusted to ensure visual harmony between 2D and 3D components.

Shadow integration is particularly important because inaccurate shadows can immediately break immersion. Character shadows are carefully positioned and adjusted according to environmental lighting conditions. This process helps reinforce spatial relationships and increases realism.

Water interaction presents an additional challenge. Characters moving through floodwaters must generate ripples, splashes, and displacement effects that correspond accurately to their actions. Achieving these interactions requires close coordination between animation and environmental systems.

Scene Review and Quality Assurance

Quality control remains an ongoing responsibility throughout production. As each scene is completed, the researchers conduct detailed reviews to identify areas requiring improvement.

These reviews evaluate multiple factors, including animation smoothness, visual consistency, emotional clarity, environmental integration, and technical accuracy. Any issues identified during evaluation are documented and addressed before the scene progresses to later stages of production.

The researchers also assess whether scenes effectively communicate their intended narrative purpose. If a sequence appears confusing or emotionally ineffective, revisions are implemented to strengthen storytelling.

Consistency checks are particularly important because production occurs over an extended period. Characters, environments, and visual effects must remain coherent throughout the film despite being created at different times. Maintaining this consistency helps ensure a seamless viewing experience.

Production Scheduling and Progress Monitoring

Because the production phase involves numerous interconnected tasks, effective scheduling is essential. The researchers establish production milestones to monitor progress and maintain project momentum.
Tasks are organized according to priority and complexity. Critical scenes are often completed first to ensure that major narrative components receive sufficient attention. Regular progress evaluations help identify potential delays and allow adjustments to be made when necessary.

The use of milestones also improves team coordination by providing clear objectives and deadlines. By breaking production into manageable stages, the researchers are able to maintain productivity while preserving quality standards.
Ongoing Development and Continuous Improvement
Although the production phase follows a structured workflow, it remains flexible enough to accommodate improvements and creative discoveries. As scenes are completed, new insights often emerge regarding storytelling, animation techniques, and visual presentation.

The researchers embrace this iterative process by continuously refining their work. Feedback from reviews, test screenings, and internal evaluations informs subsequent revisions. This commitment to continuous improvement ensures that the project evolves throughout production and benefits from ongoing learning experiences.
Ultimately, the production phase transforms the vision of Sa Likod ng Baha into a tangible cinematic experience. Through asset creation, environment development, character animation, emotional performance, technical integration, and rigorous quality control, the researchers gradually construct a film that reflects both artistic ambition and technical craftsmanship. While the process remains ongoing, each completed scene represents meaningful progress toward achieving the project's objective of delivering a visually compelling and emotionally resonant animated narrative.

POST-PRODUCTION PHASE (ONGOING)

The post-production phase serves as the final stage in the creation of Sa Likod ng Baha. After completing the pre-production and production stages, all visual, audio, and animated assets are gathered and refined during post-production to create a cohesive and emotionally engaging film. This stage is where the separate components of the project are carefully assembled into a unified narrative experience. While animation production focuses on creating individual scenes and assets, post-production focuses on improving, polishing, and enhancing these elements so that they function effectively as a complete film.
For this project, post-production is an ongoing process that continues throughout the development cycle. As new scenes are completed, they are integrated into the post-production

workflow for review and refinement. This approach allows the researchers to identify issues early and maintain consistency throughout the film. The goal of post-production is not only to improve visual quality but also to strengthen storytelling, emotional impact, and audience immersion.

Compositing and Scene Integration

The first major activity in the post-production phase is compositing. Compositing is the process of combining all visual elements of a scene into a single final image. During production, characters, backgrounds, lighting effects, shadows, environmental assets, and camera movements are often created separately. These individual components must be integrated carefully to ensure that they appear as one cohesive environment.

In Sa Likod ng Baha, compositing plays a significant role because the project combines 2D character animation with 3D environmental design. Since these elements originate from different workflows, additional adjustments are required to ensure visual harmony. The researchers carefully align characters within the environment, adjust scale relationships, and refine positioning so that every object appears naturally placed within the scene.

Depth is another important consideration during compositing. Layers are arranged according to their distance from the camera to create a stronger sense of space. Foreground elements, middle-ground objects, and background environments are positioned strategically to guide viewer attention and increase visual realism. This layered approach helps audiences perceive the environment as a believable and immersive world rather than a collection of separate assets.

Transparency and blending settings are also adjusted during compositing. Environmental effects such as fog, rain, reflections, and atmospheric particles require careful integration to ensure that they interact naturally with other visual elements. These adjustments contribute to a more polished final presentation and help maintain visual consistency throughout the film.

The compositing process also includes reviewing scene continuity. The researchers examine transitions between shots to ensure that movement, lighting, and environmental conditions remain consistent. This attention to detail prevents visual distractions and allows viewers to remain focused on the story.

Visual Effects Development

Following compositing, visual effects are added to enrich the environment and strengthen emotional storytelling. In Sa Likod ng Baha, visual effects serve both artistic and narrative purposes. Rather than functioning purely as decorative additions, these effects help communicate mood, atmosphere, and environmental conditions.

Rain is one of the most important visual effects in the project. Since flooding is a central element of the narrative, rainfall becomes a recurring visual motif throughout the film. Different rain intensities are used depending on the emotional requirements of each scene.

Gentle rainfall is often used during reflective or emotional moments, while heavier rainfall is utilized during scenes involving tension, hardship, or dramatic developments.
Water simulation is another essential component of the visual effects process. Floodwaters are animated to reflect realistic movement and interaction with the environment. Ripples, waves, and currents are carefully designed to respond naturally to character actions and environmental conditions. These details contribute to the realism of the setting and reinforce the impact of the flood on the characters' lives.
Reflections are also incorporated throughout the film. Water surfaces reflect buildings, lights, and moving objects, creating additional visual depth. These reflections help establish realism while also enhancing the film's visual style. In certain scenes, reflections are used symbolically to reinforce themes of memory, loss, and emotional introspection.

Mist and atmospheric effects are added to increase environmental depth. These elements soften distant objects and create a stronger sense of scale within the environment. Atmospheric effects are particularly useful during scenes that emphasize uncertainty, isolation, or emotional distance.
Lighting effects are refined as part of the visual effects workflow. Dynamic lighting changes are used to support storytelling and guide audience attention. Light sources such as streetlights, interior lamps, and natural sunlight are adjusted to create mood and highlight important narrative elements. By carefully controlling light and shadow, the researchers strengthen both visual composition and emotional communication.
Color Grading and Emotional Tone
Color grading is one of the most influential aspects of post-production because it directly affects audience perception and emotional response. Through color grading, the researchers adjust brightness, contrast, saturation, and overall color balance to create a consistent visual identity for the film.
The color grading process begins by establishing a baseline look for the project. Since Sa Likod ng Baha explores themes of family, loss, resilience, and social issues, the visual style must support these emotional themes without becoming distracting or overly stylized. The researchers carefully balance realism with artistic expression to achieve this goal.
In the earlier portions of the story, warmer tones are emphasized to represent stability, familiarity, and emotional connection. Soft sunlight, natural skin tones, and balanced color palettes help create a sense of comfort and normalcy. These visual characteristics establish an emotional foundation that audiences can connect with before the story transitions into more difficult circumstances.
As the narrative progresses, color palettes gradually shift toward cooler and darker tones. Blues, grays, and desaturated colors become more prominent as the effects of flooding and hardship intensify. This gradual transition mirrors the emotional journey of the characters and encourages audiences to experience these changes on a subconscious level.

Memory sequences receive special attention during the color grading process. These scenes often feature warmer highlights, softer contrasts, and slightly elevated saturation levels. Such adjustments help distinguish memories from present-day events while reinforcing feelings of nostalgia and emotional attachment.
Consistency remains a primary objective throughout color grading. Since scenes may be created at different times and under varying production conditions, color correction is performed to ensure that visual continuity is maintained across the entire film.

Sound Design and Audio Production
Sound design represents another critical component of post-production. While visuals communicate information through imagery, sound provides atmosphere, realism, and emotional depth. Effective sound design allows audiences to feel present within the world of the film and strengthens their connection to the story.
The sound design process begins with identifying the audio requirements of each scene. These requirements include dialogue, ambient sounds, environmental effects, and music. Each category serves a unique purpose within the overall soundscape.
Environmental audio is particularly important in Sa Likod ng Baha. Since the story is heavily influenced by flooding and changing weather conditions, natural sounds help establish immersion and realism. Rainfall, flowing water, distant thunder, wind, and environmental ambience are carefully selected and arranged throughout the film.
Different surfaces produce different rain sounds. Rain striking rooftops creates a distinct acoustic texture compared to rain falling on water or concrete surfaces. By incorporating these variations, the researchers create a richer and more believable auditory experience.
Water sounds are layered throughout many scenes. Gentle ripples, rushing currents, splashes, and submerged movement help reinforce the presence of flooding within the environment. These sounds contribute significantly to the film's atmosphere and thematic identity.
Ambient audio is used to provide context and depth. Distant conversations, animal sounds, traffic noise, and neighborhood activity help establish location and make environments feel inhabited. Even subtle background sounds contribute to audience immersion by preventing scenes from feeling empty or artificial.
Dialogue editing ensures that character voices remain clear and understandable. Audio recordings are cleaned to remove unwanted noise, volume levels are balanced, and timing adjustments are made when necessary. These refinements improve communication and maintain professional audio quality.

Music is used selectively throughout the project. Rather than dominating scenes, music functions as emotional support for important narrative moments. Gentle melodies and restrained arrangements allow natural environmental sounds and dialogue to remain the

primary focus. This approach aligns with the project's goal of emphasizing realism and emotional authenticity.

Quality Control and Review Process

An essential aspect of post-production involves continuous review and evaluation. Once scenes are assembled, the researchers conduct multiple screenings to identify areas requiring improvement. These reviews focus on technical quality, narrative clarity, emotional effectiveness, and overall audience experience.

During each review session, the researchers analyze pacing, scene transitions, sound balance, visual consistency, and emotional flow. Any issues identified during evaluation are documented and addressed through revision. This iterative process continues until the desired level of quality is achieved.

Particular attention is given to pacing because it significantly affects audience engagement. Scenes that feel rushed may reduce emotional impact, while overly slow scenes may weaken narrative momentum. Through repeated viewing and refinement, the researchers work to achieve a balanced rhythm that supports both storytelling and audience immersion.

Sound and image synchronization is also examined carefully. Even minor timing inconsistencies can disrupt audience engagement. Ensuring accurate synchronization improves realism and enhances overall viewing quality.

The review process extends beyond technical concerns. The researchers also evaluate whether the emotional objectives of each scene are being achieved. If a sequence fails to generate the intended response, adjustments may be made to visual effects, sound design, editing, or pacing.

Final Refinement and Preparation

The final stage of post-production involves preparing the completed film for presentation and evaluation. All project elements undergo one final quality assessment to ensure that technical standards are met and creative objectives have been achieved.

During this stage, rendering settings are optimized to produce the highest possible visual quality while maintaining compatibility with presentation requirements. Audio levels are checked one final time, and any remaining inconsistencies are corrected.

The researchers also review subtitles, credits, and supporting materials to ensure completeness and accuracy. Final exports are generated in the required formats for evaluation, presentation, and archival purposes.

Ultimately, the post-production phase transforms individual assets into a complete cinematic experience. Through compositing, visual effects, color grading, sound design, and continuous refinement, Sa Likod ng Baha evolves from a collection of animated scenes into a cohesive narrative capable of communicating its themes effectively. This stage plays a crucial role in ensuring that the final film not only meets technical standards but also delivers the emotional depth and authenticity that define the project's creative vision.

FINAL TOUCHES AND PRESENTATION (ONGOING)

The final touches and presentation stage represents the concluding phase of the development process for Sa Likod ng Baha. After completing the major tasks involved in pre-production, production, and post-production, the researchers enter a stage focused on refinement, evaluation, preparation, and presentation. While this phase may appear less demanding than the earlier stages of development, it plays a critical role in determining how effectively the finished project will be received by its intended audience. A well-produced film can lose much of its impact if it is presented poorly, contains avoidable errors, or lacks sufficient context for viewers to understand its purpose. For this reason, the researchers treat the final stage not merely as a conclusion to production but as an opportunity to strengthen the overall quality and effectiveness of the project.

The purpose of this stage is to ensure that every element of the film is functioning as intended and that the project is ready for public presentation, evaluation, and documentation. It is during this phase that all remaining adjustments are made, supporting materials are completed, and strategies for presentation are carefully planned. The researchers recognize that completing a film involves more than simply exporting the final video. It also involves preparing audiences to engage with the work, providing appropriate context, and ensuring that the project's objectives are communicated clearly.

Because Sa Likod ng Baha addresses themes related to disaster, corruption, grief, resilience, and community experiences, careful presentation becomes especially important. The researchers aim not only to showcase technical achievements but also to encourage meaningful reflection on the social and emotional themes explored throughout the narrative. As a result, the final touches and presentation phase combines technical evaluation, creative reflection, documentation, and audience preparation into a comprehensive process that supports the project's overall goals.

Comprehensive Quality Review

The first major activity within this phase is the comprehensive quality review. Before the project can be considered complete, every component of the film must undergo careful examination. The researchers conduct multiple reviews to identify any remaining issues that could affect the viewing experience. These reviews focus on both technical quality and storytelling effectiveness.

The quality review process begins with a full viewing of the film from beginning to end. Rather than focusing on individual scenes, the researchers evaluate the project as a complete narrative experience. This approach allows them to assess pacing, continuity, emotional progression, and overall flow. Watching the film in its entirety provides valuable insight into how audiences are likely to experience the story.

During each review session, special attention is given to visual consistency. Character appearances, environmental details, lighting conditions, and color grading are examined to ensure continuity throughout the film. Even small inconsistencies can distract audiences and reduce immersion, making thorough review essential.

Animation quality is another major focus of evaluation. The researchers carefully observe character movements, facial expressions, environmental interactions, and scene transitions. Any motion that appears unnatural, abrupt, or inconsistent is identified for correction. Since the project emphasizes emotional storytelling, smooth and believable animation is particularly important for maintaining audience engagement.
The review process also includes an assessment of environmental realism. Floodwaters, rainfall, atmospheric effects, and environmental interactions are examined to ensure that they function consistently across scenes. Because the environment plays a central role in the narrative, maintaining realism contributes significantly to the credibility of the story.

Lighting and shadow behavior are evaluated throughout the film. The researchers verify that light sources remain consistent and that shadows accurately correspond to environmental conditions. These details help reinforce spatial relationships and improve visual cohesion.

Audio Evaluation and Sound Quality Assurance

In addition to visual review, the researchers conduct a detailed evaluation of all audio components. Sound quality has a significant influence on audience immersion and emotional engagement. Even visually impressive scenes can lose impact if audio elements are unclear, inconsistent, or poorly balanced.

Dialogue clarity is one of the primary areas of focus during audio evaluation. The researchers listen carefully to ensure that all spoken lines remain understandable throughout the film. Audio levels are adjusted when necessary to prevent dialogue from being overshadowed by environmental sounds or music.

Environmental audio is also reviewed extensively. Since Sa Likod ng Baha relies heavily on atmospheric sound design, sounds such as rainfall, water movement, wind, distant activity,

and ambient environmental noise must function harmoniously. These sounds are evaluated not only for realism but also for their contribution to storytelling and mood.
Music placement and volume levels receive careful attention as well. The researchers evaluate whether musical cues support emotional moments without becoming distracting. Because the project's sound design prioritizes realism and subtlety, music is used selectively and must remain balanced with other audio elements.

Synchronization between audio and visual elements is another critical component of quality assurance. Sound effects, dialogue, and environmental sounds must align accurately with on-screen actions. Even minor synchronization issues can disrupt audience immersion and reduce perceived quality.

Narrative and Emotional Evaluation

Technical quality alone does not determine the success of a storytelling project. For this reason, the researchers also conduct a thorough evaluation of the film's narrative and emotional effectiveness.

The first objective of this evaluation is to determine whether the story communicates its intended themes clearly. The researchers assess whether audiences can understand the relationships between characters, recognize key narrative developments, and interpret the underlying messages presented throughout the film.

Particular attention is given to emotional progression. The researchers examine whether the emotional journey of the protagonist develops naturally and whether important moments receive sufficient emphasis. Scenes involving grief, memory, resilience, and personal reflection are reviewed to ensure that they achieve the intended emotional impact.

The pacing of emotional moments is also evaluated. Some scenes may require additional time for audiences to process important events, while others may benefit from tighter editing. By reviewing the emotional rhythm of the film, the researchers can identify opportunities to improve audience engagement.
Another aspect of narrative evaluation involves examining the balance between visual storytelling and dialogue. Since the project emphasizes the principle of "show, don't tell," the researchers assess whether visual elements successfully communicate emotions and themes without excessive verbal explanation.

Documentation and Supporting Materials

Alongside the final review process, the researchers prepare a variety of supporting materials that accompany the film. These materials provide context, explain creative decisions, and document the development process.
One of the primary supporting documents is the project summary. This summary provides a concise overview of the story, its themes, and its objectives. It serves as an introduction for audiences, evaluators, and stakeholders who may be unfamiliar with the project.

The researchers also prepare documentation describing the inspiration and motivation behind the film. This material explains why the chosen themes are significant and how they relate to real-world experiences. Providing this context helps audiences understand the purpose of the project beyond its entertainment value.

Production documentation is another important component. This documentation records the methods, techniques, and processes used throughout development. It includes information regarding pre-production planning, animation workflows, environmental design, sound production, post-production activities, and evaluation procedures.

Visual development materials are also organized during this stage. Concept art, storyboards, character designs, environmental studies, and production screenshots are compiled to illustrate the creative evolution of the project. These materials demonstrate the effort and planning involved in bringing the film to completion.

Preparation for Public Presentation

Once the film and supporting materials have been finalized, the researchers begin preparing for presentation. Presenting a capstone project requires careful planning because it involves communicating both the final output and the research process behind it.

The researchers first consider how the project will be introduced to audiences. An effective introduction provides viewers with enough context to appreciate the story without revealing important narrative developments. This balance is important because it allows audiences to engage with the film naturally while still understanding its broader purpose.

Presentation scripts and speaking notes are prepared to guide discussions before and after screenings. These materials help ensure that key points are communicated clearly and consistently. The researchers identify the most important aspects of the project that should be highlighted during presentations, including thematic objectives, technical innovations, and creative challenges.

The researchers also prepare responses to potential questions from evaluators and audience members. Since capstone presentations often involve discussions regarding methodology, creative decisions, and technical implementation, anticipating questions helps improve confidence and presentation quality.

Audience Engagement and Discussion

A significant objective of the presentation stage is encouraging meaningful audience engagement. The researchers view the film not simply as a finished product but as a starting point for conversation and reflection.

Following screenings, opportunities for discussion are provided whenever possible. Audience members are encouraged to share their interpretations, emotional reactions, and questions regarding the story. These discussions provide valuable insight into how the film is understood and experienced by different viewers.

The researchers recognize that audiences may respond to the film in diverse ways depending on their personal experiences and perspectives. By creating a welcoming environment for discussion, the presentation process becomes more inclusive and meaningful.

Audience questions may focus on technical aspects of production, narrative themes, artistic decisions, or social issues addressed within the film. The researchers prepare to engage thoughtfully with these topics while maintaining respect for differing viewpoints.
Reflection on the Creative Journey.

The final touches and presentation stage also provides an opportunity for reflection. Looking back on the development process allows the researchers to evaluate their growth, identify lessons learned, and recognize accomplishments achieved throughout the project.

The creation of Sa Likod ng Baha involves numerous challenges, including technical difficulties, artistic decisions, time management requirements, and collaborative responsibilities. Reflecting on these experiences helps the researchers better understand the strengths and weaknesses of their workflow.
This reflection extends beyond technical achievements. The researchers also consider how the project influenced their understanding of storytelling, animation, visual communication, and audience engagement. These insights contribute to professional growth and may inform future creative endeavors.

Final Preparation for Evaluation

Before formal submission and evaluation, the researchers conduct one final review of all project materials. The film, documentation, presentation materials, and supporting files are checked for completeness and accuracy.
File organization is reviewed to ensure that all required materials are accessible and properly labeled. Backup copies are created to prevent data loss and ensure readiness for presentation.

The researchers verify that technical requirements specified by the institution are satisfied, including file formats, documentation standards, and presentation guidelines. This final verification helps prevent avoidable issues during evaluation.

Conclusion
The Final Touches and Presentation phase represents the culmination of all work completed throughout the development of Sa Likod ng Baha. Although the major creative and technical tasks have already been completed, this stage remains essential because it determines how effectively the project is communicated, understood, and appreciated by its audience.

Through comprehensive quality review, audio and visual evaluation, narrative assessment, documentation preparation, audience engagement planning, and reflective analysis, the researchers ensure that the project reaches its highest possible level of quality. More importantly, this phase allows the story's themes, emotions, and messages to be presented with clarity, respect, and authenticity.
Ultimately, the goal of this stage is not simply to finish the film but to prepare it for meaningful sharing. By carefully refining every aspect of the project and thoughtfully planning its presentation, the researchers ensure that Sa Likod ng Baha is ready to be viewed, discussed, and understood as both a creative work and a reflection on important human experiences.

TANGIBLE AND INTANGIBLE BENEFITS

Every project leaves behind more than just what can be seen. Sa Likod ng Baha produces both tangible benefits—things we can hold, view, and share—and intangible benefits—things we learn, feel, and carry with us into the future. Together, they make the work far more valuable than just a school requirement.

TANGIBLE BENEFITS

Tangible benefits are the things that remain as evidence of our work. They are the outputs we produce, the skills we develop, and the materials we create. They are useful, practical, and long-lasting.

1. Something to Show for Our Work

The biggest win is simply having a finished short film. It is one thing to have an idea written down or drawn in a sketchbook; it is something entirely different to take that idea and turn it into a complete, moving, and meaningful piece of work. Many people have creative thoughts, but few have the discipline and determination to carry them through to completion. For us, finishing this film is proof that we can do exactly that.

It's a practical way to show anyone—like a future audience or a fellow community of artists—that we can actually take an idea from a rough sketch to a final, polished film. This is more than just a grade or a requirement for graduation. It is a portfolio piece that represents our ability to manage time, follow a plan, solve problems, and produce quality results. In the creative field, your work speaks louder than any description of what you can do. Having a finished animated short film gives us something concrete to present to teachers, mentors, peers, and anyone else who wants to know what we are capable of.

The film itself follows a clear structure and uses established animation and storytelling principles. It shows that we understand how to build a world, how to develop a narrative, how to use camera angles and lighting to create mood, and how to maintain consistency from one scene to the next. It also demonstrates our ability to combine different techniques—2D character animation and 3D environments—into a single, unified visual experience. This technical and creative achievement is something we can refer to and build upon for years to come.

Alongside the film, we also produce a complete set of supporting materials: the script, storyboard, design references, production notes, and final files. These documents provide a full picture of how the project was developed, making it easier for others to understand our process and for us to review our own work later. This collection of materials is a permanent record of our efforts and a valuable resource for learning and improvement.

2. Learning to Team Up

No project of this size can be completed alone. Every step requires different perspectives and different skills. One of the clearest and most useful tangible benefits is the experience we gain in working together as a team.

Since we aren't using voice actors, we're focusing on working with sound and music people. This decision changes the way we approach the story. Without dialogue to explain what is happening or what characters are feeling, the film relies entirely on what the audience sees and hears. This makes the work of sound designers and composers even more essential. It also means we have to communicate more carefully and more deeply with them to make sure their work matches the mood and pace of the visuals.

It's a great chance to practice sharing our ideas and seeing how different pieces of art come together as a team. We learn how to describe the atmosphere we want—heavy, tense, quiet, or sad—in ways that people who do not work in animation can understand. We learn how to listen when others have suggestions or different ways of seeing things. We learn how to give feedback that is honest but kind, and how to accept feedback without taking it personally. We learn that the goal is not for one person's idea to win, but for the whole project to become stronger.

Through this collaboration, we develop clear systems for communication and file sharing. We create organized folders, labeled files, and shared notes so that everyone can access what they need and understand what stage the work is in. This experience in coordination and teamwork is a skill that applies to almost any career path we choose. It is something we can take with us into any workplace, and it makes us more reliable and effective partners in future projects.

3. Getting Better Every Day

Working on our animation short film is really just a way for us to practice. There is no shortcut to becoming skilled at anything. Knowledge comes from learning, but skill comes from doing. This project gives us the chance to apply everything we have studied in class to a real, ongoing work.

Every hour we spend drawing or building 3D sets makes us a little bit faster and more confident with our tools than we were the day before. At the beginning, tasks that now take us an hour might have taken three or four. We used to struggle with managing layers, adjusting lighting, or making sure proportions stayed consistent. Now, those actions become more natural and automatic. We learn to anticipate problems before they happen, and we find solutions more quickly when they do.

The results of this practice are visible in every part of the film. Compare the first test sketches to the final character designs, or the first rough models to the finished environments, and you can see the difference clearly. The lines are cleaner, the details are more accurate, the movement is smoother, and the lighting feels more natural. We also develop our own way of working—our own rhythm and organization—that helps us stay productive and focused.

This growth is not just technical. It also applies to our understanding of storytelling. We learn what details make a scene feel real, what pacing keeps the audience engaged, and what visual choices help convey emotion. These lessons are written into the work itself, and they become part of our ability to create in the future.

4. A Simple Calling Card

We don't expect to be famous, but sharing our work lets people see our style. In the creative world, having a distinct way of seeing and creating is what helps you stand out. There are many ways to make animation, but each creator or team brings their own interests, values, and perspective to the work.

It's a small way to put our names out there so that if someone likes our "dark and rainy" vibe, they'll know who we are. Sa Likod ng Baha has a specific feeling—calm but heavy, realistic but emotional, simple but meaningful. It shows that we are interested in stories that reflect real life, that we care about atmosphere and mood, and that we use our skills to say something important rather than just show off what we can do.

This film becomes a calling card that introduces us to others. It is something we can share with teachers, potential employers, fellow creators, and audiences. It answers the question, "What kind of work do you make?" in a way that words cannot. It opens doors to conversations, collaborations, and opportunities that we might not otherwise have. Even if it is just seen by a small group, it helps us build a reputation as creators who are thoughtful, responsible, and skilled.

INTANGIBLE BENEFITS

Beyond what we can see and measure, this project gives us benefits that shape our character, our thinking, and our sense of purpose. These are the lessons that stay with us long after the files are saved and the presentation is over.

1. A Sense of Pride

Bringing the story of tragedy to life from just a small idea makes us feel incredibly proud. It is an amazing feeling to look back at where we started—with nothing but notes, observations, and a rough concept—and then see the finished film. It represents hundreds of hours of work, countless revisions, and many moments of effort when progress felt slow or difficult.

Finishing a project that deals with such real, heavy emotions shows us that we have the heart and the focus to see a difficult task through to the end. Creating a story about hardship, betrayal, and survival requires more than just technical skill. It requires the ability to feel and understand what those experiences mean, and the patience to represent them with respect. When we watch the final film, we know that we have done our best to tell the story honestly. That feeling of achievement is something that builds our confidence and strengthens our belief in our own abilities.

This pride also comes from knowing that we took this work seriously. We did not treat it as just a school assignment to rush through. We approached it with care, empathy, and responsibility. We wanted to make something that matters, and we did. That sense of purpose and accomplishment stays with us, reminding us that we are capable of handling difficult tasks and creating work that has meaning.

2. Finding Our Own Style

This film is our chance to play and experiment. In many of our classes, we are taught standard techniques and given exercises to learn the basics. But this project gives us the freedom to make our own choices—to decide what looks right, what feels right, and what serves the story best.

By mixing 2D characters with a 3D world, we are discovering our own unique “voice” as artists and learning how to tell a story that feels both dark and beautiful. This combination is not just a technical trick—it is a creative choice that defines how the audience experiences the story. The soft, hand-drawn look of the characters keeps them human and expressive, while the detailed, three-dimensional environments give them a world that feels large, heavy, and sometimes overwhelming. This balance creates a specific atmosphere that belongs to our vision.

Through every decision we make—from the color palette to the lighting, from the timing of movements to the pacing of scenes—we learn what we are drawn to. We discover that we prefer stories that are quiet but powerful, that focus on feelings rather than action, and that

look at the world with a thoughtful, honest eye. This discovery of our own style and voice is one of the most important things we can gain as creators. It guides us in every project we do from now on.

3. Making a Connection

At the end of the day, we want Sa Likod ng Baha to stay with the people who watch it. Art is not complete until it is shared. Its true power lies in its ability to cross the space between the person who makes it and the person who experiences it.

Knowing that our art might move someone or make them think about life in a new way is the most rewarding part of this whole journey. We hope that when people watch the film, they will not just see an animation—they will see a story that reflects real lives. We hope it makes them think about the communities around them, about the relationship between leaders and the people they serve, and about the difference between what is promised and what is delivered. Even if it only makes one person stop and think a little deeper, even if it only helps one person feel a little more understanding, our work has succeeded.

This ability to connect is a gift that changes how we see our role as creators. We realize that what we make can have an impact. It can raise questions, build empathy, and encourage reflection. This understanding gives our work a deeper purpose and reminds us that every story we tell carries responsibility and meaning.

The benefits of Sa Likod ng Baha go far beyond the grades or the requirements of our course. They are found in the film we created, the skills we built, the teamwork we practiced, and the confidence we gained. They are also found in the pride we feel, the style we discovered, and the connection we hope to make with those who watch it.

Together, the tangible and intangible benefits of this project turn a simple idea into a meaningful experience. They give us tools for our future careers, lessons for our personal growth, and a deeper understanding of what it means to create work that matters.

Proposed Budget:
When we start a creative project, it is easy to become so focused on the story and the art that we forget about the practical side of things. But no matter how good the idea is, it cannot be completed without resources. Even if you do not need to build big sets or hire many people,

there are still small, daily costs that come with working together, traveling, and staying focused. For Sa Likod ng Baha, we have planned our budget carefully, keeping it simple, realistic, and within our means.

This section explains everything about how we plan to spend our money, what we hope to gain from the project, and how we will share our work with others. It is written in plain language, so you can see exactly how we think, how we plan, and what we hope to achieve. We believe that being open and honest about our finances and our goals is part of making this project transparent and meaningful.

PROPOSED BUDGET

It is essential to consider a budget for the Sa Likod ng Baha animation project. A budget is not just a list of numbers written on paper—it is a roadmap that helps us make sure we have what we need when we need it. It helps us avoid wasting money, keeps us organized, and ensures that we can finish what we start. The proposed budget includes estimated costs for various elements involved in the production process, divided according to the different stages of work.

Pre-Production Phase

Pre-production is where everything begins. It is the time when we turn ideas into plans, when we write the story, draw the first sketches, design the characters, and build the world where the story will take place. During this time, we meet frequently—sometimes in classrooms, sometimes at home, sometimes in libraries or even out in the community to observe real places. All this activity comes with small but necessary costs.

- Lunch: Php 2,500 per month

Working on a project like this takes time and energy. When we gather for meetings or work sessions, we often spend four, five, or even six hours together. It is hard to stay focused, think clearly, and work well when you are hungry. So, we set aside money for simple, healthy meals. We do not go to expensive restaurants or order fancy food. We choose affordable meals that are filling and nutritious—dishes we are used to, things that give us energy without costing too much. Over the course of a month, this adds up to around Php 2,500 for the whole team. This amount ensures that no one has to worry about skipping meals or bringing heavy food from home every day. It is a small investment that helps us work better and stay healthy.

- Fare for Jeepney and Motorcycle: Php 2,000 per month
We do not all live in the same neighborhood, and we do not always work in the same place. Sometimes we need to travel to school, to the library, to areas where we can observe the environment, or to meet our advisers. In Iloilo City and nearby towns, jeepneys and motorcycles are the most common, reliable, and affordable ways to travel. We calculate this amount based on the distance we travel, the number of trips we make, and the standard fares in our area. This Php 2,000 covers all transportation costs for every member of the team each month. Knowing that we have this budget set aside means we do not have to stress about how to get to our meetings or research sites. It removes one more worry and lets us focus on what matters most: the work.

Production Phase

Once pre-production is finished and we have a clear plan, we move into production—the stage where we actually create the film. This is when we draw the characters, animate their movements, build the 3D environments, and start putting scenes together. It is the longest and most detailed part of the process, and it requires us to meet often to check progress, compare our work, fix mistakes, and make sure everything looks consistent.

- Lunch: Php 2,500 per month
During production, our work can become even more intense. We may spend longer hours drawing, adjusting models, or testing animations. Keeping our energy up is just as important as before. We continue to budget Php 2,500 per month for meals, sticking to the same simple and practical choices. This keeps us comfortable and focused, and it ensures that we can work through the day without interruptions or distractions.

- Fare for Jeepney and Motorcycle: Php 2,000 per month
Travel needs do not stop just because we are in production. We may still need to visit places to take reference photos, meet with our instructors for feedback, or gather materials. The transportation budget remains the same because we are careful to plan our trips wisely—combining multiple errands into one journey whenever possible, and choosing the shortest and cheapest routes. This keeps our travel costs predictable and helps us stay within our budget.

Post-Production Phase (Ongoing)

Post-production is where everything comes together. We take all the separate scenes, characters, and backgrounds and combine them into one smooth film. We add visual effects, adjust the lighting and colors, and design the sound that will bring the world to life. Even

though much of this work is done on computers, we still meet regularly to watch what we have done, discuss changes, and decide how to make the film feel complete. Because this phase is still ongoing, we continue to budget for food and travel in the same way, adjusting only if our schedule changes.

Final Touches and Presentation (Ongoing)

When the film is almost ready, we enter the final stage: checking every detail, fixing any errors, preparing our documentation, and getting ready to present our work. This includes reviewing the entire film many times, writing reports, designing posters or materials, and preparing for screenings or defense. We may need to travel to meet with advisers or attend events, so our budget remains active during this period, keeping costs low and focused only on what is truly needed.

Total Proposed Budget

When we add up all these estimated costs, we arrive at a total that is reasonable and easy to manage:

Total Proposed Budget: Php 9,000

This amount covers all the basic expenses we need to complete the project. It is not a large sum, and that is exactly how we planned it. We believe that good work does not require a big budget—it requires good planning, hard work, and creativity. We already have the computers, software, and tools we need. We do not need to pay for actors, studios, or expensive materials. The only costs we really have are the ones that help us work together: food and travel. This Php 9,000 is enough to cover these needs for the whole duration of the project, giving us the freedom to focus on making the best film we can.

RETURN ON INVESTMENT (ROI)

It is crucial to develop a comprehensive strategy to target the audience effectively, implement marketing efforts, and evaluate the return on investment (ROI). In business terms, ROI usually means how much money you get back compared to what you spent. But for a project like this, the return is much more than just money. It includes the skills we learn, the experience we gain, the recognition we receive, and the impact our work has on others. We

look at both sides—financial and non-financial—to understand the true value of Sa Likod ng Baha.

Targeting the Audience

To target the audience successfully, it is essential to identify the demographic and psychographic characteristics of the target market. In simple terms, this means knowing who will want to watch the film, who will understand it, and who will benefit from its message. Because our story deals with heavy themes—struggle, betrayal, disaster, and responsibility—we know exactly who we want to reach:

1. Animation and Art Lovers

These are people who appreciate animation not just as entertainment, but as a form of art. They enjoy seeing different styles, techniques, and ways of telling stories. They will notice the effort we put into combining 2D and 3D elements, the details in the drawings, and the atmosphere we have created. For them, Sa Likod ng Baha offers something different from the usual films: a work that is thoughtful, atmospheric, and visually unique.

2. Young Adults and Adults with Voting Power

This is our most important audience. These are people who are old enough to vote, to make decisions, and to understand how society works. They are the ones who choose leaders, who pay taxes, and who live with the results of policies and projects. Our film shows what happens when promises are broken and when systems fail. We hope it makes them think more carefully about their role in the community, about the leaders they support, and about the kind of future they want to build.

3. Film Enthusiasts and Students

These are people who love learning about how films are made. They are interested in the process—from the first idea to the final frame. They will appreciate the planning, the research, the problem-solving, and the teamwork that went into making this film. For them, Sa Likod ng Baha is an example of what can be done with limited resources but a lot of heart and effort.

Strategies to Reach the Audience

To reach these people, we do not need expensive advertising. We use simple, accessible, and affordable ways to share our work:

- Local Showings and Community Screenings

One of the best ways to share our film is to show it right here in our own area. We can arrange screenings in schools, barangay halls, libraries, and community centers. This lets us bring the story to the people who live in places similar to the one in the film, making it easier for them to connect with what they see. It also gives us the chance to meet viewers, hear their thoughts, and have meaningful conversations after the screening.

- Online Broadcast and Streaming

The internet makes it possible to share our work with anyone, anywhere. We can upload the film to platforms like YouTube, Facebook, and Vimeo. This costs almost nothing, and it allows people to watch it whenever they want. It also lets us reach a wider audience—other students, artists, teachers, and viewers from other parts of the country and even abroad. Online sharing makes the film accessible to everyone, regardless of where they live.

- Collaborations with Schools and Organizations

Because our film deals with social issues, community life, and responsibility, it is very useful for education. Schools, colleges, and organizations that work on disaster risk reduction, governance, and social awareness can use Sa Likod ng Baha as a teaching tool. This kind of partnership helps the film reach classrooms and seminars, turning it into something that helps people learn and understand important topics.

Marketing Strategy

To let people know about our film, we use simple and honest methods that fit our budget and our values:

- Teaser Campaign

Before releasing the full film, we will share a short teaser or trailer. This will show small parts of the film—the setting, the mood, the characters, and a hint of the story—without giving everything away. It builds curiosity and makes people want to see more. We will share this teaser online and play it before local screenings to introduce our work.

- Social Media Presence

We will use free platforms like Facebook and YouTube to share updates throughout the production process. We will post concept art, sketches, screenshots of the 3D environments, test animations, and stories about how we work. This lets people follow along as the film is made, and it helps them understand the effort behind it. It also creates a place where people can ask questions, leave comments, and share their thoughts.

Return on Investment (ROI)

Evaluating the ROI is crucial to measure the project's success and financial viability. But as we have said, this project is not just about making money—it is about making something meaningful. We break down the return into clear parts:

Monetization Opportunities

While our main goal is not profit, there are ways we can earn something from the work we have done:

- YouTube Revenue: Once the film is uploaded and gains enough views, we may qualify for revenue sharing.

- Educational Licensing: Schools or organizations that want to use the film may pay a small fee to license it for their programs.

- Support and Donations: Viewers who enjoy the film may choose to support us through small contributions.

- Future Projects: The experience and reputation we build from this film will help us find opportunities for paid work or collaborations in the future.

Engagement Metrics

We will track how well the film is doing by looking at:

- Number of views and watch time

- Likes, shares, and comments

- Attendance at screenings

- Ratings and feedback

These numbers tell us if we are reaching the people we want to reach and if the film is having the effect we hoped for.

Feedback and Reviews

The most valuable return we can get is knowing that our work has touched someone. When people tell us that the film made them think, made them feel, or helped them understand something new, that is worth more than any amount of money. Positive feedback also builds our confidence and opens doors for future work.

Breaking Down Costs and Recovery

To understand how we recover our investment, we look at all the costs clearly:

Fixed Expenses

These are the regular costs we pay every month:

- Food: Php 2,000

- Transportation: Php 2,000

- Additional Meals: Php 1,500 per member
Total: Php 5,500 per month

Variable Expenses

These are costs that change, but for this project, they are zero:

- Software, equipment, voice acting, music, marketing: Php 0

Accrued Expenses

No pending payments or services: Php 0

Total Expenses = Php 5,500 × Number of Months

With a total budget of Php 9,000, we can cover about two months of work, which is enough to complete the project.

Potential Revenue

At the start, our revenue begins at Php 0, but it grows over time as the film is shared and used. Even a small amount of income will quickly cover our costs, because our expenses are low.

Time to Recover

Using the formula:
Time to Recover = Total Expenses ÷ (Total Revenue − Variable Expenses)

We see that financially, the project will pay for itself in a short time once it starts earning. But more importantly, the real return is instant: we have a finished film, a portfolio, new skills, teamwork experience, and a message that can make a difference. In every way that matters, this project gives back far more than it costs.

The budget and return on investment for Sa Likod ng Baha are designed to be simple, realistic, and focused on what is truly important. We have kept our costs low by using what we already have and being careful with our spending. We have planned our strategy to reach the right audience, share our work freely, and measure success not just in money, but in learning, impact, and growth.

This project proves that you do not need a big budget to create something meaningful. With good planning, hard work, and a clear purpose, we can make a film that matters, that represents our community, and that gives us a return that will last long after the work is done.